q-Pochhammer Symbols and q-Binomial Coefficients

Algebra, Combinatorics, Analysis, Arithmetic, Summation Theory, and Dyadic Self-Similarity

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Abstract

The finite and infinite q-Pochhammer symbols and the Gaussian, or q-binomial, coefficients form a small set of notational primitives from which a remarkable amount of mathematics can be built. They simultaneously encode weighted subsets, partitions inside rectangles, lattice paths, subspaces over finite fields, Schubert cells in Grassmannians, noncommutative binomial expansions, q-difference operators, basic hypergeometric summations, theta products, partition identities, and congruences at roots of unity.

This monograph develops those themes from first principles. It proves the algebraic laws of shifted factorials; the polynomial, reciprocal, and cyclotomic structure of Gaussian coefficients; finite and infinite q-binomial theorems; weighted generalizations through elementary and complete symmetric functions; combinatorial and finite-geometric models; unimodality by $\mathfrak{sl}_2$ representation theory; the q-Gauss, q-Chu–Vandermonde, and q-Pfaff–Saalschütz summations; Jackson calculus and the q-gamma and q-beta functions; Jacobi's triple product, Euler's pentagonal theorem, and the Rogers–Ramanujan identities through Bailey pairs; cyclotomic factorization, q-Lucas and q-Babbage congruences; negative upper indices and q-Newton interpolation; and a recent family of central q-binomial/Lambert-series evaluations.

A separate part develops the dyadic specialization $q = 1/2$ arising in the theory of the Fabius and Rvachev functions. It proves the Thue–Morse exponential product, the geometric Gaussian coefficient-extraction stencil, translation-invariant exact formulae for every dyadic Fabius value, Appell and Hessenberg representations, Bernoulli–Bell moment formulae, a regular q-Toeplitz limit of finite uniform splines, Fourier–sinc inversion, and a geometric-uniform q-Fabius family with reciprocal germs and spectral q^2-Pochhammer products. Every displayed theorem used in the main development is accompanied by a proof, and the final appendices collect formulas, limiting transitions, and computational recipes.

Preface and source map

The aim is not merely to list identities, but to expose the mechanisms that repeatedly generate them. Three such mechanisms dominate:

1. finite products split into residue classes, blocks, or initial and tail factors;
2. coefficient extraction turns product factorizations into convolution identities;
3. functional equations determine analytic q-series from one initial value.

Once these mechanisms are visible, many formulas that initially appear unrelated become instances of the same principle.

The supplied archive was used throughout. In particular, the weighted subset framework of Konvalina motivates Chapter 4; Kupershmidt's treatment of the q-Newton binomial informs Chapters 11 and 17; Straub's seminar on q-binomial congruences is reflected in Chapter 16; the formalized q-series project of Lau, Lee, and Ono motivates the algebraic-versus-analytic distinctions in Chapter 23 and supplies a modern Bailey-pair route to the Rogers–Ramanujan identities; and the central q-binomial paper of Chern, Dilcher, and Jiu is developed, with a complete proof, in Chapter 21. The archive's reference chapters and encyclopedia snapshots were checked against the standard sources listed in the bibliography. Additional material was drawn from the classical books and articles of Andrews, Askey, Bailey, Gasper, Rahman, Rota, Goldman, Reiner, Stanton, White, and others, as well as the NIST Digital Library of Mathematical Functions.

The dyadic part synthesizes the q-relevant `.tex` corpus under `Analysis/FabiusFunction/docs` in the `ProveIt` repository, including `Fabius_q_Special_Functions.tex`, `Fabius_Dyadic_q_Connections.tex`, and `Signed_Reciprocal_q_Fabius_Frontiers.tex`, at repository snapshot `0e0bb9f067f5`. Statements that those expositions delegated to the accompanying Lean development are reproved here from weighted-simplex volumes, formal power-series coefficient extraction, and normally convergent products. The generalized signed and reciprocal material is restricted to proved identities; experimental divisor patterns and other conjectural frontier claims are not used as theorems.

Two regimes for q

Throughout the finite algebraic chapters, q may be an indeterminate and identities are understood in a polynomial or rational-function ring. In analytic chapters, unless stated otherwise,

$$q \in \mathbb{C}, \qquad 0 < |q| < 1.$$

For the Jackson integral and positivity statements we usually assume $0 < q < 1$. A capital Q is used for the cardinality of a finite field, so Q is a prime power and normally $Q > 1$.

Conventions

Empty products equal 1, empty sums equal 0, and

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = 0 \qquad \text{when } k < 0 \text{ or } k > n$$

for nonnegative integral n. Multiple parameters are compressed as

$$(a_1, \ldots, a_r; q)_n = \prod_{j=1}^{r} (a_j; q)_n.$$

The notation $(a; q)_\infty$ is analytic when $|q| < 1$; when formal power series are intended, that will be said explicitly.

Contents

Part I

The finite algebra

Chapter 1

The algebra of q-shifted factorials

1.1 Definitions and first examples

Definition 1.1. For $n \in \mathbb{N}_0$ the *q-shifted factorial*, or *q-Pochhammer symbol*, is

$$(a;q)_n = \prod_{j=0}^{n-1}(1-aq^j), \qquad (a;q)_0 = 1.$$

For parameters $a_1, \dots, a_r$ we write

$$(a_1, \dots, a_r; q)_n = \prod_{i=1}^{r}(a_i;q)_n.$$

When $0 < |q| < 1$, the infinite symbol is

$$(a;q)_\infty = \prod_{j=0}^{\infty}(1-aq^j).$$

The word "shifted" refers to the geometric progression $a, aq, aq^2, \dots$. The first few values are

$$(a;q)_1 = 1-a, \qquad (a;q)_2 = (1-a)(1-aq), \qquad (a;q)_3 = (1-a)(1-aq)(1-aq^2).$$

Important specializations include

$$(q;q)_n = \prod_{j=1}^{n}(1-q^j), \qquad (-q;q)_n = \prod_{j=1}^{n}(1+q^j), \qquad (a;q^r)_n = \prod_{j=0}^{n-1}(1-aq^{rj}).$$

1.2 Concatenation, dissection, and reversal

Proposition 1.2 (Concatenation law)**.** *For $m, n \in \mathbb{N}_0$,*

$$(a;q)_{m+n} = (a;q)_m(aq^m;q)_n.$$

Consequently, whenever the denominator is nonzero,

$$\frac{(a;q)_{m+n}}{(a;q)_m} = (aq^m;q)_n.$$

Proof. Split the defining product at $j = m$:

$$\prod_{j=0}^{m+n-1}(1-aq^j) = \left(\prod_{j=0}^{m-1}(1-aq^j)\right)\left(\prod_{j=m}^{m+n-1}(1-aq^j)\right).$$

In the second product put $j = m + r$, where $0 \le r < n$. □

Proposition 1.3 (Dissection into residue classes)**.** *For integers $r, n \ge 1$,*

$$(a;q)_{rn} = \prod_{s=0}^{r-1}(aq^s;q^r)_n.$$

If $|q| < 1$, then also

$$(a;q)_\infty = \prod_{s=0}^{r-1}(aq^s;q^r)_\infty.$$

Proof. Every integer j with $0 \le j < rn$ has a unique representation $j = s + rt$ with $0 \le s < r$ and $0 \le t < n$. Reordering the finite product gives the first identity. For the infinite identity, the factors are absolutely summably close to 1 because $\sum_{j\ge 0}|aq^j| < \infty$, so the product may be regrouped into the r residue classes. □

Proposition 1.4 (Base reversal)**.** *For $n \ge 0$ and $a \ne 0$,*

$$(a;q)_n = (-a)^n q^{\binom{n}{2}}(a^{-1};q^{-1})_n.$$

Equivalently,

$$(a;q^{-1})_n = (-a)^n q^{-\binom{n}{2}}(a^{-1};q)_n.$$

Proof. Factor $-aq^j$ from each term:

$$1 - aq^j = -aq^j\left(1 - a^{-1}q^{-j}\right).$$

Multiplying for $0 \le j < n$ gives the factor $(-a)^n q^{0+1+\cdots+(n-1)} = (-a)^n q^{\binom{n}{2}}$. The remaining product is $(a^{-1};q^{-1})_n$. □

1.3 Negative indices

The concatenation law suggests the only extension to negative integers that preserves $(a;q)_{m+n} = (a;q)_m(aq^m;q)_n$.

Definition 1.5. For $n \ge 1$, define

$$(a;q)_{-n} = \frac{1}{(aq^{-n};q)_n},$$

provided no factor in the denominator vanishes.

Proposition 1.6 (Closed form for a negative index)**.** *For $n \ge 1$ and $a \ne 0$,*

$$(a;q)_{-n} = \frac{(-1)^n a^{-n} q^{\binom{n+1}{2}}}{(q/a;q)_n}.$$

Moreover, for all integers m, n for which both sides are defined,

$$(a;q)_{m+n} = (a;q)_m(aq^m;q)_n.$$

Proof. By definition,

$$(aq^{-n};q)_n = \prod_{j=0}^{n-1}(1-aq^{-n+j}).$$

Factor $-aq^{-n+j}$ from each factor. Their product is

$$(-a)^n q^{-n^2+\binom{n}{2}} = (-a)^n q^{-\binom{n+1}{2}},$$

and the remaining product is

$$\prod_{j=0}^{n-1}(1-a^{-1}q^{n-j}) = (q/a;q)_n.$$

Taking the reciprocal proves the formula. The extended concatenation identity follows by cancelling the overlapping finite factors; the cases in which one of $m, n, m+n$ is negative reduce directly to the definition. □

Corollary 1.7 (A terminating factor)**.** *If* $0 \le k \le N$*, then*

$$(q^{-N};q)_k = (-1)^k q^{-Nk+\binom{k}{2}} \frac{(q;q)_N}{(q;q)_{N-k}}.$$

In particular, $(q^{-N};q)_k = 0$ *for* $k > N$.

Proof. Apply base reversal to $(q^{-N};q)_k$:

$$(q^{-N};q)_k = (-1)^k q^{-Nk+\binom{k}{2}} (q^{N-k+1};q)_k.$$

The last factor is $(q;q)_N/(q;q)_{N-k}$. For $k > N$, the defining product contains the factor $1-q^0$. □

1.4 Partial fractions and logarithmic derivatives

Theorem 1.8 (Finite partial-fraction decomposition)**.** *Assume* $q^i \neq q^j$ *for* $0 \le i < j \le n$*. Then*

$$\frac{1}{(z;q)_{n+1}} = \sum_{j=0}^{n} \frac{(-1)^j q^{\binom{j+1}{2}}}{(q;q)_j (q;q)_{n-j}} \frac{1}{1-zq^j}.$$

Proof. The left side has simple poles at $z = q^{-j}$, $0 \le j \le n$. The coefficient of $(1-zq^j)^{-1}$ is

$$A_j = \left.\frac{1-zq^j}{(z;q)_{n+1}}\right|_{z=q^{-j}} = \frac{1}{\prod_{r\neq j}(1-q^{r-j})}.$$

For $r < j$, put $s = j-r$ and obtain

$$\prod_{r=0}^{j-1}(1-q^{r-j}) = \prod_{s=1}^{j}(1-q^{-s}) = (-1)^j q^{-\binom{j+1}{2}}(q;q)_j.$$

The factors with $r > j$ multiply to $(q;q)_{n-j}$. Thus A_j is the displayed coefficient. Subtracting the proposed right side from the left gives a rational function with no poles and tending to 0 at infinity, hence it is identically zero. □

Proposition 1.9 (Logarithmic derivatives)**.** *Away from its zeros,*

$$\frac{\partial}{\partial a}\log(a;q)_n = -\sum_{j=0}^{n-1}\frac{q^j}{1-aq^j}.$$

If $a = a(x)$ is differentiable, then

$$\frac{\mathrm{d}}{\mathrm{d}x}\log(a(x);q)_n = -a'(x)\sum_{j=0}^{n-1}\frac{q^j}{1-a(x)q^j}.$$

Proof. Differentiate the logarithm of the finite product term by term. □

1.5 Infinite products and Lambert series

Theorem 1.10 (Convergence and zero set)**.** *Let $0 < |q| < 1$. The product $(a;q)_\infty$ converges locally uniformly as a function of $a \in \mathbb{C}$ and therefore defines an entire function. Its zeros are precisely*

$$a = q^{-j}, \qquad j = 0, 1, 2, \dots,$$

and all are simple.

Proof. On a compact set $K \subset \mathbb{C}$ let $M = \max_{a\in K}|a|$. Then

$$\sum_{j=0}^{\infty}\sup_{a\in K}|aq^j| \le \frac{M}{1-|q|} < \infty.$$

The standard convergence criterion for infinite products shows that $\prod_j(1-aq^j)$ converges uniformly on K; equivalently, one may apply uniform convergence to the tails after choosing an analytic branch of the logarithm near 1. Uniform limits of holomorphic functions are holomorphic, so the limit is entire.

If $a = q^{-j}$, one factor vanishes. Conversely, if no factor vanishes, then the tail product is nonzero because the sum of the absolute deviations from 1 converges, while the finite initial product is nonzero. At $a = q^{-j}$ exactly one factor vanishes and its derivative is $-q^j \ne 0$, whereas all other factors are nonzero; the zero is simple. □

1.5.1 Complex orders

Finite and negative integral indices extend naturally to a complex order once an infinite product is available. Fix $0 < |q| < 1$ and a branch of the logarithm at q, denoted by $\log q$, and write $q^\alpha = \exp(\alpha\log q)$.

Definition 1.11. For $\alpha \in \mathbb{C}$, define

$$(a;q)_\alpha := \frac{(a;q)_\infty}{(aq^\alpha;q)_\infty}, \tag{1.1}$$

whenever the denominator is nonzero.

Proposition 1.12 (Complex concatenation)**.** *Whenever the displayed quantities are defined,*

$$(a;q)_{\alpha+\beta} = (a;q)_\alpha(aq^\alpha;q)_\beta. \tag{1.2}$$

In particular,

$$(a;q)_{\alpha+1} = (1-aq^\alpha)(a;q)_\alpha. \tag{1.3}$$

For integral α, (1.1) agrees with the finite and negative-index definitions already given.

Proof. By definition,

$$(a;q)_\alpha (aq^\alpha;q)_\beta = \frac{(a;q)_\infty}{(aq^\alpha;q)_\infty} \frac{(aq^\alpha;q)_\infty}{(aq^{\alpha+\beta};q)_\infty} = \frac{(a;q)_\infty}{(aq^{\alpha+\beta};q)_\infty}.$$

This is (1.2). Taking $\beta = 1$ gives (1.3). If $\alpha = n \geq 0$, the tail factorization $(a;q)_\infty = (a;q)_n (aq^n;q)_\infty$ recovers the finite product. If $\alpha = -n < 0$, the same relation, applied after shifting by $-n$, gives $(a;q)_{-n} = 1/(aq^{-n};q)_n$. ◻

Proposition 1.13 (Derivative with respect to the order)**.** *On every domain avoiding the poles of* (1.1)*,*

$$\frac{\partial}{\partial \alpha} \log(a;q)_\alpha = (\log q) \sum_{j=0}^{\infty} \frac{aq^{\alpha+j}}{1-aq^{\alpha+j}}. \tag{1.4}$$

The series is locally uniformly convergent.

Proof. Only the denominator in (1.1) depends on α. Differentiate its locally uniformly convergent logarithm term by term:

$$-\frac{\partial}{\partial \alpha} \sum_{j\geq 0} \log(1-aq^{\alpha+j}) = (\log q) \sum_{j\geq 0} \frac{aq^{\alpha+j}}{1-aq^{\alpha+j}}.$$

On a compact set avoiding the poles, the denominators of the tail are bounded away from zero and the numerators form a geometric majorant, proving local uniform convergence. ◻

Theorem 1.14 (Lambert-series expansion)**.** *If* $|q| < 1$ *and* $|a| < 1$*, then*

$$\log(a;q)_\infty = -\sum_{m=1}^{\infty} \frac{a^m}{m(1-q^m)}.$$

Consequently,

$$\frac{\partial}{\partial a} \log(a;q)_\infty = -\sum_{m=1}^{\infty} \frac{a^{m-1}}{1-q^m} = -\sum_{j=0}^{\infty} \frac{q^j}{1-aq^j}.$$

All series converge locally uniformly in $|a| < 1$*.*

Proof. For $|a| < 1$, use $\log(1-u) = -\sum_{m\geq 1} u^m/m$ and absolute convergence:

$$\sum_{j\geq 0}\sum_{m\geq 1} \frac{|a|^m |q|^{jm}}{m} = \sum_{m\geq 1} \frac{|a|^m}{m(1-|q|^m)} < \infty.$$

Thus the order of summation may be exchanged:

$$\sum_{j\geq 0} \log(1-aq^j) = -\sum_{m\geq 1} \frac{a^m}{m} \sum_{j\geq 0} q^{jm}.$$

The geometric sum gives the first formula. Local uniform convergence permits termwise differentiation. Expanding $(1-aq^j)^{-1}$ geometrically gives the second form of the derivative. ◻

Corollary 1.15 (Divisor sums)**.** *As a formal power series,*

$$-q\frac{\mathrm{d}}{\mathrm{d}q} \log(q;q)_\infty = \sum_{n=1}^{\infty} \sigma_1(n) q^n,$$

where $\sigma_1(n) = \sum_{d|n} d$*.*

Proof. Logarithmic differentiation gives

$$-q\frac{\mathrm{d}}{\mathrm{d}q}\log(q;q)_\infty = \sum_{j\ge 1}\frac{jq^j}{1-q^j} = \sum_{j\ge 1}\sum_{r\ge 1} jq^{jr}.$$

The coefficient of q^n is the sum of all j dividing n. □

1.6 The dilogarithmic limit

The singular regime $q \to 1^-$ turns an infinite product into an exponential of order $(1-q)^{-1}$. The dilogarithm is the natural leading term.

Theorem 1.16 (Asymptotic expansion near $q = 1$)**.** *Fix $0 < \rho < 1$ and an integer $N \ge 1$. Uniformly for $|a| \le \rho$, as $t \to 0^+$,*

$$\begin{aligned}\log(a;\mathrm{e}^{-t})_\infty = &-\frac{\mathrm{Li}_2(a)}{t} + \frac{1}{2}\log(1-a)\\ &-\sum_{r=1}^{N-1}\frac{B_{2r}}{(2r)!}t^{2r-1}\,\mathrm{Li}_{2-2r}(a) + O_{\rho,N}(t^{2N-1}),\end{aligned}$$

where B_{2r} are Bernoulli numbers and $\mathrm{Li}_s(a) = \sum_{m\ge 1} a^m/m^s$ *for* $|a| < 1$.

Proof. By Theorem 1.14,

$$\log(a;\mathrm{e}^{-t})_\infty = -\sum_{m\ge 1}\frac{a^m}{m(1-\mathrm{e}^{-mt})}.$$

The Bernoulli expansion at the origin is

$$\frac{1}{1-\mathrm{e}^{-x}} = \frac{1}{x} + \frac{1}{2} + \sum_{r=1}^{N-1}\frac{B_{2r}}{(2r)!}x^{2r-1} + R_N(x),$$

with $R_N(x) = O_N(x^{2N-1})$ for $0 \le x \le 1$. Substitution of the displayed polynomial terms gives respectively

$$-\frac{1}{t}\sum_{m\ge 1}\frac{a^m}{m^2}, \qquad -\frac{1}{2}\sum_{m\ge 1}\frac{a^m}{m} = \frac{1}{2}\log(1-a),$$

and the polylogarithmic terms in the statement.

It remains to bound the total remainder. Split the m-sum at $m \le t^{-1}$. In this range,

$$\sum_{m\le t^{-1}}\frac{|a|^m}{m}|R_N(mt)| \le C_N t^{2N-1}\sum_{m\ge 1}\rho^m m^{2N-2} = O_{\rho,N}(t^{2N-1}).$$

For $m > t^{-1}$, subtracting finitely many asymptotic terms leaves a quantity bounded by a polynomial in mt plus $(1-\mathrm{e}^{-mt})^{-1}$; the factor ρ^m makes the tail exponentially small in $1/t$, hence smaller than every power of t. This proves the uniform remainder estimate. □

1.7 Classical shifted factorials as a limit

Proposition 1.17 (Finite classical limit)**.** *For fixed $x \in \mathbb{C}$ and $n \in \mathbb{N}_0$,*

$$\lim_{q\to 1} \frac{(q^x;q)_n}{(1-q)^n} = x(x+1)\cdots(x+n-1) = (x)_n,$$

where the limit is taken through values for which a consistent branch of q^x is chosen.

Proof. Write

$$\frac{(q^x;q)_n}{(1-q)^n} = \prod_{j=0}^{n-1} \frac{1-q^{x+j}}{1-q}.$$

Each factor tends to $x + j$, since $q^u = 1 + u(q-1) + o(q-1)$ for fixed u. □

Chapter 2

q-integers and Gaussian coefficients

2.1 Definitions

Definition 2.1. For $n \in \mathbb{N}_0$ define the q-integer and q-factorial by

$$[n]_q = \frac{1-q^n}{1-q} = 1 + q + \cdots + q^{n-1}, \qquad [n]_q! = \prod_{j=1}^{n} [j]_q.$$

For $0 \le k \le n$, the *Gaussian coefficient* is

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q;q)_n}{(q;q)_k (q;q)_{n-k}} = \frac{[n]_q!}{[k]_q! [n-k]_q!}.$$

Outside $0 \le k \le n$ it is defined to be 0.

The notation is initially a rational expression, but the recurrences below show that it is a polynomial in q with nonnegative integral coefficients.

Proposition 2.2 (Product forms)**.** *For* $0 \le k \le n$*,*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q^{n-k+1};q)_k}{(q;q)_k} = \prod_{j=1}^{k} \frac{1-q^{n-k+j}}{1-q^j}.$$

Proof. Cancel $(q;q)_{n-k}$ from $(q;q)_n$ using Theorem 1.2. □

2.2 The two Pascal recurrences

Theorem 2.3 (q-Pascal recurrences)**.** *For* $1 \le k \le n-1$*,*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q + q^k \begin{bmatrix} n-1 \\ k \end{bmatrix}_q, \tag{2.1}$$

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = q^{n-k} \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q + \begin{bmatrix} n-1 \\ k \end{bmatrix}_q. \tag{2.2}$$

Both identities remain valid at the boundary under the zero convention.

Proof. Put the two terms on the right of (2.1) over the common denominator $(q;q)_k(q;q)_{n-k}$. Their numerator is

$$(q;q)_{n-1}\big((1-q^k)+q^k(1-q^{n-k})\big)=(q;q)_{n-1}(1-q^n)=(q;q)_n.$$

The proof of (2.2) is the same, using $q^{n-k}(1-q^k)+(1-q^{n-k})=1-q^n$. □

Corollary 2.4 (Polynomiality and positivity)**.** *For all $0\le k\le n$, the Gaussian coefficient $\begin{bmatrix} n\\ k\end{bmatrix}_q$ belongs to $\mathbb{N}_0[q]$.*

Proof. Induct on n. The boundary values are 1. Either recurrence in Theorem 2.3 expresses an interior coefficient as a sum of two polynomials with nonnegative integral coefficients. □

2.3 Symmetry, reciprocity, and degree

Theorem 2.5 (Elementary structure)**.** *For $0\le k\le n$,*

(i) $\begin{bmatrix} n\\ k\end{bmatrix}_q=\begin{bmatrix} n\\ n-k\end{bmatrix}_q$;

(ii) $\begin{bmatrix} n\\ k\end{bmatrix}_{q^{-1}}=q^{-k(n-k)}\begin{bmatrix} n\\ k\end{bmatrix}_q$;

(iii) $\begin{bmatrix} n\\ k\end{bmatrix}_q$ *has degree $k(n-k)$, constant term 1, and leading coefficient 1;*

(iv) *its coefficient sequence is palindromic.*

Proof. Symmetry is immediate from the factorial definition. For reciprocity, use

$$(q^{-1};q^{-1})_m=(-1)^m q^{-\binom{m+1}{2}}(q;q)_m$$

for $m=n,k,n-k$. The signs cancel and the exponent is

$$-\binom{n+1}{2}+\binom{k+1}{2}+\binom{n-k+1}{2}=-k(n-k).$$

Since $\begin{bmatrix} n\\ k\end{bmatrix}_q$ is a polynomial with constant term 1, reciprocity says

$$\begin{bmatrix} n\\ k\end{bmatrix}_q=q^{k(n-k)}\begin{bmatrix} n\\ k\end{bmatrix}_{q^{-1}}.$$

Hence its degree is at most $k(n-k)$ and its leading coefficient equals its constant coefficient. The product form shows that the order at infinity is exactly $k(n-k)$, so the degree is exact. The same reciprocal identity equates coefficients at complementary degrees and proves palindromicity. □

Corollary 2.6 (Classical limit)**.** *For fixed n,k,*

$$\lim_{q\to 1}\begin{bmatrix} n\\ k\end{bmatrix}_q=\binom{n}{k}.$$

Proof. Use the q-factorial form and $[j]_q\to j$. □

2.4 Multinomial coefficients and flags

Definition 2.7. If $n_1, \dots, n_r \geq 0$ and $n_1 + \cdots + n_r = n$, define

$$\begin{bmatrix} n \\ n_1, \dots, n_r \end{bmatrix}_q = \frac{(q;q)_n}{(q;q)_{n_1} \cdots (q;q)_{n_r}}.$$

Proposition 2.8 (Factorization into binomials)**.** *Let $N_j = n_j + n_{j+1} + \cdots + n_r$. Then*

$$\begin{bmatrix} n \\ n_1, \dots, n_r \end{bmatrix}_q = \prod_{j=1}^{r-1} \begin{bmatrix} N_j \\ n_j \end{bmatrix}_q.$$

In particular, every q-multinomial coefficient is a polynomial with nonnegative integral coefficients.

Proof. Write out the factorial ratios. Consecutive numerator and denominator factors telescope:

$$\prod_{j=1}^{r-1} \frac{(q;q)_{N_j}}{(q;q)_{n_j}(q;q)_{N_{j+1}}} = \frac{(q;q)_n}{\prod_{j=1}^{r}(q;q)_{n_j}}.$$

Positivity follows from Theorem 2.4. □

2.5 Adjacent ratios and efficient recurrences

Proposition 2.9. *For $0 \leq k < n$,*

$$\frac{\begin{bmatrix} n \\ k+1 \end{bmatrix}_q}{\begin{bmatrix} n \\ k \end{bmatrix}_q} = \frac{1 - q^{n-k}}{1 - q^{k+1}}.$$

For $0 \leq k \leq n-1$,

$$\frac{\begin{bmatrix} n \\ k \end{bmatrix}_q}{\begin{bmatrix} n-1 \\ k \end{bmatrix}_q} = \frac{1 - q^{n}}{1 - q^{n-k}}.$$

Proof. Cancel common factors in the product or factorial definitions. □

These ratios are useful numerically, but near roots of unity they hide cancellations. The polynomial recurrences of Theorem 2.3 are then preferable.

2.6 Coefficient moments

Write

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \sum_{j=0}^{D} c_j q^j, \qquad D = k(n-k),$$

and choose an exponent X with probability $\Pr(X = j) = c_j / \binom{n}{k}$. The combinatorial meaning of X will be established in Chapter 5; its moments can already be found algebraically.

Theorem 2.10 (Mean and variance)**.** *For the preceding distribution,*

$$\mathbb{E}X = \frac{k(n-k)}{2}, \qquad \operatorname{Var}(X) = \frac{k(n-k)(n+1)}{12}.$$

Equivalently,

$$\frac{\mathrm{d}}{\mathrm{d}q} \begin{bmatrix} n \\ k \end{bmatrix}_q \Bigg|_{q=1} = \frac{k(n-k)}{2} \binom{n}{k}.$$

Proof. The palindromic identity $c_j = c_{D-j}$ immediately gives $\mathbb{E}X = D/2$ and the derivative formula.

For the variance, use the moment-generating function

$$M(t) = \frac{\left[{n \atop k}\right]_{\mathrm{e}^t}}{\binom{n}{k}} = \prod_{i=1}^{k} \frac{\mathrm{e}^{(n-k+i)t} - 1}{(n-k+i)t} \frac{it}{\mathrm{e}^{it} - 1}.$$

The local expansion

$$\log \frac{\mathrm{e}^u - 1}{u} = \frac{u}{2} + \frac{u^2}{24} + O(u^3)$$

gives

$$\log M(t) = \frac{t}{2} \sum_{i=1}^{k} (n-k) + \frac{t^2}{24} \sum_{i=1}^{k} \left((n-k+i)^2 - i^2\right) + O(t^3).$$

The first sum is $k(n-k)$. The second equals

$$k(n-k)^2 + 2(n-k)\frac{k(k+1)}{2} = k(n-k)(n+1).$$

Since the first and second derivatives of $\log M$ at 0 are the mean and variance, respectively, the result follows. $\square$

Chapter 3

Finite q-binomial theorems and inversion

3.1 Euler's finite product expansions

Theorem 3.1 (Finite q-binomial theorem)**.** *For $n \geq 0$,*

$$(z;q)_n = \sum_{k=0}^{n} (-1)^k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q z^k, \tag{3.1}$$

$$(-z;q)_n = \sum_{k=0}^{n} q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q z^k. \tag{3.2}$$

Proof. It is enough to prove the first identity; replace z by $-z$ for the second. The case $n = 0$ is clear. Suppose the formula holds for n. Then

$$(z;q)_{n+1} = (1 - zq^n)(z;q)_n.$$

The coefficient of z^k on the right is

$$(-1)^k q^{\binom{k}{2}} \left(\begin{bmatrix} n \\ k \end{bmatrix}_q + q^{n-k+1} \begin{bmatrix} n \\ k-1 \end{bmatrix}_q \right).$$

By the second q-Pascal recurrence, the parenthesis is $\begin{bmatrix} n+1 \\ k \end{bmatrix}_q$. This completes the induction. ◻

Corollary 3.2 (Alternating sums)**.** *For $n > 0$,*

$$\sum_{k=0}^{n} (-1)^k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q = 0.$$

More generally,

$$\sum_{k=0}^{n} (-1)^k q^{\binom{k}{2}+mk} \begin{bmatrix} n \\ k \end{bmatrix}_q = (q^m;q)_n.$$

Proof. Set $z = 1$ and $z = q^m$, respectively, in (3.1). ◻

Corollary 3.3 (Weighted subset sum)**.** *For $0 \leq k \leq n$,*

$$\sum_{0 \leq i_1 < \cdots < i_k \leq n-1} q^{i_1 + \cdots + i_k} = q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q.$$

Equivalently,

$$\sum_{1 \leq j_1 < \cdots < j_k \leq n} q^{j_1 + \cdots + j_k} = q^{\binom{k+1}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q.$$

Proof. Expand $\prod_{i=0}^{n-1}(1+zq^i)$ by choosing either 1 or zq^i from each factor, and compare the coefficient of z^k with (3.2). □

3.2 Reciprocal finite products

Theorem 3.4 (Reciprocal finite q-binomial theorem)**.** *For $n \geq 0$ and $|z| < 1$,*

$$\frac{1}{(z;q)_{n+1}} = \sum_{k=0}^{\infty} \begin{bmatrix} n+k \\ k \end{bmatrix}_q z^k.$$

The identity also holds formally in $\mathbb{Z}[q][[z]]$.

Proof. Let $H_n(z) = \sum_{k\geq 0} \begin{bmatrix} n+k \\ k \end{bmatrix}_q z^k$. The second q-Pascal recurrence gives

$$\begin{bmatrix} n+k \\ k \end{bmatrix}_q - q^n \begin{bmatrix} n+k-1 \\ k-1 \end{bmatrix}_q = \begin{bmatrix} n+k-1 \\ k \end{bmatrix}_q.$$

Therefore

$$(1-zq^n)H_n(z) = H_{n-1}(z),$$

where $H_0(z) = \sum_{k\geq 0} z^k = (1-z)^{-1}$. Iterating yields

$$H_n(z) = \frac{1}{(1-z)(1-zq)\cdots(1-zq^n)}.$$

All operations are valid formally; analytically they are justified for $|z| < 1$. □

3.3 q-Vandermonde convolution

Theorem 3.5 (q-Vandermonde convolution)**.** *For $m, n, r \in \mathbb{N}_0$,*

$$\begin{bmatrix} m+n \\ r \end{bmatrix}_q = \sum_{j\in\mathbb{Z}} q^{(m-j)(r-j)} \begin{bmatrix} m \\ j \end{bmatrix}_q \begin{bmatrix} n \\ r-j \end{bmatrix}_q, \tag{3.3}$$

$$= \sum_{j\in\mathbb{Z}} q^{j(n-r+j)} \begin{bmatrix} m \\ j \end{bmatrix}_q \begin{bmatrix} n \\ r-j \end{bmatrix}_q. \tag{3.4}$$

Only finitely many terms are nonzero.

Proof. Factor

$$(-z;q)_{m+n} = (-z;q)_m(-zq^m;q)_n$$

and use (3.2). The coefficient of z^r on the left is $q^{\binom{r}{2}} \begin{bmatrix} m+n \\ r \end{bmatrix}_q$. The term with j chosen from the first factor and $r-j$ from the second has exponent

$$\binom{j}{2} + m(r-j) + \binom{r-j}{2}.$$

Subtracting $\binom{r}{2}$ gives $(m-j)(r-j)$, proving (3.3). Splitting the product in the reverse order gives (3.4). □

Corollary 3.6 (Central form)**.** *For $m, n \geq 0$,*

$$\begin{bmatrix} m+n \\ m \end{bmatrix}_q = \sum_j q^{j^2} \begin{bmatrix} m \\ j \end{bmatrix}_q \begin{bmatrix} n \\ j \end{bmatrix}_q.$$

Proof. Set $r = m$ in (3.3) and put $\ell = m - j$. Then the exponent becomes ℓ^2, while symmetry gives $\begin{bmatrix} m \\ m-\ell \end{bmatrix}_q = \begin{bmatrix} m \\ \ell \end{bmatrix}_q$ and the second factor is $\begin{bmatrix} n \\ \ell \end{bmatrix}_q$. □

3.4 q-binomial inversion

Theorem 3.7 (q-binomial inversion)**.** *For sequences (a_n) and (b_n) in a commutative ring containing q, the relations*

$$b_n = \sum_{k=0}^{n} \begin{bmatrix} n \\ k \end{bmatrix}_q a_k$$

are equivalent to

$$a_n = \sum_{k=0}^{n} (-1)^{n-k} q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q b_k.$$

Proof. Substitute the first relation into the proposed inverse. The coefficient of a_j becomes

$$\sum_{k=j}^{n} (-1)^{n-k} q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q \begin{bmatrix} k \\ j \end{bmatrix}_q.$$

The factorial definition gives

$$\begin{bmatrix} n \\ k \end{bmatrix}_q \begin{bmatrix} k \\ j \end{bmatrix}_q = \begin{bmatrix} n \\ j \end{bmatrix}_q \begin{bmatrix} n-j \\ k-j \end{bmatrix}_q.$$

Put $r = n - k$. The coefficient is

$$\begin{bmatrix} n \\ j \end{bmatrix}_q \sum_{r=0}^{n-j} (-1)^r q^{\binom{r}{2}} \begin{bmatrix} n-j \\ r \end{bmatrix}_q.$$

By Theorem 3.2, this equals 0 when $n > j$ and 1 when $n = j$. Thus the composition is the identity. The same argument in the opposite order proves equivalence. □

Corollary 3.8 (Matrix orthogonality)**.** *For $0 \le j \le n$,*

$$\sum_{k=j}^{n} (-1)^{n-k} q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q \begin{bmatrix} k \\ j \end{bmatrix}_q = \delta_{n,j}.$$

Proof. This is the kernel computation in the preceding proof. □

3.5 A q-Cauchy product identity

Theorem 3.9 (q-Cauchy identity)**.** *For u, v in a commutative ring,*

$$(uv;q)_n = \sum_{k=0}^{n} \begin{bmatrix} n \\ k \end{bmatrix}_q (u;q)_k v^k (v;q)_{n-k}.$$

Proof. Induct on n. The assertion is clear at $n = 0$. Let the right side be $C_n(u, v)$. Use

$$\begin{bmatrix} n+1 \\ k \end{bmatrix}_q = q^{n+1-k} \begin{bmatrix} n \\ k-1 \end{bmatrix}_q + \begin{bmatrix} n \\ k \end{bmatrix}_q$$

and split C_{n+1} accordingly. In the first part shift $k \mapsto k+1$, use $(u;q)_{k+1} = (u;q)_k(1 - uq^k)$, and combine with the second part after using $(v;q)_{n+1-k} = (v;q)_{n-k}(1 - vq^{n-k})$. The bracket that remains is

$$(1 - vq^{n-k}) + vq^{n-k}(1 - uq^k) = 1 - uvq^n,$$

independent of k. Therefore

$$C_{n+1}(u,v) = (1 - uvq^n)C_n(u,v) = (uv;q)_{n+1}.$$

□

Remark 3.10. Taking $u = 0$ recovers the tautology $(0;q)_k = 1$ inside a weighted expansion; taking special limits of u or v produces several finite convolution formulas used later in the proof of the q-Pfaff–Saalschütz summation.

Chapter 4

Weighted binomial coefficients and symmetric functions

The ordinary and Gaussian binomial coefficients are instances of a broader construction in which each available object carries a weight. This point of view, developed systematically by Konvalina, places subset selection, multiset selection, Stirling numbers, and Gaussian coefficients in one framework.

4.1 Elementary and complete weighted coefficients

Definition 4.1. Let $\mathbf{w} = (w_1, \dots, w_n)$ be elements of a commutative ring. Define

$$C_k^n(\mathbf{w}) = \sum_{1 \le i_1 < \cdots < i_k \le n} w_{i_1} \cdots w_{i_k},$$

$$S_k^n(\mathbf{w}) = \sum_{1 \le i_1 \le \cdots \le i_k \le n} w_{i_1} \cdots w_{i_k}.$$

We set $C_0^n = S_0^n = 1$. Thus C_k^n is the elementary symmetric function $e_k(w_1, \dots, w_n)$ and S_k^n is the complete homogeneous symmetric function $h_k(w_1, \dots, w_n)$.

Theorem 4.2 (Weighted Pascal recurrences)**.** *For $n, k \ge 1$,*

$$C_k^n(\mathbf{w}) = C_k^{n-1}(\mathbf{w}) + w_n C_{k-1}^{n-1}(\mathbf{w}), \tag{4.1}$$

$$S_k^n(\mathbf{w}) = S_k^{n-1}(\mathbf{w}) + w_n S_{k-1}^{n}(\mathbf{w}). \tag{4.2}$$

Proof. A k-element subset of $\{1, \dots, n\}$ either omits n, contributing C_k^{n-1}, or contains n, in which case removal of n leaves a $(k-1)$-element subset and the weight acquires a factor w_n. This proves (4.1).

For multisets, separate those with no occurrence of n from those with at least one occurrence. Removing one occurrence of n from the latter leaves an arbitrary multiset of size $k-1$ drawn from all n types. This proves (4.2). □

Theorem 4.3 (Generating products)**.** *One has*

$$\prod_{i=1}^{n} (1 + w_i z) = \sum_{k=0}^{n} C_k^n(\mathbf{w}) z^k, \tag{4.3}$$

$$\prod_{i=1}^{n} \frac{1}{1 - w_i z} = \sum_{k=0}^{\infty} S_k^n(\mathbf{w}) z^k. \tag{4.4}$$

The second identity is formal in z, or analytic when $|w_i z| < 1$ for all i.

Proof. In the first product, choosing $w_i z$ from exactly the factors indexed by a subset I produces $z^{|I|} \prod_{i\in I} w_i$. Summing over subsets gives (4.3). For the second, expand each factor geometrically:

$$\frac{1}{1 - w_i z} = \sum_{r_i \geq 0} w_i^{r_i} z^{r_i}.$$

The coefficient of z^k sums over weak compositions $r_1 + \cdots + r_n = k$, which are equivalent to multisets of size k. □

Corollary 4.4 (Elementary–complete orthogonality)**.** *For $m \geq 0$,*

$$\sum_{j=0}^{m} (-1)^j C_j^n(\mathbf{w}) S_{m-j}^n(\mathbf{w}) = \delta_{m,0}.$$

Proof. Multiply $\prod_i (1 - w_i z)$ by its reciprocal and compare the coefficient of z^m, using Theorem 4.3 with z replaced by $-z$ in the elementary generating function. □

4.2 Classical, Stirling, and Gaussian specializations

Theorem 4.5 (A specialization table)**.** *The following identities hold.*

(i) *If $w_i = 1$, then*

$$C_k^n = \binom{n}{k}, \qquad S_k^n = \binom{n+k-1}{k}.$$

(ii) *If $w_i = q^{i-1}$, then*

$$C_k^n = q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q, \qquad S_k^n = \begin{bmatrix} n+k-1 \\ k \end{bmatrix}_q.$$

(iii) *For the unsigned Stirling numbers of the first kind $c(n,k)$ and Stirling numbers of the second kind $\left\{ {n \atop k} \right\}$,*

$$c(n+1, k+1) = C_{n-k}^n(1, 2, \ldots, n),$$
$$\left\{ {n+k \atop n} \right\} = S_k^n(1, 2, \ldots, n).$$

Proof. For $w_i = 1$, C_k^n counts k-subsets. The multiset count is the stars-and-bars number $\binom{n+k-1}{k}$.

For $w_i = q^{i-1}$, the elementary identity is Theorem 3.3. For the complete identity, Theorem 4.3 gives

$$\prod_{i=0}^{n-1} \frac{1}{1 - zq^i} = \frac{1}{(z;q)_n}.$$

The coefficient of z^k is $\begin{bmatrix} n+k-1 \\ k \end{bmatrix}_q$ by Theorem 3.4.

Finally,

$$x(x+1)\cdots(x+n) = \sum_{j=0}^{n+1} c(n+1, j) x^j.$$

The coefficient of x^{k+1} is $e_{n-k}(1, 2, \ldots, n)$. The standard ordinary generating function

$$\prod_{i=1}^{n} \frac{1}{1 - iz} = \sum_{k \geq 0} \left\{ {n+k \atop n} \right\} z^k$$

follows from the recurrence for Stirling numbers of the second kind, or by induction on n; comparison with (4.4) proves the last formula. □

4.3 Weighted inversion

Theorem 4.6 (Symmetric-function inversion)**.** *Let* (a_m) *and* (b_m) *be sequences. The transformations*

$$b_m = \sum_{j=0}^{m} S^n_{m-j}(\mathbf{w}) a_j$$

and

$$a_m = \sum_{j=0}^{m} (-1)^{m-j} C^n_{m-j}(\mathbf{w}) b_j$$

are inverse to each other.

Proof. The generating functions $A(z) = \sum a_m z^m$ and $B(z) = \sum b_m z^m$ satisfy

$$B(z) = A(z) \prod_{i=1}^{n} (1 - w_i z)^{-1}.$$

Multiplication by $\prod_i (1 - w_i z)$ gives the proposed inverse coefficient formula. Equivalently, use the orthogonality in Theorem 4.4. □

4.4 The subset–subspace analogy

The choice $w_i = q^{i-1}$ is more than a convenient specialization. It replaces the Boolean lattice of subsets by the lattice of subspaces over a finite field. Ordinary binomial coefficients count subsets by cardinality; Gaussian coefficients count subspaces by dimension. The two-term weighted recurrence mirrors the decomposition of a subspace according to its relation with a fixed hyperplane. This analogy is made exact in Chapter 6.

Chapter 5

Partitions, words, paths, and inversion statistics

5.1 Binary words

Let $\mathcal{W}_{n,k}$ be the set of binary words of length n containing k ones. Define

$$\operatorname{inv}(w) = \#\left\{(i, j) : i < j,\ w_i = 1,\ w_j = 0\right\}.$$

Theorem 5.1 (Inversion generating function)**.** *For* $0 \le k \le n$*,*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \sum_{w \in \mathcal{W}_{n,k}} q^{\operatorname{inv}(w)}.$$

Proof. Let $F_{n,k}(q)$ denote the right side. If the last letter is 1, deletion of it gives a word in $\mathcal{W}_{n-1,k-1}$ and creates no new inversion. If the last letter is 0, deletion gives a word in $\mathcal{W}_{n-1,k}$ and the final zero forms an inversion with each of the k ones, contributing a factor q^k. Hence

$$F_{n,k} = F_{n-1,k-1} + q^k F_{n-1,k}.$$

The boundary values are 1, so $F_{n,k}$ and $\left[\begin{smallmatrix} n \\ k \end{smallmatrix}\right]_q$ obey the same recurrence and initial conditions. □

Corollary 5.2. *The random variable in Theorem 2.10 is the inversion number of a uniformly random word in* $\mathcal{W}_{n,k}$.

Proof. The coefficient of q^j in Theorem 5.1 counts words with j inversions, while $|\mathcal{W}_{n,k}| = \binom{n}{k}$. □

5.2 Partitions inside a rectangle

A partition $\lambda = (\lambda_1, \ldots, \lambda_k)$ is said to fit in a $k \times m$ rectangle if

$$m \ge \lambda_1 \ge \cdots \ge \lambda_k \ge 0.$$

Its size is $|\lambda| = \lambda_1 + \cdots + \lambda_k$.

Theorem 5.3 (Rectangular partition generating function)**.** *For* $m, k \ge 0$*,*

$$\begin{bmatrix} m+k \\ k \end{bmatrix}_q = \sum_{\lambda \subseteq k \times m} q^{|\lambda|}.$$

Proof. Take a word $w \in \mathcal{W}_{m+k,k}$ and let the positions of its ones be $1 \le i_1 < \cdots < i_k \le m+k$. Define

$$\lambda_j = m + j - i_j \qquad (1 \le j \le k).$$

Then $0 \le \lambda_j \le m$ and

$$\lambda_j - \lambda_{j+1} = i_{j+1} - i_j - 1 \ge 0.$$

Thus λ fits in the rectangle. Conversely, $i_j = m + j - \lambda_j$ recovers a strictly increasing sequence of positions, so this is a bijection. The number λ_j is exactly the number of zeros to the right of the jth one, hence $|\lambda| = \mathrm{inv}(w)$. Apply Theorem 5.1. □

Corollary 5.4 (Conjugation and complement)**.** *Conjugating Ferrers diagrams proves* $\begin{bmatrix} m+k \\ k \end{bmatrix}_q = \begin{bmatrix} m+k \\ m \end{bmatrix}_q$. *Taking the complement in the rectangle proves*

$$\begin{bmatrix} m+k \\ k \end{bmatrix}_{q^{-1}} = q^{-mk} \begin{bmatrix} m+k \\ k \end{bmatrix}_q .$$

Proof. Conjugation exchanges a $k \times m$ rectangle with an $m \times k$ rectangle without changing size. Complementation sends size r to $mk - r$, so the generating polynomial is reciprocal of degree mk. □

5.3 Lattice paths and area

Encode a binary word by a path from $(0,0)$ to (m,k), reading 0 as an east step and 1 as a north step. With a suitable choice of side, the number of unit squares between the path and the boundary is the inversion number.

Corollary 5.5 (Area generating function)**.** *The polynomial* $\begin{bmatrix} m+k \\ k \end{bmatrix}_q$ *is the generating function of monotone lattice paths from* $(0,0)$ *to* (m,k) *by the area under the path:*

$$\begin{bmatrix} m+k \\ k \end{bmatrix}_q = \sum_P q^{\mathrm{area}(P)}.$$

Proof. Under the word–path encoding, every north step contributes the number of later east steps, which is precisely the number of squares in its row between the path and the chosen boundary. Summing over north steps gives $\mathrm{inv}(w)$. □

5.4 Durfee squares and the central convolution

The largest square in the northwest corner of a partition diagram is its Durfee square.

Theorem 5.6 (Durfee-square decomposition)**.** *For $m, n \ge 0$,*

$$\begin{bmatrix} m+n \\ m \end{bmatrix}_q = \sum_{j=0}^{\min(m,n)} q^{j^2} \begin{bmatrix} m \\ j \end{bmatrix}_q \begin{bmatrix} n \\ j \end{bmatrix}_q .$$

Proof. This is Theorem 3.6; we give the partition proof. A partition in an $m \times n$ rectangle with Durfee square of side j consists of the $j \times j$ square, a partition to its right fitting in a $j \times (n-j)$ rectangle, and a partition below it fitting in an $(m-j) \times j$ rectangle. By Theorem 5.3, their generating functions are

$$q^{j^2} \begin{bmatrix} n \\ j \end{bmatrix}_q \begin{bmatrix} m \\ j \end{bmatrix}_q .$$

The Durfee size is unique, so summing over j counts every partition exactly once. □

5.5 Words with several letters

Let $\mathcal{W}(n_1, \ldots, n_r)$ be the words containing n_i copies of the letter i, ordered by $1 < \cdots < r$. Let $\mathrm{inv}(w)$ count pairs $i < j$ of positions with $w_i > w_j$.

Theorem 5.7 (Multiset inversion enumerator)**.** *If $n = n_1 + \cdots + n_r$, then*

$$\begin{bmatrix} n \\ n_1, \ldots, n_r \end{bmatrix}_q = \sum_{w \in \mathcal{W}(n_1, \ldots, n_r)} q^{\mathrm{inv}(w)}.$$

Proof. Choose the positions of the largest letter r. If their binary indicator has inversion number s, then s counts precisely the inversions in which the larger letter is r. The remaining positions contain a word on $1, \ldots, r-1$, and its inversions are independent. Therefore the generating function factors as

$$\begin{bmatrix} n \\ n_r \end{bmatrix}_q \begin{bmatrix} n - n_r \\ n_1, \ldots, n_{r-1} \end{bmatrix}_q.$$

Iterating and using Theorem 2.8 proves the formula. □

5.6 Rogers–Szegő polynomials

Definition 5.8. The Rogers–Szegő polynomial is

$$H_n(z; q) = \sum_{k=0}^{n} \begin{bmatrix} n \\ k \end{bmatrix}_q z^k.$$

Proposition 5.9 (Basic recurrence)**.** *One has*

$$H_{n+1}(z; q) = zH_n(z; q) + H_n(qz; q), \qquad H_0(z; q) = 1.$$

Moreover, if $|q| < 1$, then

$$\sum_{n=0}^{\infty} \frac{H_n(z; q)}{(q; q)_n} t^n = \frac{1}{(t; q)_\infty (zt; q)_\infty}$$

for $|t| < 1$ and $|zt| < 1$.

Proof. Insert the first q-Pascal recurrence into H_{n+1}:

$$\sum_k \left(\begin{bmatrix} n \\ k-1 \end{bmatrix}_q + q^k \begin{bmatrix} n \\ k \end{bmatrix}_q \right) z^k = zH_n(z; q) + H_n(qz; q).$$

For the generating function, use

$$\frac{\begin{bmatrix} n \\ k \end{bmatrix}_q}{(q; q)_n} = \frac{1}{(q; q)_k (q; q)_{n-k}}.$$

Writing $n = k + j$ and separating the double sum gives

$$\left(\sum_{j \geq 0} \frac{t^j}{(q; q)_j} \right) \left(\sum_{k \geq 0} \frac{(zt)^k}{(q; q)_k} \right).$$

Each factor equals the reciprocal infinite product by Euler's identity, proved later in Theorem 8.2; alternatively, the equality follows already by taking $n \to \infty$ in Theorem 3.4. This gives the stated product. □

Chapter 6

Finite fields, Grassmannians, and rank enumeration

6.1 Counting subspaces

Theorem 6.1 (Gaussian coefficients count subspaces)**.** *Let Q be a prime power. The number of k-dimensional linear subspaces of $\mathbb{F}_Q^n$ is*

$$\begin{bmatrix} n \\ k \end{bmatrix}_Q .$$

Proof. Count ordered linearly independent k-tuples in $\mathbb{F}_Q^n$. The first vector may be any nonzero vector, the second any vector outside the span of the first, and so on, giving

$$(Q^n - 1)(Q^n - Q) \cdots (Q^n - Q^{k-1}).$$

Each k-dimensional subspace has

$$(Q^k - 1)(Q^k - Q) \cdots (Q^k - Q^{k-1})$$

ordered bases. Dividing and cancelling Q^i in corresponding factors yields

$$\prod_{i=0}^{k-1} \frac{Q^{n-i} - 1}{Q^{k-i} - 1} = \prod_{j=1}^{k} \frac{1 - Q^{n-k+j}}{1 - Q^j} = \begin{bmatrix} n \\ k \end{bmatrix}_Q .$$

□

Theorem 6.2 (Geometric Pascal recurrence)**.** *Let H be a fixed hyperplane in $\mathbb{F}_Q^n$. Then*

$$\begin{bmatrix} n \\ k \end{bmatrix}_Q = \begin{bmatrix} n-1 \\ k \end{bmatrix}_Q + Q^{n-k} \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_Q .$$

Proof. A k-subspace U is either contained in H or not. The contained subspaces contribute $\left[\begin{smallmatrix} n-1 \\ k \end{smallmatrix}\right]_Q$. If $U \not\subseteq H$, then $W = U \cap H$ has dimension $k-1$. Fixing W, the quotient $\mathbb{F}_Q^n/W$ has dimension $n-k+1$, and H/W is a hyperplane. The subspace U/W is a line not contained in H/W. The total number of lines in $\mathbb{F}_Q^{n-k+1}$ minus those in its hyperplane is

$$\frac{Q^{n-k+1} - 1}{Q - 1} - \frac{Q^{n-k} - 1}{Q - 1} = Q^{n-k}.$$

There are $\left[\begin{smallmatrix} n-1 \\ k-1 \end{smallmatrix}\right]_Q$ choices for W, proving the formula. □

6.2 Flags and q-multinomial coefficients

Theorem 6.3 (Partial flags)**.** *Let $d_i = n_1 + \cdots + n_i$. The number of flags*

$$0 = V_0 \subset V_1 \subset \cdots \subset V_r = \mathbb{F}_Q^n, \qquad \dim(V_i/V_{i-1}) = n_i,$$

is

$$\begin{bmatrix} n \\ n_1, \dots, n_r \end{bmatrix}_Q.$$

Proof. Choose V_1 in $\left[\begin{smallmatrix} n \\ n_1 \end{smallmatrix}\right]_Q$ ways. In the quotient $\mathbb{F}_Q^n/V_1$, choose V_2/V_1 in $\left[\begin{smallmatrix} n-n_1 \\ n_2 \end{smallmatrix}\right]_Q$ ways, and continue. The product is the factorization in Theorem 2.8. □

6.3 Matrices of fixed rank

Theorem 6.4 (Rank enumeration)**.** *The number of $m \times n$ matrices of rank r over $\mathbb{F}_Q$ is*

$$N_{m,n,r}(Q) = \begin{bmatrix} m \\ r \end{bmatrix}_Q \begin{bmatrix} n \\ r \end{bmatrix}_Q |\mathrm{GL}_r(\mathbb{F}_Q)|,$$

where

$$|\mathrm{GL}_r(\mathbb{F}_Q)| = \prod_{i=0}^{r-1} (Q^r - Q^i).$$

Equivalently,

$$N_{m,n,r}(Q) = \frac{\prod_{i=0}^{r-1}(Q^m - Q^i)(Q^n - Q^i)}{\prod_{i=0}^{r-1}(Q^r - Q^i)}.$$

Proof. Every rank-r matrix A can be factored as $A = UV$, where U is an $m \times r$ matrix of full column rank and V is an $r \times n$ matrix of full row rank. There are $\prod_{i=0}^{r-1}(Q^m - Q^i)$ choices for U and $\prod_{i=0}^{r-1}(Q^n - Q^i)$ choices for V. Two pairs (U, V) and (U', V') give the same A exactly when

$$U' = UG, \qquad V' = G^{-1}V$$

for a unique $G \in \mathrm{GL}_r(\mathbb{F}_Q)$. Thus every matrix is counted $|\mathrm{GL}_r|$ times. Substituting the subspace-count formulas gives the first expression. □

6.4 Schubert cells and a topological meaning

Let $\mathrm{Gr}(k, n)$ be the complex Grassmannian of k-planes in $\mathbb{C}^n$. Its Schubert cell decomposition is indexed by partitions inside a $k \times (n-k)$ rectangle.

Theorem 6.5 (Poincaré polynomial of the Grassmannian)**.** *The integral cohomology of* $\mathrm{Gr}(k, n)$ *is concentrated in even degrees, and its Poincaré polynomial is*

$$P_{\mathrm{Gr}(k,n)}(t) = \sum_{j \ge 0} \operatorname{rank} H^{2j}(\mathrm{Gr}(k, n); \mathbb{Z}) t^{2j} = \begin{bmatrix} n \\ k \end{bmatrix}_{t^2}.$$

Proof. Fix the standard flag in $\mathbb{C}^n$. Every k-plane has a unique row-reduced echelon matrix. Its pivot positions determine a binary word with k ones, hence a partition λ inside the $k \times (n-k)$ rectangle. Once the pivots are fixed, the entries allowed to vary freely occupy exactly the boxes of λ; the corresponding stratum is therefore an affine complex space $\mathbb{C}^{|\lambda|}$. Ordering the strata by inclusion of diagrams yields a CW decomposition with one real cell of dimension $2|\lambda|$ for each λ. Since all cells have even dimension, every cellular boundary map is zero. Thus the rank of H^{2j} is the number of such partitions of size j. Apply Theorem 5.3 with $q = t^2$. $\square$

Remark 6.6 (The "same polynomial" principle). For a prime power Q, evaluating $\begin{bmatrix} n \\ k \end{bmatrix}_q$ at $q = Q$ counts rational points of a Grassmannian over $\mathbb{F}_Q$. Evaluating it at $q = t^2$ records the even Betti numbers of the complex Grassmannian. This is an elementary instance of the broad principle that polynomial point counts over finite fields often shadow topological invariants over $\mathbb{C}$.

6.5 Growth over finite fields

Proposition 6.7 (Dimension-dominant bounds)**.** *For $Q > 1$ and $0 \le k \le n$,*

$$Q^{k(n-k)} \le \begin{bmatrix} n \\ k \end{bmatrix}_Q \le \frac{Q^{k(n-k)}}{(Q^{-1};Q^{-1})_\infty}.$$

Proof. Rewrite the product as

$$\begin{bmatrix} n \\ k \end{bmatrix}_Q = Q^{k(n-k)} \prod_{j=1}^{k} \frac{1 - Q^{-(n-k+j)}}{1 - Q^{-j}}.$$

Each ratio is at least 1 because $n - k + j \ge j$, proving the lower bound. For the upper bound, discard the numerator factors and note

$$\prod_{j=1}^{k} (1 - Q^{-j}) \ge \prod_{j=1}^{\infty} (1 - Q^{-j}) = (Q^{-1};Q^{-1})_\infty.$$

$\square$

Chapter 7

Symmetry and unimodality

Palindromicity follows immediately from reciprocity. Unimodality is subtler: the coefficients rise toward the middle and then fall. A concise conceptual proof comes from the representation theory of $\mathfrak{sl}_2$.

7.1 A representation-theoretic lemma

Let $\mathfrak{sl}_2(\mathbb{C})$ have generators E, F, H with

$$[H, E] = 2E, \qquad [H, F] = -2F, \qquad [E, F] = H.$$

Lemma 7.1 (Structure of finite-dimensional $\mathfrak{sl}_2$-modules)**.** *Every finite-dimensional complex $\mathfrak{sl}_2$-module is a direct sum of irreducibles L_d indexed by $d \in \mathbb{N}_0$. The module L_d has dimension $d+1$ and H-weights*

$$d, d-2, \dots, -d,$$

each with multiplicity one.

Proof. We include the standard argument. Restrict a representation to the compact real form $\mathfrak{su}_2$. A finite-dimensional Lie-algebra representation integrates to the simply connected compact group $\mathrm{SU}(2)$ by exponentiating the representing matrices; the Lie relations ensure consistency of the local exponentials, and simple connectedness removes monodromy. Starting from any Hermitian inner product, average it over $\mathrm{SU}(2)$ with normalized Haar measure. The result is invariant. Hence the orthogonal complement of every invariant subspace is invariant, proving complete reducibility.

It remains to classify an irreducible module V. In the invariant Hermitian structure, a suitable scalar multiple of H is self-adjoint, so H is diagonalizable. Choose an eigenvector v whose eigenvalue d has maximal real part. Since E raises an H-weight by 2, maximality gives $Ev = 0$. The commutation relations imply by induction

$$HF^j v = (d-2j)F^j v, \qquad EF^j v = j(d-j+1)F^{j-1} v.$$

Because V is finite-dimensional, let m be maximal with $F^m v \neq 0$. Applying E to $F^{m+1} v = 0$ gives

$$(m+1)(d-m)F^m v = 0,$$

so $d = m \in \mathbb{N}_0$. The vectors $v, Fv, \dots, F^d v$ have distinct weights and are linearly independent. Their span is invariant under E, F, H and is nonzero; irreducibility makes it all of V. The displayed weights and multiplicities follow. $\square$

7.2 Exterior powers and Gaussian coefficients

Let L_{n-1} have a weight basis $v_0, \dots, v_{n-1}$ with $Hv_i = (n-1-2i)v_i$.

Proposition 7.2 (Exterior-power character)**.** *For* $0 \le k \le n$*,*

$$\operatorname{ch}\left(\bigwedge^k L_{n-1}\right)(x) = x^{k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_{x^{-2}}.$$

Proof. A basis of the exterior power consists of $v_{i_1} \wedge \cdots \wedge v_{i_k}$ with $0 \le i_1 < \cdots < i_k \le n-1$. Its weight is

$$k(n-1) - 2(i_1 + \cdots + i_k).$$

Therefore

$$\operatorname{ch} = x^{k(n-1)} \sum_{i_1 < \cdots < i_k} (x^{-2})^{i_1 + \cdots + i_k}.$$

Use Theorem 3.3 with $q = x^{-2}$. The prefactor becomes

$$x^{k(n-1)} x^{-2\binom{k}{2}} = x^{k(n-k)}.$$

□

Theorem 7.3 (Unimodality of Gaussian coefficients)**.** *If*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = c_0 + c_1 q + \cdots + c_D q^D, \qquad D = k(n-k),$$

then

$$c_0 \le c_1 \le \cdots \le c_{\lfloor D/2 \rfloor}$$

and $c_j = c_{D-j}$*. Thus the coefficient sequence is symmetric and unimodal.*

Proof. Symmetry was proved in Theorem 2.5. By Theorem 7.1, decompose

$$\bigwedge^k L_{n-1} \cong \bigoplus_{d \ge 0} m_d L_d.$$

All weights in this exterior power have the same parity as D. The character of one summand L_d, placed inside the ambient sequence of weights

$$D, D-2, \dots, -D,$$

contributes a consecutive block of ones centered at weight 0: it is zero until weight d is reached, then one at every step down to $-d$, then zero again. As one moves from weight D toward 0, such a block can turn on but cannot turn off. A nonnegative sum of these centered blocks therefore has weakly increasing multiplicities toward the middle.

By Theorem 7.2, the multiplicity of weight $D - 2j$ is exactly c_j. Hence $c_0 \le c_1 \le \cdots \le c_{\lfloor D/2 \rfloor}$. □

Remark 7.4. The proof explains more than the inequality: the successive differences $c_j - c_{j-1}$ record multiplicities of irreducible summands with prescribed highest weight. Stronger strict-unimodality results require additional arguments and have a small family of exceptional cases; ordinary unimodality is exactly what the $\mathfrak{sl}_2$ decomposition supplies for free.

Part II

Infinite series and basic hypergeometric summation

Chapter 8

The infinite q-binomial theorem

8.1 A functional-equation proof

Theorem 8.1 (Infinite q-binomial theorem)**.** *Let* $|q| < 1$ *and* $|z| < 1$*. Then*

$$\sum_{n=0}^{\infty} \frac{(a;q)_n}{(q;q)_n} z^n = \frac{(az;q)_\infty}{(z;q)_\infty}.$$

The convergence is locally uniform in (a, z) *on compact subsets of* $\mathbb{C} \times \{|z| < 1\}$*.*

Proof. Set

$$F(z) = \sum_{n \geq 0} c_n z^n, \qquad c_n = \frac{(a;q)_n}{(q;q)_n}.$$

Since $c_n/c_{n-1} = (1 - aq^{n-1})/(1 - q^n) \to 1$, the radius of convergence is 1 unless the series terminates; in every case it converges for $|z| < 1$. The coefficient recurrence is equivalent to

$$(1 - q^n)c_n = (1 - aq^{n-1})c_{n-1}.$$

Comparing coefficients shows that

$$(1 - z)F(z) = (1 - az)F(qz), \qquad F(0) = 1.$$

Now put $G(z) = (az;q)_\infty/(z;q)_\infty$. Using $(z;q)_\infty = (1 - z)(qz;q)_\infty$ gives the same functional equation and $G(0) = 1$.

To prove uniqueness, iterate the functional equation:

$$F(z) = \prod_{j=0}^{N-1} \frac{1 - azq^j}{1 - zq^j} F(q^N z).$$

As $N \to \infty$, $q^N z \to 0$, so $F(q^N z) \to 1$. The finite product tends to $G(z)$. Hence $F = G$. □

Corollary 8.2 (Euler's two infinite identities)**.** *For* $|q| < 1$*,*

$$(z;q)_\infty = \sum_{n=0}^{\infty} \frac{(-1)^n q^{\binom{n}{2}}}{(q;q)_n} z^n, \qquad z \in \mathbb{C}, \tag{8.1}$$

$$\frac{1}{(z;q)_\infty} = \sum_{n=0}^{\infty} \frac{z^n}{(q;q)_n}, \qquad |z| < 1. \tag{8.2}$$

Proof. Set $a = 0$ in Theorem 8.1 to obtain (8.2). For (8.1), start with the finite identity

$$(z;q)_N = \sum_{n=0}^{N} (-1)^n q^{\binom{n}{2}} \begin{bmatrix} N \\ n \end{bmatrix}_q z^n.$$

For fixed n, $\left[\begin{smallmatrix} N \\ n \end{smallmatrix}\right]_q \to 1/(q;q)_n$ as $N \to \infty$. The limiting series converges for every z because its successive-term ratio has magnitude asymptotic to $|z||q|^n$. To justify passage to the limit uniformly on $|z| \le R$, note that

$$\left| \begin{bmatrix} N \\ n \end{bmatrix}_q \right| \le \frac{\prod_{j\ge1}(1+|q|^j)}{\inf_{r\ge0}|(q;q)_r|},$$

where the denominator is positive because $(q;q)_r \to (q;q)_\infty \ne 0$. Thus the summands, extended by zero for $n > N$, are dominated by a constant multiple of $|q|^{\binom{n}{2}} R^n$. Dominated convergence, together with local uniform convergence of the products, now permits passage to the limit. □

Corollary 8.3 (A ratio expansion)**.** *For $|z| < 1$,*

$$\frac{(a_1 z;q)_\infty \cdots (a_r z;q)_\infty}{(b_1 z;q)_\infty \cdots (b_s z;q)_\infty}$$

can be expanded by multiplying r copies of (8.1) *and s copies of* (8.2)*. In particular, every coefficient is a finite convolution of q-Pochhammer ratios.*

Proof. Each numerator factor has an entire series expansion and each reciprocal denominator factor converges for $|b_j z| < 1$. Cauchy multiplication is valid in their common disk of absolute convergence. □

8.2 Formal and analytic interpretations

Proposition 8.4 (Formal version)**.** *If q is an indeterminate, then*

$$\frac{(az;q)_\infty}{(z;q)_\infty} = \sum_{n\ge0} \frac{(a;q)_n}{(q;q)_n} z^n$$

holds in the q-adically complete ring $\mathbb{Q}(a)[[q]][[z]]$. More generally, it remains valid after any specialization into a complete topological ring for which the displayed products and series converge.

Proof. The products are well defined q-adically: apart from their initial factors, the jth factors differ from 1 by a multiple of q^j. For a formal power series $F(z) = \sum c_n z^n$, the equation

$$(1-z)F(z) = (1-az)F(qz), \qquad c_0 = 1,$$

determines c_n recursively by $(1-q^n)c_n = (1-aq^{n-1})c_{n-1}$, because $1-q^n$ is a unit in $\mathbb{Q}(a)[[q]]$. Both sides satisfy this equation, so their coefficients agree. The same uniqueness argument works in every stated complete specialization. □

8.3 Limits in the number of factors

Proposition 8.5. *For fixed k and $|q| < 1$,*

$$\lim_{n\to\infty} \begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{1}{(q;q)_k}.$$

More generally,

$$\lim_{n\to\infty} \begin{bmatrix} n+k \\ k \end{bmatrix}_q = \frac{1}{(q;q)_k}.$$

Proof. Use

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q^{n-k+1};q)_k}{(q;q)_k}.$$

Every numerator factor tends to 1. □

Remark 8.6. The two Euler identities are therefore the stable limits of the finite product and reciprocal-product formulas. This stability is one reason finite q-binomial identities so often serve as polynomial approximations to infinite q-series identities.

Chapter 9

Basic hypergeometric series

9.1 Definition and convergence

Definition 9.1. The basic hypergeometric series is

$$ {}_r\phi_s\left[\begin{matrix} a_1, \dots, a_r \\ b_1, \dots, b_s \end{matrix}; q, z\right] = \sum_{n=0}^{\infty} \frac{(a_1, \dots, a_r; q)_n}{(q, b_1, \dots, b_s; q)_n} \left((-1)^n q^{\binom{n}{2}}\right)^{1+s-r} z^n, $$

provided no denominator factor vanishes. If one numerator parameter is q^{-N}, the series terminates at $n = N$.

Proposition 9.2 (Elementary convergence classification)**.** *Assume the series does not terminate and the denominator parameters avoid poles.*

(i) *If $r \le s$, the series converges for every z.*

(ii) *If $r = s + 1$, its radius of convergence is* 1.

(iii) *If $r > s + 1$, the terms generally fail to tend to zero for $z \neq 0$; such series are normally interpreted only when terminating or formally.*

Proof. Let T_n be the nth summand. Since $(1 - aq^n) \to 1$ for each fixed parameter,

$$ \frac{T_{n+1}}{T_n} = z \, \frac{\prod_{i=1}^{r}(1 - a_i q^n)}{(1 - q^{n+1}) \prod_{j=1}^{s}(1 - b_j q^n)} (-1)^{1+s-r} q^{n(1+s-r)}. $$

The ratio tends to 0 when $r \le s$, to z when $r = s + 1$, and has unbounded magnitude when $r > s + 1$ and $z \neq 0$. Apply the ratio test. $\square$

The infinite q-binomial theorem is the evaluation

$$ {}_1\phi_0\left[\begin{matrix} a \\ - \end{matrix}; q, z\right] = \frac{(az; q)_\infty}{(z; q)_\infty}. $$

9.2 Heine's transformation

Theorem 9.3 (Heine transformation)**.** *Initially for $|z| < 1$ and $|b| < 1$,*

$$ {}_2\phi_1\left[\begin{matrix} a, b \\ c \end{matrix}; q, z\right] = \frac{(b, az; q)_\infty}{(c, z; q)_\infty} {}_2\phi_1\left[\begin{matrix} c/b, z \\ az \end{matrix}; q, b\right]. $$

The identity extends wherever both sides are defined by analytic continuation.

Proof. Use the tail-factor identity to write

$$\frac{(b;q)_n}{(c;q)_n} = \frac{(b;q)_\infty}{(c;q)_\infty}\frac{(cq^n;q)_\infty}{(bq^n;q)_\infty}.$$

By the infinite q-binomial theorem,

$$\frac{(cq^n;q)_\infty}{(bq^n;q)_\infty} = \sum_{k\geq 0}\frac{(c/b;q)_k}{(q;q)_k}b^k q^{nk}.$$

Insert this into the ${}_2\phi_1$ series and interchange the absolutely convergent sums:

$$\frac{(b;q)_\infty}{(c;q)_\infty}\sum_{k\geq 0}\frac{(c/b;q)_k}{(q;q)_k}b^k\sum_{n\geq 0}\frac{(a;q)_n}{(q;q)_n}(zq^k)^n.$$

The inner sum is

$$\frac{(azq^k;q)_\infty}{(zq^k;q)_\infty} = \frac{(az;q)_\infty}{(z;q)_\infty}\frac{(z;q)_k}{(az;q)_k}.$$

Collecting factors yields the claimed transformation. Both sides are meromorphic functions of the parameters, so equality extends from the initial open domain by the identity theorem. □

9.3 The q-Gauss sum

Theorem 9.4 (q-Gauss summation)**.** *If* $|c/(ab)| < 1$, *then*

$${}_2\phi_1\left[\begin{matrix}a, b\\ c\end{matrix}; q, \frac{c}{ab}\right] = \frac{(c/a, c/b;q)_\infty}{(c, c/(ab);q)_\infty}.$$

Proof. Set $z = c/(ab)$ in Theorem 9.3. Then $az = c/b$, so one numerator parameter of the transformed series cancels its denominator parameter:

$${}_2\phi_1\left[\begin{matrix}c/b, c/(ab)\\ c/b\end{matrix}; q, b\right] = {}_1\phi_0\left[\begin{matrix}c/(ab)\\ -\end{matrix}; q, b\right] = \frac{(c/a;q)_\infty}{(b;q)_\infty}.$$

Multiplication by the prefactor in Heine's transformation gives the result. The initially stronger restrictions are removed by analytic continuation within the convergence domain. □

9.4 A second finite Cauchy identity

Lemma 9.5 (Cauchy convolution II)**.** *For* a, b, c *in a commutative ring,*

$$(ac;q)_n(bc;q)_n = \sum_{k=0}^{n}\begin{bmatrix}n\\ k\end{bmatrix}_q (a;q)_k(b;q)_k c^k (c;q)_{n-k}(abcq^k;q)_{n-k}.$$

Proof. Apply the q-Cauchy identity Theorem 3.9 with $u = bq^k$, $v = ac$, and length $n-k$:

$$(abcq^k;q)_{n-k} = \sum_{r=0}^{n-k}\begin{bmatrix}n-k\\ r\end{bmatrix}_q (bq^k;q)_r(ac)^r(ac;q)_{n-k-r}.$$

Insert this in the right side of the proposed identity. Use

$$\begin{bmatrix} n \\ k \end{bmatrix}_q \begin{bmatrix} n-k \\ r \end{bmatrix}_q = \begin{bmatrix} n \\ k+r \end{bmatrix}_q \begin{bmatrix} k+r \\ k \end{bmatrix}_q, \qquad (b;q)_k (bq^k;q)_r = (b;q)_{k+r}.$$

Put $m = k + r$ and use

$$(c;q)_{n-k} = (c;q)_{n-m}(cq^{n-m};q)_{m-k}.$$

The double sum becomes

$$\sum_{m=0}^{n} \begin{bmatrix} n \\ m \end{bmatrix}_q (b;q)_m c^m (c;q)_{n-m}(ac;q)_{n-m} \sum_{k=0}^{m} \begin{bmatrix} m \\ k \end{bmatrix}_q (a;q)_k a^{m-k}(cq^{n-m};q)_{m-k}.$$

In the inner sum replace $j = m - k$ and apply Theorem 3.9 with $u = cq^{n-m}$ and $v = a$; it equals $(acq^{n-m};q)_m$. Hence

$$(ac;q)_{n-m}(acq^{n-m};q)_m = (ac;q)_n$$

can be factored out. The remaining sum is

$$(ac;q)_n \sum_{m=0}^{n} \begin{bmatrix} n \\ m \end{bmatrix}_q (b;q)_m c^m (c;q)_{n-m} = (ac;q)_n (bc;q)_n$$

by one final application of the q-Cauchy identity. □

9.5 The q-Pfaff–Saalschütz sum

Theorem 9.6 (q-Pfaff–Saalschütz summation)**.** *For $n \in \mathbb{N}_0$,*

$${}_3\phi_2\left[\begin{matrix} a, b, q^{-n} \\ c, abq^{1-n}/c \end{matrix}; q, q\right] = \frac{(c/a, c/b;q)_n}{(c, c/(ab);q)_n}.$$

Proof. In Theorem 9.5, replace its parameters by

$$a \mapsto a, \qquad b \mapsto b, \qquad c \mapsto \frac{c}{ab}.$$

This gives

$$\frac{(c/a, c/b;q)_n}{(c, c/(ab);q)_n} = \sum_{k=0}^{n} \begin{bmatrix} n \\ k \end{bmatrix}_q (a,b;q)_k \left(\frac{c}{ab}\right)^k \frac{(c/(ab);q)_{n-k}(cq^k;q)_{n-k}}{(c, c/(ab);q)_n}.$$

We show that the kth summand is the kth term of the stated ${}_3\phi_2$. The tail identities give

$$\frac{(cq^k;q)_{n-k}}{(c;q)_n} = \frac{1}{(c;q)_k}.$$

Also, reversal of the two finite products gives

$$(q^{-n};q)_k = (-1)^k q^{-nk+\binom{k}{2}}(q^{n-k+1};q)_k,$$

$$(abq^{1-n}/c;q)_k = \left(-\frac{ab}{c}\right)^k q^{k(1-n)+\binom{k}{2}}(cq^{n-k}/(ab);q)_k.$$

Therefore

$$\begin{aligned}
&\frac{(q^{-n};q)_k q^k}{(q;q)_k (abq^{1-n}/c;q)_k} \frac{(c/(ab);q)_n}{(c/(ab);q)_{n-k}} \\
&= \left(\frac{c}{ab}\right)^k \frac{(q^{n-k+1};q)_k}{(q;q)_k} \frac{(c/(ab);q)_n}{(c/(ab);q)_{n-k}(cq^{n-k}/(ab);q)_k} \\
&= \begin{bmatrix} n \\ k \end{bmatrix}_q \left(\frac{c}{ab}\right)^k .
\end{aligned}$$

Substitution yields exactly

$$\sum_{k=0}^{n} \frac{(a,b,q^{-n};q)_k}{(q,c,abq^{1-n}/c;q)_k} q^k .$$

□

9.6 The two q-Chu–Vandermonde sums

Corollary 9.7 (q-Chu–Vandermonde)**.** *For $n \in \mathbb{N}_0$,*

$${}_2\phi_1\left[\begin{matrix} q^{-n}, a \\ c \end{matrix}; q, \frac{cq^n}{a}\right] = \frac{(c/a;q)_n}{(c;q)_n}, \tag{9.1}$$

$${}_2\phi_1\left[\begin{matrix} q^{-n}, a \\ c \end{matrix}; q, q\right] = a^n \frac{(c/a;q)_n}{(c;q)_n}. \tag{9.2}$$

Proof. For (9.1), put $b = q^{-n}$ in the q-Gauss sum. The infinite products telescope:

$$\frac{(cq^n, c/a;q)_\infty}{(c, cq^n/a;q)_\infty} = \frac{(c/a;q)_n}{(c;q)_n}.$$

Because the series terminates, no convergence restriction remains.

For (9.2), let $b \to 0$ in Theorem 9.6, with its other numerator parameter equal to a. Then $(b;q)_k \to 1$ and $(abq^{1-n}/c;q)_k \to 1$, so the left side becomes the displayed ${}_2\phi_1$. On the right,

$$\frac{(c/b;q)_n}{(c/(ab);q)_n} \longrightarrow a^n$$

by comparing the leading factors of the two degree-n polynomials in $1/b$. □

9.7 A hierarchy of mechanisms

The proofs in this chapter form a hierarchy:

finite q-Pascal $\Longrightarrow$ q-Cauchy $\Longrightarrow$ q-Pfaff–Saalschütz,

while

infinite q-binomial $\Longrightarrow$ Heine $\Longrightarrow$ q-Gauss $\Longrightarrow$ first q-Chu–Vandermonde.

The finite chain is valid in a commutative ring after denominators are cleared. The infinite chain requires convergence or a suitable complete formal topology. This distinction becomes important in computer formalization; see Chapter 23.

Chapter 10

Bilateral series and Ramanujan's ${}_1\psi_1$ sum

10.1 Definition

For integral n, negative q-shifted factorials are understood as in Theorem 1.6. Define

$$ {}_1\psi_1\left[\begin{matrix}a\\b\end{matrix};q,z\right] = \sum_{n\in\mathbb{Z}} \frac{(a;q)_n}{(b;q)_n} z^n. $$

The positive tail behaves like a geometric series in z, while the negative tail behaves like a geometric series in $b/(az)$.

Proposition 10.1 (Convergence annulus)**.** *If denominator factors do not vanish, the ${}_1\psi_1$ series converges absolutely in*

$$ |b/a| < |z| < 1. $$

Proof. For $n \to +\infty$, the ratio of consecutive terms tends to z. For the negative tail put $n = -m$ and use the negative-index formula. The ratio of successive terms in m tends to $b/(az)$. The ratio test gives the annulus. □

Theorem 10.2 (Ramanujan's ${}_1\psi_1$ summation)**.** *For $|b/a| < |z| < 1$,*

$$ {}_1\psi_1\left[\begin{matrix}a\\b\end{matrix};q,z\right] = \frac{(q, b/a, az, q/(az);q)_\infty}{(b, q/a, z, b/(az);q)_\infty}. $$

Proof. We first prove the formula for $b = q^{m+1}$, where $m \in \mathbb{N}_0$. In that case $1/(b;q)_n = 0$ for $n < -m$ in the limiting sense, so shift $n = k - m$:

$$ \begin{aligned} \sum_{n\in\mathbb{Z}} \frac{(a;q)_n}{(q^{m+1};q)_n} z^n &= z^{-m} \frac{(a;q)_{-m}}{(q^{m+1};q)_{-m}} \sum_{k=0}^{\infty} \frac{(aq^{-m};q)_k}{(q;q)_k} z^k \\ &= z^{-m} \frac{(q;q)_m}{(aq^{-m};q)_m} \frac{(aq^{-m}z;q)_\infty}{(z;q)_\infty}. \end{aligned} $$

The second equality is the infinite q-binomial theorem. Reverse the finite products:

$$ (aq^{-m};q)_m = (-a)^m q^{-m(m+1)/2} (q/a;q)_m, $$

$$ (aq^{-m}z;q)_m = (-az)^m q^{-m(m+1)/2} (q/(az);q)_m. $$

Since $(aq^{-m}z;q)_\infty = (aq^{-m}z;q)_m(az;q)_\infty$, we obtain

$$\frac{(q;q)_m}{(q/a;q)_m}\frac{(q/(az);q)_m(az;q)_\infty}{(z;q)_\infty}.$$

On the other hand, substituting $b = q^{m+1}$ in the proposed product and cancelling tails gives exactly the same expression:

$$\frac{(q;q)_\infty}{(q^{m+1};q)_\infty}\frac{(q^{m+1}/a;q)_\infty}{(q/a;q)_\infty}\frac{(q/(az);q)_\infty}{(q^{m+1}/(az);q)_\infty}\frac{(az;q)_\infty}{(z;q)_\infty}.$$

For fixed a, z with $|z| < 1$, both sides are analytic functions of b in a neighborhood of 0 contained in $|b| < |az|$. Absolute convergence is locally uniform there by Theorem 10.1. For all sufficiently large m, the points $b = q^{m+1}$ lie in that neighborhood, and they accumulate at 0. The identity theorem therefore proves equality near 0. Both sides are meromorphic functions of b throughout the disk $|b| < |az|$: the series gives the meromorphic continuation termwise on compacta avoiding denominator zeros, and the product quotient is visibly meromorphic. Uniqueness of meromorphic continuation proves the identity everywhere in that disk where both sides are defined, which is precisely the asserted annulus for fixed a and z. □

Remark 10.3. Ramanujan's sum contains the infinite q-binomial theorem and several theta-product identities as limiting or specialized cases. It is a model example of a bilateral identity: the second inequality in the convergence annulus is controlled by the positive tail, while the first is controlled by the negative tail.

Chapter 11

q-difference calculus and the quantum binomial

11.1 The q-derivative

Assume $q \neq 1$. For $x \neq 0$ define

$$D_q f(x) = \frac{f(x) - f(qx)}{(1-q)x}.$$

At $x = 0$ one uses the continuous extension when it exists.

Proposition 11.1 (Elementary rules)**.** *For $n \geq 1$,*

$$D_q x^n = [n]_q x^{n-1}.$$

The product rules are

$$\begin{aligned} D_q(fg)(x) &= g(x) D_q f(x) + f(qx) D_q g(x), \\ &= f(x) D_q g(x) + g(qx) D_q f(x). \end{aligned}$$

Moreover,

$$D_q\big(f(q^r x)\big) = q^r (D_q f)(q^r x)$$

for integral r.

Proof. The monomial identity follows from $(x^n - q^n x^n)/((1-q)x) = [n]_q x^{n-1}$. For the first product rule, insert and subtract $f(qx)g(x)$ in $f(x)g(x) - f(qx)g(qx)$. For the second, insert and subtract $f(x)g(qx)$. The scaling rule follows directly from the definition. □

Theorem 11.2 (q-Leibniz rule)**.** *For $n \in \mathbb{N}_0$,*

$$D_q^n(fg)(x) = \sum_{k=0}^{n} \begin{bmatrix} n \\ k \end{bmatrix}_q (D_q^k f)(q^{n-k} x)(D_q^{n-k} g)(x).$$

Proof. Induct on n. The assertion is the ordinary product at $n = 0$. Apply D_q to the kth term at level n. By the product and scaling rules, it contributes

$$q^{n-k} \begin{bmatrix} n \\ k \end{bmatrix}_q (D_q^{k+1} f)(q^{n-k} x)(D_q^{n-k} g)(x)$$

to index $k+1$, and

$$\begin{bmatrix} n \\ k \end{bmatrix}_q (D_q^k f)(q^{n-k+1}x)(D_q^{n-k+1}g)(x)$$

to index k. The total coefficient of index k is

$$\begin{bmatrix} n \\ k \end{bmatrix}_q + q^{n-k+1}\begin{bmatrix} n \\ k-1 \end{bmatrix}_q = \begin{bmatrix} n+1 \\ k \end{bmatrix}_q$$

by the second q-Pascal recurrence. □

11.2 q-Taylor and Newton interpolation

Define the q-falling power based at a by

$$(x-a)_q^m = \prod_{j=0}^{m-1}(x-q^j a), \qquad (x-a)_q^0 = 1.$$

Lemma 11.3. *For $m \geq 1$,*

$$D_q(x-a)_q^m = [m]_q (x-a)_q^{m-1}.$$

Consequently,

$$D_q^j (x-a)_q^m = \frac{[m]_q!}{[m-j]_q!}(x-a)_q^{m-j}$$

for $0 \leq j \leq m$.

Proof. Let $P_m(x) = (x-a)_q^m$. Then

$$P_m(qx) = q^m (x - q^{-1}a)\prod_{j=0}^{m-2}(x-q^j a).$$

Subtracting from $P_m(x) = (x-q^{m-1}a)\prod_{j=0}^{m-2}(x-q^j a)$ gives

$$P_m(x) - P_m(qx) = (1-q^m)xP_{m-1}(x).$$

Divide by $(1-q)x$. Iteration proves the second formula. □

Theorem 11.4 (q-Taylor formula)**.** *Assume $[j]_q \neq 0$ for $1 \leq j \leq N$. If f is a polynomial of degree at most N, then*

$$f(x) = \sum_{k=0}^{N} \frac{(D_q^k f)(a)}{[k]_q!}(x-a)_q^k.$$

Proof. The polynomials $(x-a)_q^k$, $0 \leq k \leq N$, are monic of distinct degrees and hence form a basis. Write $f = \sum c_k (x-a)_q^k$. Apply D_q^j and set $x = a$. By Theorem 11.3, every term with $k > j$ still contains the factor $x - a$, terms with $k < j$ vanish after too many derivatives, and the $k = j$ term equals $c_j [j]_q!$. Thus $c_j = (D_q^j f)(a)/[j]_q!$. □

Remark 11.5. When the nodes $a, aq, aq^2, \ldots$ are distinct, this is a Newton interpolation formula on that geometric grid. In every case covered by the theorem, its basis zeros follow a multiplicative rather than an additive progression.

11.3 The quantum plane

Theorem 11.6 (Noncommutative q-binomial theorem)**.** *Let X, Y belong to an associative algebra and satisfy*

$$YX = qXY.$$

Then

$$(X+Y)^n = \sum_{k=0}^{n} \begin{bmatrix} n \\ k \end{bmatrix}_q X^{n-k} Y^k.$$

Proof. Induct on n and multiply on the right by $X+Y$. Since $Y^k X = q^k X Y^k$, the coefficient of $X^{n+1-k} Y^k$ becomes

$$q^k \begin{bmatrix} n \\ k \end{bmatrix}_q + \begin{bmatrix} n \\ k-1 \end{bmatrix}_q = \begin{bmatrix} n+1 \\ k \end{bmatrix}_q$$

by the first q-Pascal recurrence. □

Corollary 11.7 (Factorization of q-exponentials)**.** *Assume $|q| < 1$. Define*

$$e_q(x) = \sum_{n=0}^{\infty} \frac{x^n}{[n]_q!} = \frac{1}{((1-q)x; q)_\infty}, \qquad |(1-q)x| < 1,$$

and

$$E_q(x) = \sum_{n=0}^{\infty} \frac{q^{\binom{n}{2}} x^n}{[n]_q!} = (-(1-q)x; q)_\infty, \qquad x \in \mathbb{C}.$$

Then $e_q(x)E_q(-x) = 1$ wherever the first series is defined. If $YX = qXY$, then, as an identity of formal series in total degree,

$$e_q(X+Y) = e_q(X)e_q(Y).$$

Proof. The product formulas and the stated convergence domains follow from Euler's identities after replacing z by $(1-q)x$. Their product is 1. For the noncommutative factorization, apply Theorem 11.6 degree by degree and use

$$\frac{1}{[n]_q!} \begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{1}{[n-k]_q! [k]_q!}.$$

Reindex with $r = n-k$ to obtain the Cauchy product of $e_q(X)$ and $e_q(Y)$, with all X factors preceding all Y factors. Since every total degree receives only finitely many contributions, the formal rearrangement is valid. □

Proposition 11.8 (Eigenfunction properties)**.** *Under the hypotheses and within the convergence domains of Theorem 11.7, for scalar λ,*

$$D_q e_q(\lambda x) = \lambda e_q(\lambda x), \qquad D_q E_q(\lambda x) = \lambda E_q(q\lambda x).$$

Proof. Differentiate the series term by term and use $D_q x^n = [n]_q x^{n-1}$. □

11.4 Jackson's q-integral

Assume $0 < q < 1$. Define

$$\int_0^a f(t)\, d_q t = (1-q)a \sum_{n=0}^{\infty} q^n f(aq^n),$$

whenever the series converges.

Theorem 11.9 (Fundamental theorem of q-calculus)**.** *Let*

$$F(x) = \int_0^x f(t)\, d_q t.$$

Whenever the defining series converges locally uniformly,

$$D_q F(x) = f(x).$$

Conversely, if $G(aq^n) \to G(0)$, then

$$\int_0^a D_q G(t)\, d_q t = G(a) - G(0).$$

Proof. From the definition,

$$F(x) = (1-q)x \sum_{n \geq 0} q^n f(xq^n),$$

while

$$F(qx) = (1-q)x \sum_{n \geq 1} q^n f(xq^n).$$

Their difference is $(1-q)xf(x)$, proving the first identity. For the second,

$$\begin{aligned}\int_0^a D_q G(t)\, d_q t &= (1-q)a \sum_{n \geq 0} q^n \frac{G(aq^n) - G(aq^{n+1})}{(1-q)aq^n} \\ &= \sum_{n \geq 0} \big(G(aq^n) - G(aq^{n+1})\big),\end{aligned}$$

which telescopes to $G(a) - G(0)$. □

Corollary 11.10 (Jackson integration by parts)**.** *Under convergence assumptions,*

$$\int_0^a f(t) D_q g(t)\, d_q t = f(a)g(a) - f(0)g(0) - \int_0^a g(qt) D_q f(t)\, d_q t.$$

Proof. Use the product rule $D_q(fg) = f D_q g + g(q\,\cdot) D_q f$ and integrate with Theorem 11.9. □

Example 11.11 (Monomials)**.** For $\Re(s) > -1$,

$$\int_0^a t^s\, d_q t = (1-q)a^{s+1} \sum_{n \geq 0} q^{n(s+1)} = \frac{a^{s+1}}{[s+1]_q},$$

where $[u]_q = (1-q^u)/(1-q)$ for complex u.

Chapter 12

The q-gamma and q-beta functions

Throughout this chapter $0 < q < 1$ unless stated otherwise.

12.1 Definition and functional equation

Definition 12.1. For $x > 0$, define

$$\Gamma_q(x) = (1-q)^{1-x} \frac{(q;q)_\infty}{(q^x;q)_\infty}.$$

With $q^z = \exp(z \log q)$, the same product quotient defines a meromorphic function of $z \in \mathbb{C}$; its possible poles occur where $q^{z+j} = 1$ for some $j \in \mathbb{N}_0$. Whenever complex arguments of Γ_q occur below, this meromorphic continuation is intended.

Theorem 12.2 (Basic properties)**.** *For $x > 0$,*

$$\Gamma_q(1) = 1, \qquad \Gamma_q(x+1) = [x]_q \Gamma_q(x).$$

Consequently, for $n \in \mathbb{N}$,

$$\Gamma_q(n) = [n-1]_q!.$$

The recurrence extends as an identity of meromorphic functions of a complex argument.

Proof. The value at 1 is immediate. The tail identity $(q^x;q)_\infty = (1-q^x)(q^{x+1};q)_\infty$ gives

$$\frac{\Gamma_q(x+1)}{\Gamma_q(x)} = (1-q)^{-1}(1-q^x) = [x]_q.$$

Iteration gives the integral values. The same tail identity is valid for complex x, and hence proves the meromorphic recurrence wherever both sides are finite; equality elsewhere follows in the sense of meromorphic continuation. □

Proposition 12.3 (Logarithmic derivatives and convexity)**.** *For $x > 0$,*

$$\frac{\mathrm{d}}{\mathrm{d}x} \log \Gamma_q(x) = -\log(1-q) + (\log q) \sum_{n=0}^{\infty} \frac{q^{n+x}}{1-q^{n+x}},$$

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2} \log \Gamma_q(x) = (\log q)^2 \sum_{n=0}^{\infty} \frac{q^{n+x}}{(1-q^{n+x})^2} > 0.$$

Thus Γ_q is strictly log-convex.

Proof. Differentiate the locally uniformly convergent logarithm of the defining infinite product. A second differentiation gives the positive series. □

12.2 A q-Bohr–Mollerup theorem

Theorem 12.4 (q-Bohr–Mollerup characterization). *The function* Γ_q *is the unique function* $f : (0,\infty) \to (0,\infty)$ *satisfying*

$$f(1) = 1, \qquad f(x+1) = [x]_q f(x),$$

and such that $\log f$ *is convex.*

Proof. Existence follows from Theorems 12.2 and 12.3. For uniqueness let f have the stated properties and fix $0 < x \le 1$. Convexity on the interval from n to $n+1$ gives

$$f(n+x) \le f(n)^{1-x} f(n+1)^x.$$

A comparison of the adjacent secant slopes on $[n+x, n+1]$ and $[n+1, n+2]$ gives

$$f(n+x) \ge \frac{f(n+1)}{[n+1]_q^{1-x}}.$$

Using the recurrence,

$$f(n+x) = f(x)\prod_{j=0}^{n-1}[x+j]_q, \quad f(n) = [n-1]_q!, \quad f(n+1) = [n]_q!,$$

we obtain

$$\frac{[n]_q!}{[n+1]_q^{1-x}\prod_{j=0}^{n-1}[x+j]_q} \le f(x) \le \frac{[n-1]_q!\,[n]_q^x}{\prod_{j=0}^{n-1}[x+j]_q}.$$

The ratio of the upper bound to the lower bound is

$$\left(\frac{[n+1]_q}{[n]_q}\right)^{1-x} \longrightarrow 1.$$

Hence the bounds determine $f(x)$ uniquely for $0 < x \le 1$, and the recurrence determines it on all of $(0,\infty)$. Since Γ_q satisfies the same conditions, $f = \Gamma_q$. □

12.3 Classical limit

Theorem 12.5 (Limit to Euler's gamma function). *For every* $x > 0$*,*

$$\lim_{q\to 1^-}\Gamma_q(x) = \Gamma(x).$$

The convergence is locally uniform on $(0,\infty)$.

Proof. It suffices by the functional equations to consider $0 < x \le 1$. The bounds in the proof of Theorem 12.4 hold for $\Gamma_q(x)$. Fix n and let $q \to 1^-$. Since $[u]_q \to u$, they become

$$\frac{n!}{(n+1)^{1-x}x(x+1)\cdots(x+n-1)} \le \liminf_{q\to 1^-}\Gamma_q(x)$$

$$\le \limsup_{q\to 1^-}\Gamma_q(x) \le \frac{(n-1)!\,n^x}{x(x+1)\cdots(x+n-1)}.$$

Both outer expressions tend to $\Gamma(x)$ as $n \to \infty$. To see this directly, use Euler's beta integral:

$$B(x,n) = \int_0^1 t^{x-1}(1-t)^{n-1}\,dt = \frac{(n-1)!}{x(x+1)\cdots(x+n-1)}.$$

After $t = u/n$, dominated convergence gives $n^x B(x,n) \to \int_0^\infty u^{x-1}\mathrm{e}^{-u}\,du = \Gamma(x)$. The lower expression differs from the upper by a factor tending to 1. This proves pointwise convergence. We record the uniformity argument. On a compact interval $x \in [\delta, 1]$, the upper outer bound can be written as

$$n^x B(x,n) = \int_0^n u^{x-1}\left(1-\frac{u}{n}\right)^{n-1}\,\mathrm{d}u.$$

For $n \geq 2$, its integrand is dominated uniformly in x by $u^{\delta-1}$ on $(0,1]$ and by $e^{-u/2}$ on $[1,\infty)$. More explicitly, after extending the kernel by 0 for $u > n$,

$$\sup_{\delta\leq x\leq 1}\left|u^{x-1}\left(1-\frac{u}{n}\right)_+^{n-1} - u^{x-1}e^{-u}\right| \longrightarrow 0$$

for every $u > 0$, while this supremum is bounded by an integrable multiple of $u^{\delta-1}\mathbf{1}_{(0,1]}(u) + e^{-u/2}\mathbf{1}_{[1,\infty)}(u)$. The dominated convergence theorem therefore gives

$$\sup_{\delta\leq x\leq 1}|n^x B(x,n) - \Gamma(x)| \longrightarrow 0.$$

The lower outer bound differs by the factor $(n/(n+1))^{1-x}$, which also tends uniformly to 1. For fixed n, the original q-bounds converge uniformly as $q \to 1^-$ because they contain only finitely many continuous factors. A uniform squeeze proves local uniform convergence on $(0,1]$, and the recurrence extends it to arbitrary compact subsets of $(0,\infty)$. □

12.4 The Jackson q-beta integral

Definition 12.6. For $x, y > 0$, define

$$B_q(x,y) = \int_0^1 t^{x-1}\frac{(qt;q)_\infty}{(q^y t;q)_\infty}\,d_q t.$$

Theorem 12.7 (q-beta evaluation)**.** *For $x, y > 0$,*

$$B_q(x,y) = \frac{\Gamma_q(x)\Gamma_q(y)}{\Gamma_q(x+y)}.$$

Proof. Expand the Jackson integral:

$$\begin{aligned} B_q(x,y) &= (1-q)\sum_{n\geq 0} q^{nx}\frac{(q^{n+1};q)_\infty}{(q^{n+y};q)_\infty} \\ &= (1-q)\frac{(q;q)_\infty}{(q^y;q)_\infty}\sum_{n\geq 0}\frac{(q^y;q)_n}{(q;q)_n}q^{nx}. \end{aligned}$$

The infinite q-binomial theorem evaluates the sum as $(q^{x+y};q)_\infty/(q^x;q)_\infty$. Thus

$$B_q(x,y) = (1-q)\frac{(q;q)_\infty(q^{x+y};q)_\infty}{(q^x;q)_\infty(q^y;q)_\infty},$$

which is exactly the gamma quotient after substituting the definition of Γ_q. □

Corollary 12.8 (Recurrences)**.** *For $x, y > 0$,*

$$B_q(x+1, y) = \frac{[x]_q}{[x+y]_q} B_q(x, y), \qquad B_q(x, y+1) = \frac{[y]_q}{[x+y]_q} B_q(x, y).$$

Proof. Apply the functional equation for Γ_q to the quotient in Theorem 12.7. □

12.5 Multiplication and reflection-type formulas

Theorem 12.9 (q-multiplication formula)**.** *For an integer $m \geq 1$ and $x > 0$,*

$$\Gamma_q(mx) \prod_{j=1}^{m-1} \Gamma_{q^m}\left(\frac{j}{m}\right) = [m]_q^{mx-1} \prod_{j=0}^{m-1} \Gamma_{q^m}\left(x + \frac{j}{m}\right).$$

Proof. Multiply the definitions on the right. By dissection,

$$\prod_{j=0}^{m-1} (q^{mx+j}; q^m)_\infty = (q^{mx}; q)_\infty.$$

Hence

$$\prod_{j=0}^{m-1} \Gamma_{q^m}\left(x + \frac{j}{m}\right) = (1 - q^m)^{(m+1)/2 - mx} \frac{(q^m; q^m)_\infty^m}{(q^{mx}; q)_\infty}.$$

Similarly,

$$\prod_{j=1}^{m-1} \Gamma_{q^m}\left(\frac{j}{m}\right) = (1 - q^m)^{(m-1)/2} \frac{(q^m; q^m)_\infty^m}{(q; q)_\infty}.$$

Taking the ratio and using $[m]_q = (1 - q^m)/(1 - q)$ produces the formula. □

Proposition 12.10 (Reflection-type product)**.** *For $0 < x < 1$,*

$$\Gamma_q(x)\Gamma_q(1-x) = (1-q) \frac{(q; q)_\infty^2}{(q^x, q^{1-x}; q)_\infty}.$$

Proof. Multiply the two defining products; the powers of $1 - q$ combine to one. □

Remark 12.11. Unlike Euler's reflection formula, the right side is naturally a theta quotient rather than an elementary trigonometric function. Jacobi's triple product in the next part makes that theta structure explicit.

Part III

Theta products and partition identities

Chapter 13

Jacobi's triple product

13.1 A bilateral theta series

For $z \neq 0$ and $|q| < 1$, define

$$\Theta(z;q) = \sum_{n\in\mathbb{Z}} q^{\binom{n}{2}} z^n.$$

The series converges locally uniformly on $\mathbb{C}^*$ because its terms decay quadratically in both directions.

Theorem 13.1 (Jacobi triple product)**.** *For $z \neq 0$ and $|q| < 1$,*

$$\sum_{n\in\mathbb{Z}} q^{\binom{n}{2}} z^n = (-z, -q/z, q; q)_\infty, \tag{13.1}$$

$$\sum_{n\in\mathbb{Z}} (-1)^n q^{\binom{n}{2}} z^n = (z, q/z, q; q)_\infty. \tag{13.2}$$

Proof. It suffices to prove (13.1); replace z by $-z$ for (13.2). Put

$$F(z) = (-z;q)_\infty(-q/z;q)_\infty(q;q)_\infty.$$

The products converge locally uniformly on $\mathbb{C}^*$, so F has a Laurent expansion $F(z) = \sum_{n\in\mathbb{Z}} c_n z^n$. The shift identities give

$$F(qz) = \frac{1+z^{-1}}{1+z} F(z) = z^{-1}F(z).$$

Comparing Laurent coefficients yields

$$c_{n+1} = q^n c_n, \qquad c_n = q^{\binom{n}{2}} c_0$$

for every integer n.

It remains to find c_0. Euler's product expansions give

$$(-z;q)_\infty = \sum_{r\geq 0} \frac{q^{\binom{r}{2}} z^r}{(q;q)_r}, \qquad (-q/z;q)_\infty = \sum_{s\geq 0} \frac{q^{\binom{s+1}{2}} z^{-s}}{(q;q)_s}.$$

Hence

$$c_0 = (q;q)_\infty \sum_{r\geq 0} \frac{q^{r^2}}{(q;q)_r^2}.$$

The sum is the generating function obtained by decomposing an arbitrary partition according to the side r of its Durfee square: the square contributes q^{r^2}, while the diagrams to its right and below are independent partitions with at most r parts, each having generating function $1/(q;q)_r$. Therefore

$$\sum_{r\geq 0}\frac{q^{r^2}}{(q;q)_r^2}=\frac{1}{(q;q)_\infty},$$

so $c_0 = 1$. Thus $F(z)=\sum_{n\in\mathbb{Z}} q^{\binom{n}{2}} z^n$. □

Corollary 13.2 (Quasi-periodicity and zeros)**.** *The theta function satisfies*

$$\Theta(qz;q)=z^{-1}\Theta(z;q).$$

Its zeros are the simple points $z=-q^m$, $m\in\mathbb{Z}$.

Proof. The functional equation follows either from reindexing the bilateral series or from the proof above. The product in (13.1) vanishes exactly when $1+zq^j=0$ or $1+q^{j+1}/z=0$, which together give $z=-q^m$ for $m\in\mathbb{Z}$. Exactly one factor vanishes at each such point. □

13.2 Euler's pentagonal number theorem

Theorem 13.3 (Pentagonal number theorem)**.** *For $|q|<1$,*

$$(q;q)_\infty=\sum_{n\in\mathbb{Z}}(-1)^n q^{n(3n-1)/2}=1+\sum_{n=1}^{\infty}(-1)^n\left(q^{n(3n-1)/2}+q^{n(3n+1)/2}\right).$$

Proof. In (13.2), replace the base q by q^3 and set $z=q$. The product side is

$$(q;q^3)_\infty(q^2;q^3)_\infty(q^3;q^3)_\infty=(q;q)_\infty$$

by dissection. The series exponent is

$$3\binom{n}{2}+n=\frac{n(3n-1)}{2}.$$

Pair the terms n and $-n$ to obtain the second form. □

Corollary 13.4 (Euler's partition recurrence)**.** *Let $p(0)=1$ and $p(n)=0$ for $n<0$. Then for $n\geq 1$,*

$$p(n)=\sum_{j=1}^{\infty}(-1)^{j-1}\left(p\left(n-\frac{j(3j-1)}{2}\right)+p\left(n-\frac{j(3j+1)}{2}\right)\right).$$

Proof. The generating function for partitions is $\sum_{n\geq 0}p(n)q^n=1/(q;q)_\infty$. Multiply this by the pentagonal expansion and compare the coefficient of q^n. □

13.3 Jacobi's cubic identity

Theorem 13.5 (Jacobi identity)**.** *For $|q|<1$,*

$$(q;q)_\infty^3=\sum_{n=0}^{\infty}(-1)^n(2n+1)q^{n(n+1)/2}.$$

Proof. Differentiate (13.2) with respect to z and set $z = 1$. Since

$$(z; q)_\infty = (1 - z)(qz; q)_\infty,$$

the derivative of the product side at $z = 1$ is $-(q; q)_\infty^3$. On the series side it is

$$\sum_{m \in \mathbb{Z}} m(-1)^m q^{\binom{m}{2}}.$$

Pair $m = -n$ with $m = n + 1$ for $n \geq 0$. Their sum is

$$-(2n + 1)(-1)^n q^{n(n+1)/2}.$$

Cancelling the common minus sign proves the identity. □

13.4 A theta form of the q-gamma reflection product

Corollary 13.6. *For* $0 < x < 1$,

$$\Gamma_q(x)\Gamma_q(1 - x) = (1 - q)\frac{(q; q)_\infty^3}{\sum_{n \in \mathbb{Z}} (-1)^n q^{\binom{n}{2} + xn}}.$$

Proof. Set $z = q^x$ in (13.2):

$$\sum_{n \in \mathbb{Z}} (-1)^n q^{\binom{n}{2} + xn} = (q^x, q^{1-x}, q; q)_\infty.$$

Substitute this in Theorem 12.10. □

Chapter 14

Partitions and product–sum identities

14.1 Unrestricted, distinct, and odd parts

Theorem 14.1 (Basic partition products)**.** *Let $p(n)$ count unrestricted partitions, $d(n)$ partitions into distinct parts, and $o(n)$ partitions into odd parts. Then*

$$\begin{aligned}\sum_{n\geq 0} p(n)q^n &= \frac{1}{(q;q)_\infty},\\ \sum_{n\geq 0} d(n)q^n &= (-q;q)_\infty,\\ \sum_{n\geq 0} o(n)q^n &= \frac{1}{(q;q^2)_\infty}.\end{aligned}$$

Moreover, $d(n) = o(n)$ for every n.

Proof. For unrestricted partitions, the part j may occur $0, 1, 2, \ldots$ times and contributes the geometric factor $(1 - q^j)^{-1}$. For distinct partitions, the part j occurs zero or one time and contributes $1 + q^j$. For odd parts, retain only the odd geometric factors. Finally,

$$(-q;q)_\infty = \prod_{j\geq 1} \frac{1 - q^{2j}}{1 - q^j} = \frac{(q^2;q^2)_\infty}{(q;q)_\infty} = \frac{1}{(q;q^2)_\infty},$$

so the two generating functions agree coefficientwise. □

Remark 14.2 (A bijection). The equality $d(n) = o(n)$ also has a direct proof. In a partition into distinct parts, write each part uniquely as $2^r u$ with u odd. Replace that part by 2^r copies of u. Distinct powers of two ensure that the process is reversible by binary expansion of the multiplicity of every odd part.

14.2 Bounded partitions

Proposition 14.3. *The following generating functions are equal:*

$$\frac{1}{(q;q)_r} = \sum_{n\geq 0} p_{\leq r}(n)q^n,$$

where $p_{\leq r}(n)$ counts partitions with largest part at most r, or equivalently partitions with at most r parts.

Proof. The product allows parts $1, \dots, r$ with arbitrary multiplicity. Conjugation of Ferrers diagrams exchanges the condition "largest part at most r" with "at most r parts." □

Corollary 14.4 (Durfee decomposition of all partitions)**.**

$$\frac{1}{(q;q)_\infty} = \sum_{r=0}^{\infty} \frac{q^{r^2}}{(q;q)_r^2}.$$

Proof. Decompose every partition by the side r of its Durfee square. The diagrams to the right and below are independent partitions with at most r parts. This is the identity used in the proof of Jacobi's triple product. □

14.3 Finite approximations and stabilization

Proposition 14.5. *For fixed N, the coefficient of q^N in $\begin{bmatrix} m+k \\ k \end{bmatrix}_q$ equals $p(N)$ whenever $m, k \geq N$.*

Proof. The Gaussian coefficient counts partitions inside a $k \times m$ rectangle. Any partition of N has at most N parts and largest part at most N, so for $m, k \geq N$ the rectangle restriction is vacuous. □

Remark 14.6. Thus $\begin{bmatrix} m+k \\ k \end{bmatrix}_q$ converges coefficientwise to $1/(q;q)_\infty$ as both sides of the rectangle tend to infinity. This finite-to-infinite passage is a recurring bridge between Gaussian polynomials and partition identities.

Chapter 15

Bailey pairs and the Rogers–Ramanujan identities

15.1 Bailey pairs

For brevity, write $(a)_n = (a;q)_n$ in this chapter.

Definition 15.1. Sequences $(\alpha_n)_{n\geq 0}$ and $(\beta_n)_{n\geq 0}$ form a *Bailey pair relative to a* if

$$\beta_n = \sum_{r=0}^{n} \frac{\alpha_r}{(q)_{n-r}(aq)_{n+r}} \qquad (n \geq 0).$$

15.2 Bailey inversion

Lemma 15.2 (q-difference annihilation)**.** *If $P(x)$ is a polynomial of degree less than N, then*

$$\sum_{s=0}^{N} (-1)^{N-s} q^{\binom{N-s}{2}} \begin{bmatrix} N \\ s \end{bmatrix}_q P(q^s) = 0.$$

Proof. By linearity it suffices to take $P(x) = x^m$ with $0 \leq m < N$. Put $t = N - s$ and use the finite q-binomial theorem:

$$\begin{aligned} \sum_{s=0}^{N} (-1)^{N-s} q^{\binom{N-s}{2}} \begin{bmatrix} N \\ s \end{bmatrix}_q q^{ms} &= q^{mN} \sum_{t=0}^{N} (-1)^t q^{\binom{t}{2}} \begin{bmatrix} N \\ t \end{bmatrix}_q q^{-mt} \\ &= q^{mN} (q^{-m}; q)_N = 0, \end{aligned}$$

for the product contains the factor $1 - q^0$. □

Theorem 15.3 (Bailey inversion)**.** *Assume $a \neq 1$ initially. The Bailey-pair relation is equivalent to*

$$\alpha_n = \frac{1 - aq^{2n}}{1 - a} \sum_{j=0}^{n} \frac{(a)_{n+j}}{(q)_{n-j}} (-1)^{n-j} q^{\binom{n-j}{2}} \beta_j.$$

At $a = 1$, the right side is interpreted by its removable limit.

Proof. Substitute the definition of β_j into the proposed inverse. The coefficient of α_r is

$$\frac{1-aq^{2n}}{1-a}\sum_{j=r}^{n}\frac{(a)_{n+j}}{(q)_{n-j}(q)_{j-r}(aq)_{j+r}}(-1)^{n-j}q^{\binom{n-j}{2}}.$$

If $n=r$, only $j=n$ occurs, and

$$\frac{1-aq^{2n}}{1-a}\frac{(a)_{2n}}{(aq)_{2n}}=1.$$

Suppose $N=n-r>0$ and put $j=r+s$. Then

$$\frac{(a)_{n+r+s}}{(aq)_{2r+s}}=(1-a)(aq^{2r+s+1};q)_{N-1}.$$

The coefficient becomes

$$\frac{1-aq^{2n}}{(q)_N}\sum_{s=0}^{N}(-1)^{N-s}q^{\binom{N-s}{2}}\begin{bmatrix}N\\ s\end{bmatrix}_q P(q^s),$$

where $P(x)=(aq^{2r+1}x;q)_{N-1}$ has degree $N-1$. It vanishes by Theorem 15.2. Thus the two triangular matrices are inverse. Since every expression is rational in a and the singularity at $a=1$ is removable, the limiting statement follows. □

Corollary 15.4 (Two unit Bailey pairs)**.** *Let* $\beta_n=\delta_{n,0}$.

(i) *Relative to* $a=1$,

$$\alpha_0=1,\qquad \alpha_n=(-1)^nq^{\binom{n}{2}}(1+q^n)\quad (n>0).$$

(ii) *Relative to* $a=q$,

$$\alpha_n=(-1)^nq^{\binom{n}{2}}[2n+1]_q\quad (n\ge 0).$$

Proof. With $\beta_j=\delta_{j,0}$, Bailey inversion gives

$$\alpha_n=\frac{1-aq^{2n}}{1-a}\frac{(a)_n}{(q)_n}(-1)^nq^{\binom{n}{2}}.$$

For $a\to 1$ and $n>0$, $(a)_n/(1-a)\to(q)_{n-1}$, so the prefactor reduces to $(1-q^{2n})/(1-q^n)=1+q^n$. For $a=q$, the ratio $(a)_n/(q)_n$ is 1 and the remaining factor is $[2n+1]_q$. □

15.3 The limiting Bailey lemma

Lemma 15.5 (Auxiliary finite identity)**.** *For* $N\ge 0$,

$$\sum_{k=0}^{N}z^kq^{k(k-1)}\begin{bmatrix}N\\ k\end{bmatrix}_q(zq^k;q)_{N-k}=1.$$

Proof. Call the left side $S_N(z)$. Using the second q-Pascal recurrence and splitting the last factor of $(zq^k;q)_{N+1-k}$ gives

$$S_{N+1}(z)=(1-zq^N)S_N(z)+zq^NS_N(qz).$$

The identity is immediate for $N=0$. If S_N is identically 1, the recurrence gives $S_{N+1}=1-zq^N+zq^N=1$. Induction completes the proof. □

Theorem 15.6 (Limiting Bailey lemma, finite form)**.** *If (α_n, β_n) is a Bailey pair relative to a, define*

$$\alpha'_n = a^n q^{n^2} \alpha_n, \qquad \beta'_n = \sum_{j=0}^{n} \frac{a^j q^{j^2}}{(q)_{n-j}} \beta_j .$$

Then (α'_n, β'_n) is also a Bailey pair relative to a.

Proof. Substitute the Bailey relation for β_j and interchange the finite sums. The coefficient of α_r in β'_n is

$$a^r q^{r^2} \sum_{s=0}^{N} \frac{a^s q^{s^2+2rs}}{(q)_{N-s}(q)_s(aq)_{2r+s}}, \qquad N = n - r.$$

Let $A = aq^{2r+1}$. Since

$$a^s q^{s^2+2rs} = A^s q^{s(s-1)}, \qquad (aq)_{2r+s} = (aq)_{2r}(A)_s,$$

the desired Bailey coefficient $a^r q^{r^2}/((q)_N (aq)_{n+r})$ is equivalent, after clearing factors, to

$$\sum_{s=0}^{N} \begin{bmatrix} N \\ s \end{bmatrix}_q A^s q^{s(s-1)} (Aq^s)_{N-s} = 1.$$

This is Theorem 15.5. □

Corollary 15.7 (Limiting Bailey lemma, infinite form)**.** *If the series converge absolutely, then every Bailey pair relative to a satisfies*

$$\sum_{n=0}^{\infty} a^n q^{n^2} \beta_n = \frac{1}{(aq;q)_\infty} \sum_{n=0}^{\infty} a^n q^{n^2} \alpha_n .$$

Proof. The transformed pair in Theorem 15.6 satisfies

$$\beta'_N = \sum_{r=0}^{N} \frac{\alpha'_r}{(q)_{N-r}(aq)_{N+r}} .$$

Let $N \to \infty$. On the left,

$$\beta'_N \longrightarrow \frac{1}{(q)_\infty} \sum_{j \ge 0} a^j q^{j^2} \beta_j .$$

On the right,

$$\beta'_N \longrightarrow \frac{1}{(q)_\infty (aq)_\infty} \sum_{r \ge 0} a^r q^{r^2} \alpha_r .$$

Absolute convergence permits dominated convergence. Cancel $(q)_\infty^{-1}$. □

15.4 The Rogers–Ramanujan identities

Theorem 15.8 (Rogers–Ramanujan)**.** *For $|q| < 1$,*

$$\sum_{n=0}^{\infty} \frac{q^{n^2}}{(q;q)_n} = \frac{1}{(q;q^5)_\infty (q^4;q^5)_\infty}, \tag{15.1}$$

$$\sum_{n=0}^{\infty} \frac{q^{n(n+1)}}{(q;q)_n} = \frac{1}{(q^2;q^5)_\infty (q^3;q^5)_\infty}. \tag{15.2}$$

Proof. For the first identity, begin with the unit pair relative to $a = 1$ in Theorem 15.4. Apply the finite limiting lemma once. The new pair is

$$\alpha'_n = q^{n^2}\alpha_n, \qquad \beta'_n = \frac{1}{(q)_n}.$$

Applying the infinite limiting lemma to this new pair gives

$$\begin{aligned}\sum_{n\geq 0}\frac{q^{n^2}}{(q)_n} &= \frac{1}{(q)_\infty}\left(1+\sum_{r\geq 1}(-1)^r q^{2r^2+\binom{r}{2}}(1+q^r)\right)\\ &= \frac{1}{(q)_\infty}\sum_{r\in\mathbb{Z}}(-1)^r q^{(5r^2-r)/2}.\end{aligned}$$

By Jacobi's triple product with base q^5 and $z = q^2$,

$$\sum_{r\in\mathbb{Z}}(-1)^r q^{(5r^2-r)/2} = (q^2, q^3, q^5; q^5)_\infty.$$

Dissection gives

$$(q;q)_\infty = (q, q^2, q^3, q^4, q^5; q^5)_\infty,$$

which proves (15.1).

For the second identity use the unit pair relative to $a = q$. After one finite limiting step,

$$\alpha'_n = (-1)^n q^{n^2+n+\binom{n}{2}}[2n+1]_q, \qquad \beta'_n = \frac{1}{(q)_n}.$$

The infinite limiting lemma gives

$$\begin{aligned}\sum_{n\geq 0}\frac{q^{n^2+n}}{(q)_n} &= \frac{1}{(q^2;q)_\infty}\sum_{r\geq 0}q^{r^2+r}\alpha'_r\\ &= \frac{1}{(q;q)_\infty}\sum_{r\geq 0}(-1)^r q^{(5r^2+3r)/2}(1-q^{2r+1})\\ &= \frac{1}{(q;q)_\infty}\sum_{r\in\mathbb{Z}}(-1)^r q^{(5r^2+3r)/2}.\end{aligned}$$

In the last step, the subtracted term indexed by $r \geq 0$ is the term with bilateral index $-r-1$. Jacobi's triple product with base q^5 and $z = q^4$ gives

$$\sum_{r\in\mathbb{Z}}(-1)^r q^{(5r^2+3r)/2} = (q, q^4, q^5; q^5)_\infty.$$

Divide by the fivefold dissection of $(q;q)_\infty$ to obtain (15.2). □

Remark 15.9 (Partition interpretation). The product side of (15.1) counts partitions into parts congruent to 1 or 4 modulo 5; the product side of (15.2) counts parts congruent to 2 or 3 modulo 5. The sum sides count partitions with successive parts differing by at least 2, with an additional smallest-part condition in the second identity. Establishing the latter combinatorial interpretations requires a separate bijective or finite-polynomial argument; the proof above instead reveals the hypergeometric engine.

Part IV

Cyclotomic arithmetic and extended indices

Chapter 16

Cyclotomic factorization and congruences

The arithmetic of Gaussian coefficients is governed by cyclotomic polynomials. This is the precise sense in which a primitive root of unity plays the role of a prime. The point of this chapter is that the essential congruences do not require mysterious cancellation: they follow from product factorization, coefficient extraction, and a careful accounting of blocks.

16.1 Cyclotomic valuations and carries

Let $\Phi_d(q)$ denote the dth cyclotomic polynomial, normalized by

$$q^n - 1 = \prod_{d|n} \Phi_d(q).$$

For a nonzero polynomial $F \in \mathbb{Z}[q]$, write $\nu_d(F)$ for the exponent of $\Phi_d(q)$ in its factorization.

Theorem 16.1 (Cyclotomic factorization of a shifted factorial)**.** *For every $n \geq 0$,*

$$(q;q)_n = (-1)^n \prod_{d=1}^{n} \Phi_d(q)^{\lfloor n/d \rfloor}.$$

Consequently,

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \prod_{d=1}^{n} \Phi_d(q)^{e_d(n,k)}, \qquad e_d(n,k) = \lfloor n/d \rfloor - \lfloor k/d \rfloor - \lfloor (n-k)/d \rfloor. \tag{16.1}$$

Moreover $e_d(n,k)$ is always either 0 or 1.

Proof. Since $1 - q^j = -(q^j - 1)$ and $q^j - 1 = \prod_{d|j} \Phi_d(q)$,

$$(q;q)_n = \prod_{j=1}^{n}(1-q^j) = (-1)^n \prod_{j=1}^{n} \prod_{d|j} \Phi_d(q).$$

For fixed d, the multiples of d among $1, \ldots, n$ are $d, 2d, \ldots, \lfloor n/d \rfloor d$, so the exponent of Φ_d is $\lfloor n/d \rfloor$. Dividing the factorizations of $(q;q)_n$, $(q;q)_k$, and $(q;q)_{n-k}$ gives (16.1); the signs cancel.

Write $k = ad + r$ and $n - k = bd + s$ with $0 \leq r, s < d$. Then

$$e_d(n,k) = \lfloor (r+s)/d \rfloor,$$

which is 0 or 1. □

Corollary 16.2 (Square-free cyclotomic structure)**.** *Every Gaussian polynomial is a product of distinct cyclotomic polynomials. More precisely,*

$$\Phi_d(q) \mid \begin{bmatrix} n \\ k \end{bmatrix}_q \iff (k \bmod d) > (n \bmod d). \tag{16.2}$$

Equivalently, Φ_d occurs exactly when adding k and $n-k$ produces a carry in the units place in base d.

Proof. The first assertion follows because every exponent in (16.1) belongs to $\{0,1\}$. If $n = cd + t$ and $k = ad + r$ with $0 \le r, t < d$, then the residue of $n - k$ is $t - r$ when $r \le t$ and $t - r + d$ when $r > t$. Thus the two residues add to t in the first case and to $t + d$ in the second. The latter is precisely the carry case, and by the proof of Theorem 16.1 it is equivalent to $e_d(n,k) = 1$. □

Example 16.3. For $n = 10$ and $k = 4$, the values of d for which $4 \bmod d > 10 \bmod d$ are $d = 5, 7, 8, 9$. Hence

$$\begin{bmatrix} 10 \\ 4 \end{bmatrix}_q = \Phi_5(q)\Phi_7(q)\Phi_8(q)\Phi_9(q).$$

The degrees add to $4 + 6 + 4 + 6 = 20 = 4(10 - 4)$, as required by reciprocity.

16.2 The q-Lucas congruence

The following theorem is often called the q-Lucas theorem. It is naturally proved by grouping a finite product into complete root-of-unity blocks.

Lemma 16.4 (A complete root-of-unity block)**.** *If ζ is a primitive dth root of unity, then*

$$\prod_{j=0}^{d-1} (1 + x\zeta^j) = 1 - (-x)^d. \tag{16.3}$$

Proof. The roots in x of the left side are $-\zeta^{-j}$, which are exactly the roots of $1 - (-x)^d$. Both polynomials have constant term 1, so they are equal. □

Theorem 16.5 (q-Lucas)**.** *Let $a, r \ge 0$, let $d \ge 1$, and let $0 \le b, s < d$. Then*

$$\begin{bmatrix} ad + b \\ rd + s \end{bmatrix}_q \equiv \binom{a}{r} \begin{bmatrix} b \\ s \end{bmatrix}_q \pmod{\Phi_d(q)}. \tag{16.4}$$

Here the right side is understood as 0 when $r > a$ or $s > b$.

Proof. It suffices to prove equality after substituting an arbitrary primitive dth root ζ for q, because Φ_d is the minimal polynomial of ζ over $\mathbb{Q}$.

The finite q-binomial theorem gives

$$\prod_{j=0}^{ad+b-1} (1 + x\zeta^j) = \sum_{h=0}^{ad+b} \zeta^{\binom{h}{2}} \begin{bmatrix} ad + b \\ h \end{bmatrix}_\zeta x^h. \tag{16.5}$$

Group the product into a complete blocks and one incomplete block. By Theorem 16.4,

$$\prod_{j=0}^{ad+b-1} (1 + x\zeta^j) = \left(1 - (-x)^d\right)^a \prod_{j=0}^{b-1} (1 + x\zeta^j).$$

The coefficient of x^{rd+s} on the right is

$$\binom{a}{r}(-1)^{r(d+1)}\zeta^{\binom{s}{2}}\begin{bmatrix} b \\ s \end{bmatrix}_{\zeta}. \tag{16.6}$$

Comparing (16.5) and (16.6) leaves only the phase. We claim that

$$(-1)^{r(d+1)}\zeta^{\binom{s}{2}-\binom{rd+s}{2}} = 1. \tag{16.7}$$

Indeed,

$$2\left(\binom{rd+s}{2}-\binom{s}{2}\right) = rd(rd+2s-1).$$

If d is odd, $r(rd+2s-1)$ is even, so the exponent difference is a multiple of d, while $r(d+1)$ is even. If d is even, then $\zeta^{d/2} = -1$ and $rd+2s-1$ is odd; hence the second factor in (16.7) equals $(-1)^r$, as does the first. Their product is 1. The coefficient comparison therefore gives (16.4) at $q = \zeta$, completing the proof. □

Corollary 16.6 (Evaluation at $q = -1$)**.** *For* $0 \le k \le n$*,*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q\bigg|_{q=-1} = \begin{cases} \dbinom{n/2}{k/2}, & n \text{ even and } k \text{ even}, \\ 0, & n \text{ even and } k \text{ odd}, \\ \dbinom{(n-1)/2}{\lfloor k/2 \rfloor}, & n \text{ odd}. \end{cases}$$

Proof. Apply Theorem 16.5 with $d = 2$. Write $n = 2a + b$ and $k = 2r + s$, where $b, s \in \{0, 1\}$. Since $\left[\begin{smallmatrix} b \\ s \end{smallmatrix}\right]_{-1}$ is 1 when $s \le b$ and 0 otherwise, the three cases follow. □

16.3 Cyclic sieving for subsets

A triple $(X, C, F(q))$, where a finite cyclic group $C = \langle c \rangle$ acts on a finite set X and $F(q) \in \mathbb{Z}[q]$, exhibits the *cyclic sieving phenomenon* when

$$\#\operatorname{Fix}(c^j) = F(\omega^j)$$

for every j, where ω is a primitive $|C|$th root of unity.

Theorem 16.7 (Cyclic sieving for k-subsets)**.** *Let X be the set of k-element subsets of $\mathbb{Z}/n\mathbb{Z}$, and let C_n act by translation. Then*

$$\left(X, C_n, \begin{bmatrix} n \\ k \end{bmatrix}_q\right)$$

exhibits the cyclic sieving phenomenon.

Proof. Fix j, and let d be the order of translation by j. Its orbits on $\mathbb{Z}/n\mathbb{Z}$ all have size d, and there are n/d such orbits. A subset is fixed precisely when it is a union of these orbits. Thus

$$\#\operatorname{Fix}(c^j) = \begin{cases} \dbinom{n/d}{k/d}, & d \mid k, \\ 0, & d \nmid k. \end{cases} \tag{16.8}$$

The number ω^j is a primitive dth root. Write $n = (n/d)d$ and $k = rd + s$ with $0 \le s < d$. By q-Lucas,

$$\begin{bmatrix} n \\ k \end{bmatrix}_q\bigg|_{q=\omega^j} = \binom{n/d}{r}\begin{bmatrix} 0 \\ s \end{bmatrix}_{\omega^j}.$$

This is 0 if $s > 0$, and it is $\binom{n/d}{k/d}$ if $s = 0$, exactly as in (16.8). □

16.4 A q-Babbage congruence modulo a cyclotomic square

The modulus $\Phi_n(q)^2$ records not only the value at a primitive nth root, but also first-order contact there. The next congruence is a particularly transparent example. Its block proof is also a useful template for stronger q-congruences; see Andrews [3] and Straub [35].

Theorem 16.8 (q-Babbage congruence)**.** *For integers $a \geq b \geq 0$ and $n \geq 1$,*

$$\begin{bmatrix} an \\ bn \end{bmatrix}_q \equiv \begin{bmatrix} a \\ b \end{bmatrix}_{q^{n^2}} \pmod{\Phi_n(q)^2}. \tag{16.9}$$

Proof. Split the finite product into a consecutive blocks of length n:

$$\prod_{j=0}^{an-1}(1+xq^j) = \prod_{i=0}^{a-1}\prod_{j=0}^{n-1}(1+xq^{in+j}).$$

Apply the finite q-binomial theorem to each block. Comparing the coefficient of x^{bn} gives

$$q^{\binom{bn}{2}}\begin{bmatrix} an \\ bn \end{bmatrix}_q = \sum_{\substack{0\leq k_0,\ldots,k_{a-1}\leq n \\ k_0+\cdots+k_{a-1}=bn}} q^{\sum_i \binom{k_i}{2}+n\sum_i ik_i}\prod_{i=0}^{a-1}\begin{bmatrix} n \\ k_i \end{bmatrix}_q. \tag{16.10}$$

If $0 < k_i < n$, then $\Phi_n(q)$ divides $\begin{bmatrix} n \\ k_i \end{bmatrix}_q$ by Theorem 16.2. A tuple contributing to (16.10) cannot have exactly one such intermediate entry: all other entries would be 0 or n, making the sum of the k_i nonzero modulo n. Therefore every tuple containing an intermediate entry contributes a multiple of $\Phi_n(q)^2$.

Modulo $\Phi_n(q)^2$, only tuples with each $k_i \in \{0, n\}$ survive. Such a tuple is specified by a b-element subset $S \subseteq \{0, 1, \ldots, a-1\}$. After division by $q^{\binom{bn}{2}}$, its exponent is

$$n^2\sum_{i\in S} i + b\binom{n}{2} - \binom{bn}{2} = n^2\left(\sum_{i\in S} i - \binom{b}{2}\right).$$

Hence

$$\begin{bmatrix} an \\ bn \end{bmatrix}_q \equiv \sum_{\substack{S\subseteq\{0,\ldots,a-1\} \\ |S|=b}} (q^{n^2})^{\sum_{i\in S} i-\binom{b}{2}} \pmod{\Phi_n(q)^2}.$$

The subset form of the finite q-binomial theorem identifies the sum as $\begin{bmatrix} a \\ b \end{bmatrix}_{q^{n^2}}$. □

Remark 16.9 (Why the square appears). The decisive observation in the proof is combinatorial: because the target degree is a multiple of the block length, nontrivial block choices occur in pairs. Each nontrivial block supplies one factor $\Phi_n(q)$, so two of them supply the square. Congruences modulo Φ_n^3 require genuinely finer first- and second-order information; the q-Ljunggren congruences of Straub and subsequent refinements belong to that deeper layer.

Corollary 16.10 (Primitive-root value and derivative)**.** *Let ζ be a primitive nth root of unity. Then*

$$\begin{bmatrix} an \\ bn \end{bmatrix}_\zeta = \binom{a}{b},$$

and the derivatives at $q = \zeta$ of the two sides of (16.9) *are equal.*

Proof. The first assertion follows either from q-Lucas or from (16.9), because $\zeta^{n^2} = 1$. If a polynomial is divisible by Φ_n^2, it has a zero of order at least two at every primitive nth root, so both its value and first derivative vanish there. □

Chapter 17

Negative upper indices and geometric Newton interpolation

Ordinary binomial coefficients acquire a useful extension to negative upper arguments. Gaussian coefficients do as well, but the powers of q are essential: omitting them destroys the Pascal recurrences. The same algebra leads naturally to interpolation on a geometric grid.

17.1 Generalized Gaussian coefficients

Definition 17.1. For an integer N and $k \geq 0$, define

$$\begin{bmatrix} N \\ k \end{bmatrix}_q := \frac{(q^{N-k+1};q)_k}{(q;q)_k}. \tag{17.1}$$

For $N \geq 0$ this agrees with the usual Gaussian coefficient, including the value 0 when $k > N$.

Proposition 17.2 (Negative-index reflection)**.** *For $m \geq 1$ and $k \geq 0$,*

$$\begin{bmatrix} -m \\ k \end{bmatrix}_q = (-1)^k q^{-mk-\binom{k}{2}} \begin{bmatrix} m+k-1 \\ k \end{bmatrix}_q. \tag{17.2}$$

Proof. From the definition,

$$\begin{aligned}
(q^{-m-k+1};q)_k &= \prod_{j=0}^{k-1} \left(1 - q^{-(m+k-1-j)}\right) \\
&= (-1)^k q^{-\sum_{j=0}^{k-1}(m+k-1-j)} \prod_{j=0}^{k-1} (1 - q^{m+k-1-j}) \\
&= (-1)^k q^{-mk-\binom{k}{2}} (q^m;q)_k.
\end{aligned}$$

After division by $(q;q)_k$, the final ratio is $\begin{bmatrix} m+k-1 \\ k \end{bmatrix}_q$. □

Theorem 17.3 (Pascal recurrences for all integral upper indices)**.** *For every integer N and every $k \geq 1$,*

$$\begin{aligned}
\begin{bmatrix} N \\ k \end{bmatrix}_q &= \begin{bmatrix} N-1 \\ k-1 \end{bmatrix}_q + q^k \begin{bmatrix} N-1 \\ k \end{bmatrix}_q, \qquad (17.3) \\
&= q^{N-k} \begin{bmatrix} N-1 \\ k-1 \end{bmatrix}_q + \begin{bmatrix} N-1 \\ k \end{bmatrix}_q. \qquad (17.4)
\end{aligned}$$

Both formulas are identities in $\mathbb{Q}(q)$.

Proof. Put the two terms on the right of (17.3) over the common denominator $(q;q)_k$. Since

$$\begin{bmatrix} N-1 \\ k-1 \end{bmatrix}_q = \frac{(q^{N-k+1};q)_{k-1}(1-q^k)}{(q;q)_k}$$

the numerator of the right side is

$$\begin{aligned} &(q^{N-k+1};q)_{k-1}\left[(1-q^k)+q^k(1-q^{N-k})\right] \\ &\quad = (q^{N-k+1};q)_{k-1}(1-q^N) = (q^{N-k+1};q)_k, \end{aligned}$$

which proves (17.3). For (17.4), the corresponding bracket is

$$q^{N-k}(1-q^k)+(1-q^{N-k}) = 1-q^N.$$

The same common-denominator calculation applies. □

Corollary 17.4 (Reciprocal product series)**.** *For $m \geq 1$ and $|z| < 1$,*

$$\frac{1}{(z;q)_m} = \sum_{k=0}^{\infty} \begin{bmatrix} m+k-1 \\ k \end{bmatrix}_q z^k = \sum_{k=0}^{\infty} (-1)^k q^{mk+\binom{k}{2}} \begin{bmatrix} -m \\ k \end{bmatrix}_q z^k. \tag{17.5}$$

The first equality also holds in the formal power-series ring $\mathbb{Z}[q][[z]]$.

Proof. The reciprocal finite q-binomial theorem, already proved in Theorem 3.4, gives the first expansion. Substitute (17.2) to obtain the second. Formal validity follows because each coefficient of z^k is a polynomial and multiplication by the finite product $(z;q)_m$ can be checked coefficientwise. □

17.2 An arbitrary upper parameter

When a branch of q^α has been fixed, it is often convenient to define

$$\begin{bmatrix} \alpha \\ k \end{bmatrix}_q = \frac{(q^{\alpha-k+1};q)_k}{(q;q)_k}. \tag{17.6}$$

The proofs of the Pascal recurrences are purely algebraic in q^α and therefore remain valid. A related, and usually more analytic, expansion is the following.

Theorem 17.5 (Generalized upper-parameter series)**.** *For $|q| < 1$, $|z| < 1$, and any parameter α for which the factors are defined,*

$$\frac{(q^{\alpha+1}z;q)_\infty}{(z;q)_\infty} = \sum_{k=0}^{\infty} \begin{bmatrix} \alpha+k \\ k \end{bmatrix}_q z^k. \tag{17.7}$$

Proof. Apply the infinite q-binomial theorem with $a = q^{\alpha+1}$:

$$\frac{(q^{\alpha+1}z;q)_\infty}{(z;q)_\infty} = \sum_{k\geq 0} \frac{(q^{\alpha+1};q)_k}{(q;q)_k} z^k.$$

The coefficient is exactly $\begin{bmatrix} \alpha+k \\ k \end{bmatrix}_q$ by (17.6). □

17.3 Two complex arguments and generalized binomial series

The extension in Theorem 1.11 also permits both arguments of a Gaussian coefficient to be complex. Throughout this section take $0 < q < 1$ and set $q^z = \exp(z \log q)$, using the real logarithm of q.

Definition 17.6. For $\alpha, \beta \in \mathbb{C}$, define

$$\begin{bmatrix}\alpha\\ \beta\end{bmatrix}_q := \frac{(q^{\beta+1}, q^{\alpha-\beta+1}; q)_\infty}{(q, q^{\alpha+1}; q)_\infty}, \tag{17.8}$$

whenever the denominator is nonzero.

Theorem 17.7 (Gamma representation and Pascal laws)**.** *Wherever the expressions are defined,*

$$\begin{bmatrix}\alpha\\ \beta\end{bmatrix}_q = \frac{\Gamma_q(\alpha+1)}{\Gamma_q(\beta+1)\Gamma_q(\alpha-\beta+1)}, \tag{17.9}$$

$$\begin{bmatrix}\alpha\\ \beta\end{bmatrix}_q = \begin{bmatrix}\alpha\\ \alpha-\beta\end{bmatrix}_q, \tag{17.10}$$

$$\begin{bmatrix}\alpha+1\\ \beta\end{bmatrix}_q = q^\beta \begin{bmatrix}\alpha\\ \beta\end{bmatrix}_q + \begin{bmatrix}\alpha\\ \beta-1\end{bmatrix}_q, \tag{17.11}$$

$$= \begin{bmatrix}\alpha\\ \beta\end{bmatrix}_q + q^{\alpha+1-\beta}\begin{bmatrix}\alpha\\ \beta-1\end{bmatrix}_q. \tag{17.12}$$

If $\beta = k \in \mathbb{N}_0$, this definition agrees with (17.6)*.*

Proof. Substitution of the product definition of Γ_q proves (17.9); all powers of $1-q$ cancel. Symmetry is immediate either from that quotient or directly from (17.8). For $\beta = k \in \mathbb{N}_0$, tail cancellation gives

$$\frac{(q^{k+1}; q)_\infty}{(q; q)_\infty} = \frac{1}{(q; q)_k}, \qquad \frac{(q^{\alpha-k+1}; q)_\infty}{(q^{\alpha+1}; q)_\infty} = (q^{\alpha-k+1}; q)_k,$$

which is exactly (17.6).

For the first Pascal law, divide every term by $\left[\begin{smallmatrix}\alpha\\ \beta\end{smallmatrix}\right]_q$. The gamma functional equation gives

$$\frac{\left[\begin{smallmatrix}\alpha+1\\ \beta\end{smallmatrix}\right]_q}{\left[\begin{smallmatrix}\alpha\\ \beta\end{smallmatrix}\right]_q} = \frac{[\alpha+1]_q}{[\alpha-\beta+1]_q}, \qquad \frac{\left[\begin{smallmatrix}\alpha\\ \beta-1\end{smallmatrix}\right]_q}{\left[\begin{smallmatrix}\alpha\\ \beta\end{smallmatrix}\right]_q} = \frac{[\beta]_q}{[\alpha-\beta+1]_q}.$$

The identity $[\alpha+1]_q = q^\beta[\alpha-\beta+1]_q + [\beta]_q$ proves (17.11). The alternative identity $[\alpha+1]_q = [\alpha-\beta+1]_q + q^{\alpha+1-\beta}[\beta]_q$ proves (17.12). □

Corollary 17.8 (Classical generalized binomial)**.** *If $\alpha > \beta > -1$ are real, then*

$$\lim_{q\to 1^-} \begin{bmatrix}\alpha\\ \beta\end{bmatrix}_q = \frac{\Gamma(\alpha+1)}{\Gamma(\beta+1)\Gamma(\alpha-\beta+1)}.$$

Proof. Apply the locally uniform q-gamma limit in Theorem 12.5 to (17.9). □

Theorem 17.9 (Generalized finite and reciprocal q-binomial theorems)**.** *Let $0 < q < 1$. If $|zq^\alpha| < 1$, then*

$$(z; q)_\alpha = \sum_{k=0}^{\infty} (-1)^k q^{\binom{k}{2}} \begin{bmatrix}\alpha\\ k\end{bmatrix}_q z^k. \tag{17.13}$$

If $|z| < 1$*, then*

$$\frac{1}{(z;q)_\alpha} = \sum_{k=0}^{\infty} \begin{bmatrix} \alpha + k - 1 \\ k \end{bmatrix}_q z^k. \tag{17.14}$$

For a nonnegative integral α*, the first series terminates and becomes Euler's finite theorem; in that case the restriction* $|zq^\alpha| < 1$ *can be removed because both sides are polynomials. For a positive integral* α*, the second becomes the reciprocal finite theorem.*

Proof. Apply the infinite q-binomial theorem with $a = q^{-\alpha}$ and argument zq^α:

$$\frac{(z;q)_\infty}{(zq^\alpha;q)_\infty} = \sum_{k\geq 0} \frac{(q^{-\alpha};q)_k}{(q;q)_k} (zq^\alpha)^k.$$

The left side is $(z;q)_\alpha$. Reversing the finite numerator product gives

$$\frac{(q^{-\alpha};q)_k}{(q;q)_k} q^{\alpha k} = (-1)^k q^{\binom{k}{2}} \begin{bmatrix} \alpha \\ k \end{bmatrix}_q,$$

which proves (17.13).

For the reciprocal formula, apply the infinite theorem with $a = q^\alpha$ and argument z:

$$\frac{(q^\alpha z;q)_\infty}{(z;q)_\infty} = \sum_{k\geq 0} \frac{(q^\alpha;q)_k}{(q;q)_k} z^k.$$

The left side is $(z;q)_\alpha^{-1}$ and the coefficient is $\begin{bmatrix} \alpha+k-1 \\ k \end{bmatrix}_q$. The integral specializations follow from the consistency results already proved. □

17.4 Newton interpolation on the grid $1, q, q^2, \ldots$

The usual Newton basis is adapted to an arbitrary sequence of distinct nodes. At geometric nodes its basis polynomials are q-Pochhammer symbols.

Theorem 17.10 (Geometric Newton interpolation)**.** *Assume* $q^i \neq q^j$ *whenever* $0 \leq i < j \leq n$*. For arbitrary values* $y_0, \ldots, y_n$ *there is a unique polynomial* P *of degree at most* n *satisfying*

$$P(q^j) = y_j \qquad (0 \leq j \leq n).$$

It has the Newton expansion

$$P(x) = \sum_{k=0}^{n} c_k \prod_{r=0}^{k-1} (x - q^r) = \sum_{k=0}^{n} c_k (-1)^k q^{\binom{k}{2}} (x;q^{-1})_k, \tag{17.15}$$

where

$$c_k = \sum_{j=0}^{k} \frac{y_j}{\prod_{\substack{0\leq r\leq k \\ r\neq j}} (q^j - q^r)} \tag{17.16}$$

$$= \sum_{j=0}^{k} \frac{(-1)^j q^{-jk+\binom{j+1}{2}}}{(q;q)_j (q;q)_{k-j}} y_j. \tag{17.17}$$

Proof. For distinct nodes $x_0, \dots, x_n$, Lagrange interpolation gives existence and uniqueness. The coefficient of the Newton basis polynomial $\prod_{r=0}^{k-1}(x - x_r)$ is the divided difference

$$y[x_0, \dots, x_k] = \sum_{j=0}^{k} \frac{y_j}{\prod_{r \neq j}(x_j - x_r)};$$

this follows by comparing the coefficient of x^k in the Lagrange interpolant through the first $k + 1$ nodes. Taking $x_j = q^j$ proves (17.16).

It remains to simplify the denominator. Splitting the factors below and above j gives

$$\begin{aligned}\prod_{\substack{0 \leq r \leq k \\ r \neq j}} (q^j - q^r) &= \left[\prod_{r=0}^{j-1} -q^r(1 - q^{j-r})\right]\left[\prod_{r=j+1}^{k} q^j(1 - q^{r-j})\right] \\ &= (-1)^j q^{\binom{j}{2} + j(k-j)} (q;q)_j (q;q)_{k-j} \\ &= (-1)^j q^{jk - \binom{j+1}{2}} (q;q)_j (q;q)_{k-j}.\end{aligned}$$

Taking reciprocals yields (17.17). Finally,

$$\prod_{r=0}^{k-1}(x - q^r) = (-1)^k q^{\binom{k}{2}} \prod_{r=0}^{k-1}(1 - xq^{-r}) = (-1)^k q^{\binom{k}{2}} (x; q^{-1})_k,$$

which proves the second form of (17.15). □

Corollary 17.11 (Triangular reconstruction)**.** *The coefficients in* (17.15) *can be recovered successively by*

$$c_0 = y_0, \qquad c_k = \frac{y_k - \sum_{r=0}^{k-1} c_r \prod_{j=0}^{r-1}(q^k - q^j)}{\prod_{j=0}^{k-1}(q^k - q^j)}. \tag{17.18}$$

Proof. Evaluate (17.15) at $x = q^k$. Every term with index greater than k vanishes, while the coefficient multiplying c_k is nonzero by the distinct-node hypothesis. Solving for c_k gives the formula. □

Remark 17.12. Kupershmidt's q-Newton viewpoint emphasizes that the classical binomial theorem, Newton interpolation, and q-difference calculus are not separate constructions: each is controlled by a basis whose successive factors form a geometric progression. The finite formula above is algebraic and needs no convergence hypothesis; infinite Newton series require additional growth conditions on the data.

Part V

Dyadic q-calculus and self-similar distributions

Chapter 18

Thue–Morse cancellation and exact Fabius values

The preceding chapters develop q-shifted factorials and Gaussian coefficients as general algebraic objects. This chapter explains a much more specialized occurrence in which the base $q = 1/2$ is forced by a dilation functional equation. The resulting formulae link four structures:

1. the Fabius distribution, obtained from an infinite geometrically weighted sum of uniform random variables;
2. finite Thue–Morse blocks, whose signed moments vanish to a prescribed order;
3. the polynomial $(z; 1/2)_n$, whose roots are the dyadic nodes $1, 2, \ldots, 2^{n-1}$;
4. Gaussian coefficients at $1/2$, which form the unique geometric finite-difference stencil attached to those nodes.

The main result is an exact formula for every dyadic value of the Fabius function. All analytic and algebraic steps are proved below; no appeal to a computer algebra system or proof assistant is needed.

18.1 The probabilistic construction

Let $U_1, U_2, \ldots$ be independent random variables, each uniformly distributed on $[0, 1]$, and define

$$X = \sum_{j=1}^{\infty} \frac{U_j}{2^j}, \qquad \mathsf{F}(x) = \mathbb{P}(X \le x). \tag{18.1}$$

The series converges uniformly in the sample point and $0 \le X \le 1$. We extend F by 0 on $(-\infty, 0]$ and by 1 on $[1, \infty)$.

Proposition 18.1 (Symmetry and dyadic self-similarity)**.** *The random variable X satisfies*

$$X \overset{d}{=} 1 - X \tag{18.2}$$

and, if X' is an independent copy of X,

$$X \overset{d}{=} \frac{U_1 + X'}{2}. \tag{18.3}$$

Consequently,

$$\mathsf{F}(1 - x) = 1 - \mathsf{F}(x) \qquad (0 \le x \le 1), \tag{18.4}$$

and

$$\mathsf{F}(x) = \int_0^1 \mathsf{F}(2x - u)\ \mathrm{d}u. \tag{18.5}$$

In particular, for $0 \le x \le 1/2$*,*

$$\mathsf{F}'(x) = 2\mathsf{F}(2x). \tag{18.6}$$

Proof. Replacing every U_j by $1 - U_j$ preserves their joint distribution, and

$$\sum_{j \ge 1} \frac{1 - U_j}{2^j} = \sum_{j \ge 1} 2^{-j} - X = 1 - X,$$

which proves (18.2). Splitting off the first term and reindexing the tail gives

$$X = \frac{U_1}{2} + \frac{1}{2} \sum_{j \ge 1} \frac{U_{j+1}}{2^j},$$

and the reindexed sum has the same law as X and is independent of U_1. This proves (18.3).

The symmetry of the distribution gives (18.4); there are no atoms because, conditional on X', the variable $(U_1 + X')/2$ is uniform on an interval. Conditioning on $U_1 = u$ in (18.3) gives

$$\mathsf{F}(x) = \int_0^1 \mathbb{P}(X' \le 2x - u)\ \mathrm{d}u,$$

which is (18.5). When $0 \le x \le 1/2$, the integrand vanishes for $u > 2x$, so

$$\mathsf{F}(x) = \int_0^{2x} \mathsf{F}(2x - u)\ \mathrm{d}u = \int_0^{2x} \mathsf{F}(v)\ \mathrm{d}v.$$

Differentiation proves (18.6). □

The distribution function in (18.1) is the classical Fabius function. Its compactly supported density, after the affine change of variable $Y = 2X - 1$, is Rvachev's up function; this will be proved in Section 19.7.

18.2 Moment product, Mahler equation, and inverse dyadic values

Put

$$\Phi(z) = \frac{\mathrm{e}^z - 1}{z}, \qquad \Phi(0) = 1, \tag{18.7}$$

and define moments and normalized moments by

$$d_n = \mathbb{E}X^n, \qquad a_n = \frac{d_n}{n!}. \tag{18.8}$$

Theorem 18.2 (Moment product and Mahler equation)**.** *The moment generating function*

$$G(z) = \mathbb{E}\mathrm{e}^{zX} = \sum_{n=0}^{\infty} a_n z^n \tag{18.9}$$

is entire and has the locally uniformly convergent product

$$G(z) = \prod_{j=1}^{\infty} \Phi\Big(\frac{z}{2^j}\Big). \tag{18.10}$$

It satisfies

$$G(2z) = \Phi(z)G(z), \tag{18.11}$$

and hence

$$(2^n - 1)a_n = \sum_{k=0}^{n-1} \frac{a_k}{(n-k+1)!} \qquad (n \geq 1), \qquad a_0 = 1. \tag{18.12}$$

Proof. Independence gives

$$\mathbb{E}\exp\left(z\sum_{j=1}^{N} 2^{-j}U_j\right) = \prod_{j=1}^{N} \mathbb{E}e^{z2^{-j}U_j} = \prod_{j=1}^{N} \Phi(z/2^j).$$

On every compact set $K \subset \mathbb{C}$ one has $\Phi(w) = 1 + O_K(w)$ as $w \to 0$, and $\sum_j \sup_{z\in K} |z|2^{-j} < \infty$. The standard convergence criterion for infinite products therefore gives local uniform convergence. The partial random sums converge uniformly to X, so dominated convergence identifies the product with G. Since X is bounded, G is entire.

Replacing z by $2z$ in the product separates the first factor:

$$G(2z) = \prod_{j\geq 1} \Phi(z/2^{j-1}) = \Phi(z)G(z).$$

Finally, compare the coefficient of z^n in

$$\sum_{n\geq 0} 2^n a_n z^n = \left(\sum_{r\geq 0} \frac{z^r}{(r+1)!}\right)\left(\sum_{k\geq 0} a_k z^k\right).$$

The term $r = 0$ on the right is a_n; moving it to the left gives (18.12). □

The next identity is the geometric source of all exact inverse-dyadic values.

Lemma 18.3 (A weighted cube cut)**.** *Let* $c_1, \ldots, c_n > 0$*, let* $(y)_+ = \max(y, 0)$*, and let*

$$V(t) = \mathrm{vol}\left\{u \in [0,1]^n : \sum_{j=1}^{n} c_j u_j \leq t\right\}.$$

Then

$$V(t) = \frac{1}{n!\prod_{j=1}^{n} c_j} \sum_{S\subseteq\{1,\ldots,n\}} (-1)^{|S|}\left(t - \sum_{j\in S} c_j\right)_+^n. \tag{18.13}$$

Proof. First ignore the upper bounds $u_j \leq 1$ and put $y_j = c_j u_j$. The simplex $y_j \geq 0$, $\sum y_j \leq t$ has volume $t_+^n/(n!\prod c_j)$ in the u-coordinates. To impose all upper bounds, use inclusion–exclusion. For a fixed set S of violated inequalities $u_j > 1$, translate $u_j = v_j + 1$ for $j \in S$. The remaining region is a simplex with threshold $t - \sum_{j\in S} c_j$, hence has volume

$$\frac{(t - \sum_{j\in S} c_j)_+^n}{n!\prod c_j}.$$

Alternating over S gives (18.13). □

Theorem 18.4 (Inverse-dyadic moment identity)**.** *For every* $n \geq 0$*,*

$$\boxed{\mathsf{F}(2^{-n}) = 2^{-\binom{n}{2}} a_n = \frac{d_n}{n!\,2^{\binom{n}{2}}}.} \tag{18.14}$$

Proof. The case $n = 0$ is $\mathsf{F}(1) = 1$. For $n \ge 1$, split the random series after n terms and scale by 2^n:

$$2^n X = \sum_{j=1}^{n} 2^{n-j} U_j + X', \tag{18.15}$$

where X' is an independent copy of X. Conditional on $X' = x$, the event $2^n X \le 1$ cuts the cube by

$$\sum_{j=1}^{n} 2^{n-j} U_j \le 1 - x.$$

Because every coefficient 2^{n-j} is at least 1 and the threshold is at most 1, none of the translated terms in the inclusion–exclusion formula survives. Equivalently, the admissible region is simply a scaled simplex. Its volume is

$$\frac{(1-x)^n}{n! \prod_{j=1}^{n} 2^{n-j}} = 2^{-\binom{n}{2}} \frac{(1-x)^n}{n!}.$$

Average over X'. By symmetry $1 - X' \overset{d}{=} X$, so the expectation is d_n. This proves (18.14). □

The first values are

$$a_0 = 1, \quad a_1 = \frac{1}{2}, \quad a_2 = \frac{5}{36}, \quad a_3 = \frac{1}{36}, \quad a_4 = \frac{143}{32400},$$

and therefore

$$\mathsf{F}(1) = 1, \quad \mathsf{F}(1/2) = \frac{1}{2}, \quad \mathsf{F}(1/4) = \frac{5}{72}, \quad \mathsf{F}(1/8) = \frac{1}{288}, \quad \mathsf{F}(1/16) = \frac{143}{2073600}.$$

18.3 The half-base product and its arithmetic

Write

$$\Pi_n = (1/2; 1/2)_n = \prod_{j=1}^{n} (1 - 2^{-j}). \tag{18.16}$$

The elementary base-reversal laws from Chapters 1 and 2 give the following useful dictionary.

Proposition 18.5 (Mersenne and finite-field normalizations)**.** *For $0 \le k \le n$,*

$$\Pi_n = 2^{-\binom{n+1}{2}} \prod_{j=1}^{n} (2^j - 1), \tag{18.17}$$

$$\begin{bmatrix} n \\ k \end{bmatrix}_{1/2} = 2^{-k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_2. \tag{18.18}$$

In particular, $\left[\begin{smallmatrix} n \\ k \end{smallmatrix}\right]_2$ is the number of k-dimensional subspaces of $\mathbb{F}_2^n$, whereas $\left[\begin{smallmatrix} n \\ k \end{smallmatrix}\right]_{1/2}$ is that integer divided by $2^{k(n-k)}$.

Proof. For the first identity, write $1 - 2^{-j} = 2^{-j}(2^j - 1)$ and sum the exponents $1 + \cdots + n$. For the second, apply base reversal to each factor in

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q;q)_n}{(q;q)_k (q;q)_{n-k}}.$$

The signs cancel, while the net power is

$$-\binom{n+1}{2} + \binom{k+1}{2} + \binom{n-k+1}{2} = -k(n-k).$$

The finite-field interpretation was proved in Chapter 6. □

Thus the half-base notation does not hide irrational constants. It packages Mersenne products and normalized binary Grassmannian counts in precisely the form required by the dyadic interpolation grid.

18.4 Thue–Morse blocks and their forced zero

Let $s_2(r)$ be the number of 1's in the binary expansion of r, and put

$$\varepsilon_r = (-1)^{s_2(r)}. \tag{18.19}$$

For $k, d \geq 0$ and a common translation $\tau \in \mathbb{C}$, define

$$T_{k,d}(\tau) = \sum_{r=0}^{2^k-1} \varepsilon_r (r - 2^k + \tau)^d. \tag{18.20}$$

Theorem 18.6 (Thue–Morse exponential product)**.** *For every $k \geq 0$,*

$$\sum_{r=0}^{2^k-1} \varepsilon_r \mathrm{e}^{rz} = \prod_{j=0}^{k-1} (1 - \mathrm{e}^{2^j z}). \tag{18.21}$$

Consequently,

$$T_{k,d}(\tau) = 0 \quad (0 \leq d < k), \qquad T_{k,k}(\tau) = (-1)^k 2^{\binom{k}{2}} k!. \tag{18.22}$$

Proof. Every $r < 2^k$ is uniquely $r = \sum_{j=0}^{k-1} b_j 2^j$ with $b_j \in \{0, 1\}$. Since $\varepsilon_r = (-1)^{b_0 + \cdots + b_{k-1}}$,

$$\sum_{r<2^k} \varepsilon_r \mathrm{e}^{rz} = \sum_{b_0,\ldots,b_{k-1} \in \{0,1\}} \prod_{j=0}^{k-1} (-\mathrm{e}^{2^j z})^{b_j} = \prod_{j=0}^{k-1} (1 - \mathrm{e}^{2^j z}).$$

Multiplying by $\mathrm{e}^{(-2^k+\tau)z}$ gives the exponential generating function of the $T_{k,d}(\tau)$. The product has a zero of exact order k at 0; its leading term is

$$\prod_{j=0}^{k-1} (-2^j z) = (-1)^k 2^{\binom{k}{2}} z^k.$$

Comparison with $\sum_d T_{k,d}(\tau) z^d / d!$ proves (18.22). □

Using $1 - \mathrm{e}^u = -u\Phi(u)$ and

$$\Phi(-u) = \mathrm{e}^{-u} \Phi(u), \tag{18.23}$$

we obtain

$$\sum_{d \geq 0} \frac{T_{k,d}(0)}{d!} z^d = (-1)^k 2^{\binom{k}{2}} z^k \Psi_k(z), \tag{18.24}$$

where

$$\Psi_k(z) = e^{-z} \prod_{j=0}^{k-1} \Phi(-2^j z) = e^{-2^k z} \prod_{j=0}^{k-1} \Phi(2^j z). \tag{18.25}$$

Iterating the Mahler equation (18.11) at negative arguments yields the bridge

$$\boxed{\Psi_k(z)G(-z) = e^{-z} G(-2^k z).} \tag{18.26}$$

This is where Thue–Morse cancellation meets the moment product.

18.5 The Gaussian stencil on a geometric grid

Specializing the finite q-binomial theorem to $q = 1/2$ gives

$$(z; 1/2)_n = \sum_{k=0}^{n} (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2} z^k. \tag{18.27}$$

Define

$$w_{n,k} = (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2}, \qquad x_k = -2^k, \qquad C_n = 2^{\binom{n+1}{2}} \Pi_n. \tag{18.28}$$

Proposition 18.7 (Dyadic nodal annihilation)**.** *For* $0 \le d \le n$*,*

$$\sum_{k=0}^{n} w_{n,k} x_k^d = \begin{cases} 0, & d < n, \\ C_n, & d = n. \end{cases} \tag{18.29}$$

Hence the functional $L_n[p] = \sum_{k=0}^{n} w_{n,k} p(-2^k)$ *annihilates polynomials of degree less than n and extracts* C_n *times the leading coefficient of every polynomial of degree at most n.*

Proof. By (18.27),

$$\sum_{k=0}^{n} w_{n,k} x_k^d = (-1)^d (2^d; 1/2)_n.$$

If $d < n$, the factor indexed by $j = d$ is $1 - 2^d 2^{-d} = 0$. If $d = n$,

$$(-1)^n (2^n; 1/2)_n = (-1)^n \prod_{j=0}^{n-1} (1 - 2^{n-j}) = \prod_{r=1}^{n} (2^r - 1) = C_n.$$

Linearity gives the final assertion. □

The next theorem is the abstract coefficient extractor behind both the inverse-dyadic and arbitrary-dyadic formulae.

Theorem 18.8 (Geometric coefficient extraction)**.** *Let*

$$U(z) = \sum_{r \ge 0} u_r z^r, \qquad u_0 = 1, \qquad V(z) = \sum_{r \ge 0} v_r z^r,$$

and, for $0 \le k \le n$*, let* $H_k(z) = \sum_{r \ge 0} h_{k,r} z^r$ *satisfy*

$$H_k(z)U(z) = e^{-z} V(-2^k z). \tag{18.30}$$

Then

$$\sum_{k=0}^{n} w_{n,k} h_{k,d} = 0 \qquad (0 \le d < n) \tag{18.31}$$

and

$$\sum_{k=0}^{n} w_{n,k} h_{k,n} = C_n v_n. \tag{18.32}$$

Proof. Take the coefficient of z^d in (18.30):

$$\sum_{i=0}^{d} h_{k,i} u_{d-i} = \sum_{j=0}^{d} \frac{(-1)^j}{j!} v_{d-j} (-2^k)^{d-j}. \tag{18.33}$$

Multiply by $w_{n,k}$ and sum over k. If $d < n$, every node power on the right has degree at most $d < n$, so it vanishes by Theorem 18.7. Induct on d on the left: the terms $i < d$ vanish by the induction hypothesis, and the remaining term is $u_0 \sum_k w_{n,k} h_{k,d}$. Since $u_0 = 1$, this proves (18.31).

For $d = n$, every term on the right with $j > 0$ again has node degree below n. The $j = 0$ term is $v_n \sum_k w_{n,k} (-2^k)^n = C_n v_n$. On the left, all terms $i < n$ vanish by (18.31), leaving $\sum_k w_{n,k} h_{k,n}$. This proves (18.32). □

18.6 The exact inverse-dyadic formula

From (18.24), if $\Psi_k(z) = \sum_{r \ge 0} c_{k,r} z^r$, then

$$c_{k,r} = (-1)^k 2^{-\binom{k}{2}} \frac{T_{k,r+k}(0)}{(r+k)!}. \tag{18.34}$$

Apply Theorem 18.8 to $U(z) = G(-z)$, $V(z) = G(z)$, and $H_k(z) = \Psi_k(z)$, using (18.26). Since $v_n = a_n$, we obtain the central identity.

Theorem 18.9 (Centered q-binomial formula)**.** *For every $n \ge 0$,*

$$\boxed{\mathsf{F}(2^{-n}) = \frac{1}{2^{n^2} \Pi_n} \sum_{k=0}^{n} \frac{\binom{n}{k}_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{2^k - 1} \varepsilon_r (r - 2^k)^{n+k}.} \tag{18.35}$$

Proof. Multiply (18.34) by $w_{n,k}$ and sum over k. The signs cancel and the two triangular powers combine to $4^{-\binom{k}{2}}$, so

$$\sum_{k=0}^{n} \frac{\binom{n}{k}_{1/2}}{4^{\binom{k}{2}} (n+k)!} T_{k,n+k}(0) = C_n a_n.$$

Divide by $2^{n^2} \Pi_n$. Because $C_n = 2^{\binom{n+1}{2}} \Pi_n$,

$$\frac{C_n a_n}{2^{n^2} \Pi_n} = 2^{-\binom{n}{2}} a_n = \mathsf{F}(2^{-n})$$

by Theorem 18.4. □

For $n = 2$, $\Pi_2 = 3/8$ and $\left(\binom{2}{0}_{1/2}, \binom{2}{1}_{1/2}, \binom{2}{2}_{1/2}\right) = (1, 3/2, 1)$. The three signed power sums are $1, -7, 160$, so the numerator is

$$\frac{1}{2} - \frac{7}{4} + \frac{5}{3} = \frac{5}{12},$$

and (18.35) gives $(5/12)/(16 \cdot 3/8) = 5/72$.

18.7 Common translations and raw coordinates

Translation of the inner powers multiplies their exponential generating function by $e^{\tau z}$. The lower coefficient vanishings in (18.31) therefore imply a stronger invariance.

Theorem 18.10 (Translation-invariant formula)**.** *For every $n \geq 0$ and every $\tau \in \mathbb{C}$,*

$$\boxed{\mathsf{F}(2^{-n}) = \frac{1}{2^{n^2}\Pi_n} \sum_{k=0}^{n} \frac{\left[{n \atop k}\right]_{1/2}}{4^{\binom{k}{2}}(n+k)!} \sum_{r=0}^{2^k-1} \varepsilon_r (r - 2^k + \tau)^{n+k}.} \tag{18.36}$$

The same translation τ must be used in every k-block.

Proof. Write $\Psi_k(z) = \sum_d c_{k,d} z^d$. After translation, its normalized form is $e^{\tau z}\Psi_k(z)$. The weighted coefficient of degree n is

$$\sum_{j=0}^{n} \frac{\tau^j}{j!} \sum_{k=0}^{n} w_{n,k} c_{k,n-j}.$$

Every term with $j > 0$ vanishes by (18.31); the term $j = 0$ is the centered numerator. The normalization is therefore unchanged, and Theorem 18.9 applies. □

Corollary 18.11 (Raw-coordinate formula)**.** *For every $n \geq 0$ and every $\tau \in \mathbb{C}$,*

$$\boxed{\mathsf{F}(2^{-n}) = \frac{1}{(-2)^{n^2}\Pi_n} \sum_{k=0}^{n} \frac{\left[{n \atop k}\right]_{1/2}}{4^{\binom{k}{2}}(n+k)!} \sum_{r=0}^{2^k-1} \varepsilon_r (r + \tau)^{n+k}.} \tag{18.37}$$

Proof. In (18.36) replace the common translation by $1 - \tau$ and reflect each block by $r \mapsto 2^k - 1 - r$. Complementing the k binary digits changes ε_r by $(-1)^k$, while

$$(2^k - 1 - r) - 2^k + 1 - \tau = -(r + \tau).$$

For degree $n + k$ the total sign is $(-1)^k(-1)^{n+k} = (-1)^n$, independent of k. Since $n^2 \equiv n \pmod 2$, multiplying the numerator by $(-1)^n$ changes 2^{n^2} into $(-2)^{n^2}$. □

18.8 Every dyadic numerator

The Gaussian row depends only on the resolution 2^{-n}, not on the dyadic numerator. To make this precise, for integers $m, k, d \geq 0$ define

$$T_{m;k,d}(\tau) = \sum_{r=0}^{m2^k-1} \varepsilon_r (r - m2^k + \tau)^d, \tag{18.38}$$

with the empty sum understood as zero.

Theorem 18.12 (Exact formula at an arbitrary dyadic point)**.** *Let $n \geq 0$ and $0 \leq m \leq 2^n$. For every common translation $\tau \in \mathbb{C}$,*

$$\boxed{\mathsf{F}\left(\frac{m}{2^n}\right) = \frac{1}{2^{n^2}\Pi_n} \sum_{k=0}^{n} \frac{\left[{n \atop k}\right]_{1/2}}{4^{\binom{k}{2}}(n+k)!} \sum_{r=0}^{m2^k-1} \varepsilon_r (r - m2^k + \tau)^{n+k}.} \tag{18.39}$$

Proof. The cases $n = 0$ are immediate, so assume $n \geq 1$. Define

$$H_m(z) = \sum_{h=0}^{m-1} \varepsilon_h \mathrm{e}^{(m-h-1)z}, \qquad B_m(z) = G(z)H_m(z) = \sum_{d \geq 0} b_{m,d} z^d. \tag{18.40}$$

First we identify $b_{m,n}$. From the definition,

$$b_{m,n} = \frac{1}{n!} \sum_{h=0}^{m-1} \varepsilon_h \mathbb{E}(X + m - h - 1)^n. \tag{18.41}$$

Scale the decomposition (18.15). Conditional on the tail $X' = x$, apply Theorem 18.3 with $c_j = 2^{n-j}$ and threshold $m - x$. The subset sums of the c_j are exactly the integers $r = 0, \ldots, 2^n - 1$, and their inclusion–exclusion signs are ε_r. Since $0 \leq x \leq 1$ and $m \leq 2^n$, only $r < m$ can contribute. Thus

$$\begin{aligned} \mathsf{F}(m/2^n) &= \frac{2^{-\binom{n}{2}}}{n!} \sum_{r=0}^{m-1} \varepsilon_r \mathbb{E}(m - r - X')^n \\ &= \frac{2^{-\binom{n}{2}}}{n!} \sum_{r=0}^{m-1} \varepsilon_r \mathbb{E}(X + m - r - 1)^n = 2^{-\binom{n}{2}} b_{m,n}, \end{aligned}$$

where symmetry $1 - X' \overset{d}{=} X$ was used in the second line.

It remains to recover $b_{m,n}$ from the long Thue–Morse blocks. Write $r = h2^k + s$, with $0 \leq h < m$ and $0 \leq s < 2^k$. Disjoint binary digits give $\varepsilon_{h2^k+s} = \varepsilon_h \varepsilon_s$. The exponential generating function of (18.38) therefore factors as

$$\begin{aligned} \sum_{d \geq 0} T_{m;k,d}(0) \frac{z^d}{d!} &= \mathrm{e}^{-m2^k z} \left(\sum_{h=0}^{m-1} \varepsilon_h \mathrm{e}^{h2^k z} \right) \prod_{j=0}^{k-1} (1 - \mathrm{e}^{2^j z}) \\ &= (-1)^k 2^{\binom{k}{2}} z^k \Psi_{m,k}(z), \end{aligned}$$

where

$$\Psi_{m,k}(z) = H_m(-2^k z) \Psi_k(z).$$

Using (18.26),

$$\Psi_{m,k}(z) G(-z) = \mathrm{e}^{-z} H_m(-2^k z) G(-2^k z) = \mathrm{e}^{-z} B_m(-2^k z).$$

Apply Theorem 18.8 with $U(z) = G(-z)$, $V(z) = B_m(z)$, and $H_k(z) = \Psi_{m,k}(z)$. Exactly as in (18.34), its degree-n coefficient is

$$(-1)^k 2^{-\binom{k}{2}} \frac{T_{m;k,n+k}(0)}{(n+k)!}.$$

The weighted sum is therefore $C_n b_{m,n}$. A common translation multiplies all normalized blocks by $\mathrm{e}^{\tau z}$ and disappears by the lower-coefficient vanishing, exactly as in Theorem 18.10. Dividing by $2^{n^2} \Pi_n$ gives $2^{-\binom{n}{2}} b_{m,n} = \mathsf{F}(m/2^n)$, completing the proof. □

The formula automatically gives 0 at $m = 0$ and 1 at $m = 2^n$. Its architecture is now transparent: the long Thue–Morse block contains the numerator m, whereas the outer Gaussian row records only the resolution and the geometric dilation.

Chapter 19

Appell polynomials, Bell moments, and a q-Toeplitz limit

The exact Gaussian–Thue–Morse formula is one coordinate system for the Fabius moments. This chapter develops four complementary coordinate systems: Appell polynomials, ordered compositions, Hessenberg determinants, and Bernoulli–Bell cumulants. It then shows that the same half-base Gaussian row acts as a regular Toeplitz kernel on finite uniform splines.

19.1 Moment Appell polynomials

Define

$$A_n(\theta) = \mathbb{E}(X+\theta)^n = \sum_{k=0}^{n} \binom{n}{k} d_{n-k}\theta^k. \tag{19.1}$$

Proposition 19.1 (Appell and reflection laws)**.** *The exponential generating function is*

$$\sum_{n\geq 0} A_n(\theta)\frac{z^n}{n!} = \mathrm{e}^{\theta z}G(z). \tag{19.2}$$

Moreover,

$$A_n'(\theta) = nA_{n-1}(\theta), \tag{19.3}$$

$$A_n(\theta+c) = \sum_{j=0}^{n} \binom{n}{j} A_{n-j}(\theta)c^j, \tag{19.4}$$

$$A_n(\theta) = (-1)^n A_n(-1-\theta). \tag{19.5}$$

Proof. Taking expectation in $\mathrm{e}^{z(X+\theta)} = \mathrm{e}^{\theta z}\mathrm{e}^{zX}$ proves (19.2). Differentiating with respect to θ and comparing coefficients gives (19.3); multiplying by e^{cz} gives (19.4). Finally, $X \overset{d}{=} 1-X$, so

$$A_n(\theta) = \mathbb{E}(1-X+\theta)^n = (-1)^n\mathbb{E}(X-1-\theta)^n.$$

This is (19.5). □

The Taylor-reduction polynomials occurring naturally in the dyadic functional equation are

$$R_n(y) = \sum_{k=0}^{n} 2^{\binom{k+1}{2}-\binom{n-k}{2}} \frac{d_{n-k}}{(n-k)!}\frac{y^k}{k!}. \tag{19.6}$$

Their apparently irregular powers of 2 collapse completely after the natural rescaling.

Theorem 19.2 (Appell compression)**.** *For all $n \ge 0$ and $y \in \mathbb{C}$,*

$$\boxed{R_n(y) = 2^{-\binom{n}{2}} \frac{A_n(2^n y)}{n!}.} \tag{19.7}$$

In particular, $R_n(0) = \mathsf{F}(2^{-n})$.

Proof. Substitute the expansion (19.1) into the right side. Its kth term is

$$2^{-\binom{n}{2}} \frac{d_{n-k}(2^n y)^k}{(n-k)!k!}.$$

The exponent identity

$$-\binom{n}{2} + nk = \binom{k+1}{2} - \binom{n-k}{2}$$

turns this into the kth summand of (19.6). At $y = 0$ use Theorem 18.4. □

For $0 \le \theta \le 1$, bounded support gives $0 \le A_n(\theta) \le 2^n$, hence

$$0 \le R_n(2^{-n}\theta) \le \frac{2^{n-\binom{n}{2}}}{n!}. \tag{19.8}$$

This superexponential bound is useful in binary reduction series.

19.2 A finite composition formula

Expanding the moment product by the set of scales at which a nonconstant term is selected gives an exact formula involving ordered compositions. Write $(r_1, \dots, r_\ell) \vDash n$ when the r_i are positive and sum to n, and put $s_j = r_1 + \cdots + r_j$.

Lemma 19.3 (Geometric simplex sum)**.** *If $r_1, \dots, r_\ell \ge 1$ and $t_h = r_h + \cdots + r_\ell$, then*

$$\sum_{1 \le j_1 < \cdots < j_\ell} 2^{-\sum_{i=1}^{\ell} j_i r_i} = \prod_{h=1}^{\ell} \frac{1}{2^{t_h} - 1}. \tag{19.9}$$

Proof. Write $j_i = i + g_1 + \cdots + g_i$ with $g_h \ge 0$. Then

$$\sum_{i=1}^{\ell} j_i r_i = \sum_{i=1}^{\ell} i r_i + \sum_{h=1}^{\ell} g_h t_h.$$

Summing the independent geometric series gives

$$2^{-\sum i r_i} \prod_{h=1}^{\ell} \frac{1}{1 - 2^{-t_h}} = 2^{-\sum i r_i + \sum t_h} \prod_{h=1}^{\ell} \frac{1}{2^{t_h} - 1}.$$

Since $\sum_h t_h = \sum_i i r_i$, the powers of 2 cancel. □

Theorem 19.4 (Composition representation)**.** *For $n \ge 1$,*

$$\boxed{\mathsf{F}(2^{-n}) = 2^{-\binom{n}{2}} \sum_{(r_1,\dots,r_\ell) \vDash n} \prod_{j=1}^{\ell} \frac{1}{(2^{s_j} - 1)(r_j + 1)!}.} \tag{19.10}$$

Proof. Expand each factor in (18.10) as

$$\Phi(z/2^j) = \sum_{r\geq 0} \frac{z^r}{2^{jr}(r+1)!}.$$

To obtain z^n, choose nonconstant terms at $1 \leq j_1 < \cdots < j_\ell$ with positive degrees $r_1 + \cdots + r_\ell = n$. Thus

$$a_n = \sum_{\ell=1}^{n} \sum_{r_1+\cdots+r_\ell=n} \frac{1}{\prod_i (r_i+1)!} \sum_{j_1<\cdots<j_\ell} 2^{-\sum_i j_i r_i}.$$

Apply Theorem 19.3. Its denominators use tail sums; reversing every composition turns them into the prefix sums s_j without changing the set of all compositions. Finally multiply by $2^{-\binom{n}{2}}$ using Theorem 18.4. □

The Mersenne factors $2^{s_j} - 1$ are therefore nothing but closed geometric sums over the infinitely many binary scales in the moment product.

19.3 A lower-Hessenberg determinant

Put

$$\beta_{n,k} = \frac{1}{(2^n-1)(n-k+1)!} \qquad (0 \leq k < n), \tag{19.11}$$

and define the $n \times n$ lower-Hessenberg matrix

$$H_n = \begin{pmatrix} \beta_{1,0} & -1 & 0 & \cdots & 0 \\ \beta_{2,0} & \beta_{2,1} & -1 & \ddots & \vdots \\ \beta_{3,0} & \beta_{3,1} & \beta_{3,2} & \ddots & 0 \\ \vdots & \vdots & \vdots & \ddots & -1 \\ \beta_{n,0} & \beta_{n,1} & \beta_{n,2} & \cdots & \beta_{n,n-1} \end{pmatrix}. \tag{19.12}$$

Proposition 19.5 (Determinant representation)**.** *For $n \geq 1$,*

$$\boxed{\mathsf{F}(2^{-n}) = 2^{-\binom{n}{2}} \det H_n.} \tag{19.13}$$

Proof. Let $D_0 = 1$ and $D_n = \det H_n$. Expand along the last row. For the entry in column $k+1$, the remaining upper-right block is triangular with -1 on its diagonal. Its determinant and the cofactor sign cancel, leaving

$$D_n = \sum_{k=0}^{n-1} \beta_{n,k} D_k.$$

By (18.12), the sequence a_n obeys the same recurrence with $a_0 = 1$. Induction gives $D_n = a_n$, and Theorem 18.4 finishes the proof. □

Expanding this determinant by admissible Hessenberg paths reproduces the composition formula in Theorem 19.4; the two expressions are standard expansions of the same triangular resolvent.

19.4 Bernoulli cumulants and Bell-polynomial reconstruction

Let the Bernoulli numbers be defined by

$$\frac{z}{e^z - 1} = \sum_{n=0}^{\infty} B_n \frac{z^n}{n!}, \qquad B_1 = -\frac{1}{2}. \tag{19.14}$$

Lemma 19.6 (Logarithm of the uniform moment factor)**.** *Near $z = 0$,*

$$\log \Phi(z) = \frac{z}{2} + \sum_{n=2}^{\infty} \frac{B_n}{n} \frac{z^n}{n!}. \tag{19.15}$$

Proof. Differentiate:

$$\begin{aligned} \frac{d}{dz} \log \Phi(z) &= \frac{e^z}{e^z - 1} - \frac{1}{z} \\ &= 1 + \frac{1}{z}\left(\frac{z}{e^z - 1} - 1\right) = \frac{1}{2} + \sum_{n=2}^{\infty} B_n \frac{z^{n-1}}{n!}. \end{aligned}$$

Integrate termwise and use $\log \Phi(0) = 0$. □

Let κ_n be the cumulants of X, so $\log G(z) = \sum_{n \geq 1} \kappa_n z^n / n!$.

Theorem 19.7 (Cumulants and Bell moments)**.** *The cumulants are*

$$\kappa_1 = \frac{1}{2}, \qquad \kappa_n = \frac{B_n}{n(2^n - 1)} \quad (n \geq 2). \tag{19.16}$$

Thus $\kappa_{2m+1} = 0$ for $m \geq 1$. If $\mathcal{B}_n$ denotes the complete exponential Bell polynomial, defined by

$$\exp\left(\sum_{j \geq 1} x_j \frac{z^j}{j!}\right) = \sum_{n \geq 0} \mathcal{B}_n(x_1, \ldots, x_n) \frac{z^n}{n!},$$

then

$$\boxed{\mathsf{F}(2^{-n}) = \frac{2^{-\binom{n}{2}}}{n!} \mathcal{B}_n(\kappa_1, \ldots, \kappa_n).} \tag{19.17}$$

Proof. Take logarithms in (18.10), use Theorem 19.6, and sum the geometric series:

$$\begin{aligned} \log G(z) &= \sum_{j \geq 1} \log \Phi(z/2^j) \\ &= \frac{z}{2} \sum_{j \geq 1} 2^{-j} + \sum_{n \geq 2} \frac{B_n}{n} \frac{z^n}{n!} \sum_{j \geq 1} 2^{-jn} \\ &= \frac{z}{2} + \sum_{n \geq 2} \frac{B_n}{n(2^n - 1)} \frac{z^n}{n!}. \end{aligned}$$

This proves (19.16). Exponentiating and using the defining generating function of $\mathcal{B}_n$ gives $d_n = \mathcal{B}_n(\kappa_1, \ldots, \kappa_n)$. Substitute into Theorem 18.4. □

Equivalently,

$$\mathsf{F}(2^{-n}) = 2^{-\binom{n}{2}} \sum_{\substack{\nu_1, \ldots, \nu_n \geq 0 \\ \nu_1 + 2\nu_2 + \cdots + n\nu_n = n}} \prod_{j=1}^{n} \frac{1}{\nu_j!} \left(\frac{\kappa_j}{j!}\right)^{\nu_j}. \tag{19.18}$$

Only κ_1 and the even cumulants are nonzero.

19.5 Finite Thue–Morse splines

Let

$$X_p = \sum_{j=1}^{p} \frac{U_j}{2^j}, \qquad F_p(x) = \mathbb{P}(X_p \le x). \tag{19.19}$$

The distribution of X_p is a compactly supported box spline.

Theorem 19.8 (Exact finite-spline formula)**.** *For $p \ge 1$ and every real x,*

$$\boxed{F_p(x) = \frac{1}{2^{\binom{p}{2}} p!} \sum_{r=0}^{2^p-1} \varepsilon_r (2^p x - r)_+^p.} \tag{19.20}$$

Proof. Apply Theorem 18.3 with $c_j = 2^{p-j}$ and threshold $2^p x$. The subset sums of the c_j are exactly $r = 0, \dots, 2^p - 1$, and the parity of the chosen subset is $s_2(r)$. Moreover, $\prod_{j=1}^{p} 2^{p-j} = 2^{\binom{p}{2}}$. The cube volume is exactly $\mathbb{P}(2^p X_p \le 2^p x)$, giving (19.20). □

Proposition 19.9 (Uniform convergence of the finite splines)**.** *For every $p \ge 1$,*

$$\sup_{x\in\mathbb{R}} |F_p(x) - \mathsf{F}(x)| \le 2^{1-p}. \tag{19.21}$$

Proof. Conditional on an independent copy X' in $X = (U_1 + X')/2$, the variable X is uniform on an interval of length $1/2$ and therefore has conditional density at most 2. Averaging shows that its unconditional density is at most 2, so F is 2-Lipschitz.

The tail satisfies $0 \le X - X_p \le \sum_{j>p} 2^{-j} = 2^{-p}$. Hence

$$\mathsf{F}(x) \le F_p(x) \le \mathsf{F}(x + 2^{-p}).$$

The Lipschitz bound gives $0 \le F_p(x) - \mathsf{F}(x) \le 2^{1-p}$ uniformly in x. □

19.6 A regular half-base Gaussian Toeplitz transform

For $0 \le j \le n$, define

$$\omega_{n,j} = \frac{(-1)^j \begin{bmatrix} n \\ j \end{bmatrix}_{1/2} 2^{-\binom{j+1}{2}}}{\Pi_n}, \qquad W_n(z) = \sum_{j=0}^{n} \omega_{n,j} z^j. \tag{19.22}$$

Theorem 19.10 (Row polynomial, mass, and variation)**.** *One has*

$$\boxed{W_n(z) = \frac{(z/2; 1/2)_n}{(1/2; 1/2)_n}.} \tag{19.23}$$

Consequently,

$$\sum_{j=0}^{n} \omega_{n,j} = 1, \tag{19.24}$$

and

$$V_n := \sum_{j=0}^{n} |\omega_{n,j}| = \frac{(-1/2; 1/2)_n}{(1/2; 1/2)_n} < 16. \tag{19.25}$$

Proof. The finite q-binomial theorem at $q = 1/2$ gives

$$(z/2;1/2)_n = \sum_{j=0}^{n} (-1)^j 2^{-\binom{j}{2}} \begin{bmatrix} n \\ j \end{bmatrix}_{1/2} (z/2)^j,$$

which is (19.23). At $z = 1$ the numerator is Π_n, proving unit mass. Since all Gaussian coefficients at $1/2$ are positive, the signs of the row alternate, and replacing z by -1 sums the absolute values. This gives the product in (19.25).

For the bound, the first two factors of the quotient multiply to $3 \cdot (5/3) = 5$. For $0 \le x \le 1/2$,

$$\log \frac{1+x}{1-x} = 2 \operatorname{artanh} x \le 4x.$$

Therefore the remaining factors, beginning with $x = 2^{-3}$, have logarithm at most $4 \sum_{\ell \ge 3} 2^{-\ell} = 1$. Hence $V_n \le 5\mathrm{e} < 15 < 16$. □

Define the Toeplitz average

$$D_n(x) = \sum_{j=0}^{n} \omega_{n,j} F_{2n-j}(x). \tag{19.26}$$

Theorem 19.11 (Quantitative q-Toeplitz limit)**.** *For every $n \ge 1$,*

$$\boxed{\sup_{x\in\mathbb{R}} |D_n(x) - \mathsf{F}(x)| \le 16\, 2^{1-n}.} \tag{19.27}$$

In particular, $D_n \to \mathsf{F}$ uniformly.

Proof. By unit row mass,

$$D_n(x) - \mathsf{F}(x) = \sum_{j=0}^{n} \omega_{n,j} \big(F_{2n-j}(x) - \mathsf{F}(x)\big).$$

Every index $2n - j$ is at least n, so Theorem 19.9 bounds each parenthesis by 2^{1-n}. Sum against absolute weights and use $V_n < 16$. □

Substitution of Theorem 19.8 gives a completely explicit q-binomial–Thue–Morse approximation.

Corollary 19.12 (Explicit discrete q-sum)**.** *For $n \ge 1$,*

$$\begin{aligned} D_n(x) &= \frac{1}{2^{n^2}\Pi_n} \sum_{k=0}^{n} \frac{(-1)^{n-k} \left[\begin{smallmatrix} n \\ k \end{smallmatrix}\right]_{1/2}}{4^{\binom{k}{2}} (n+k)!} \\ &\quad \times \sum_{r=0}^{2^{n+k}-1} \varepsilon_r (2^{n+k}x - r)_+^{n+k}. \end{aligned} \tag{19.28}$$

Proof. In (19.26), put $j = n - k$ and $p = 2n - j = n + k$, then use Theorem 19.8. Gaussian symmetry gives $\left[\begin{smallmatrix} n \\ n-k \end{smallmatrix}\right]_{1/2} = \left[\begin{smallmatrix} n \\ k \end{smallmatrix}\right]_{1/2}$, and

$$\binom{n-k+1}{2} + \binom{n+k}{2} = n^2 + 2\binom{k}{2}.$$

Collecting powers of 2 and signs gives (19.28). □

The exact dyadic formula and the Toeplitz limit use the same Gaussian row for different reasons. In Theorem 18.9 it extracts one coefficient exactly from complete Thue–Morse blocks. In Theorem 19.11 it is a bounded-variation row of unit mass averaging already-convergent finite splines.

19.7 Fourier–sinc product and smoothness

Use the unnormalized convention

$$\operatorname{sinc} z = \begin{cases} \sin z/z, & z \neq 0, \\ 1, & z = 0. \end{cases} \tag{19.29}$$

Let $Y = 2X - 1$.

Theorem 19.13 (Rvachev density and Fourier product)**.** *The law of Y has a compactly supported C^∞ density* up *on* $[-1, 1]$*. Its Fourier transform is*

$$\widehat{\mathrm{up}}(t) = \mathbb{E}e^{itY} = \prod_{j=1}^{\infty} \operatorname{sinc}\left(\frac{t}{2^j}\right), \tag{19.30}$$

and

$$\mathrm{up}(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-itx} \prod_{j=1}^{\infty} \operatorname{sinc}(t/2^j)\, dt. \tag{19.31}$$

Moreover,

$$\mathsf{F}(x) = \int_{-1}^{2x-1} \mathrm{up}(y)\, dy \qquad (0 \leq x \leq 1). \tag{19.32}$$

Proof. Since $\sum_{j \geq 1} 2^{-j} = 1$,

$$Y = \sum_{j \geq 1} \frac{2U_j - 1}{2^j}.$$

The variables $2U_j - 1$ are independent and uniform on $[-1, 1]$, whose characteristic function is $\operatorname{sinc} t$. Independence proves (19.30); local uniform convergence follows as for the moment product.

For every fixed N and $|t| \geq 2^N$,

$$\left| \prod_{j \geq 1} \operatorname{sinc}(t/2^j) \right| \leq \prod_{j=1}^{N} \frac{2^j}{|t|} = 2^{N(N+1)/2} |t|^{-N}.$$

Thus the characteristic function decays faster than every power. In particular, $t^m \widehat{\mathrm{up}}(t) \in L^1(\mathbb{R})$ for every m. Fourier inversion therefore produces a C^∞ density, with derivatives obtained by multiplying the integrand by $(-it)^m$. Its support lies in $[-1, 1]$ because $|Y| \leq 1$. Finally, $Y \leq 2x - 1$ is equivalent to $X \leq x$, proving (19.32). □

On $[-1, 0]$, differentiating (19.32) and using (18.6) shows $\mathrm{up}(y) = \mathsf{F}(y + 1)$; evenness gives the usual Rvachev normalization on the whole support.

Chapter 20

The geometric-uniform q-Fabius family

The dyadic construction admits a natural one-parameter deformation. It separates which identities are special to $q = 1/2$ from those caused merely by geometric scaling. It also produces a direct spectral factorization by infinite q^2-Pochhammer products.

20.1 The contracting family and its Mahler equation

For $q \in \mathbb{C}$ with $|q| < 1$, define the bounded complex random series

$$X_q = (1-q)\sum_{j=0}^{\infty} q^j U_j \tag{20.1}$$

and its entire moment generating function

$$M_q(t) = \mathbb{E}e^{tX_q} = \sum_{n=0}^{\infty} \mu_n(q)\frac{t^n}{n!}. \tag{20.2}$$

For real $0 \le q < 1$, let F_q be the distribution function of X_q. The classical Fabius function is $F_{1/2}$ after shifting the index in (18.1).

Theorem 20.1 (Product and universal Mahler equation)**.** *For $|q| < 1$,*

$$\boxed{M_q(t) = \prod_{j=0}^{\infty} \Phi\big((1-q)q^j t\big),} \tag{20.3}$$

and

$$\boxed{M_q(t) = \Phi((1-q)t)M_q(qt).} \tag{20.4}$$

Writing $a_n(q) = \mu_n(q)/n!$, the coefficients satisfy

$$(1-q^n)a_n(q) = \sum_{k=0}^{n-1} \frac{q^k(1-q)^{n-k}}{(n-k+1)!}a_k(q) \qquad (n \ge 1). \tag{20.5}$$

Proof. Independence gives the finite products, and local uniform convergence follows from $\sum_j |(1-q)q^j| < \infty$, exactly as in Theorem 18.2. Splitting off the factor $j = 0$ and reindexing proves (20.4). To extract coefficients, write

$$\sum_{n\ge 0} a_n(q)t^n = \left(\sum_{r\ge 0}\frac{(1-q)^r t^r}{(r+1)!}\right)\left(\sum_{k\ge 0} a_k(q)q^k t^k\right).$$

Move the $r = 0$ term $q^n a_n(q)$ to the left and put $k = n - r$ in the remaining sum. $\square$

Proposition 20.2 (Reflection and support). *For every real q with $|q| < 1$,*

$$X_q \overset{d}{=} 1 - X_q, \qquad \mathbb{E}X_q = \frac{1}{2}. \tag{20.6}$$

For $0 \le q < 1$, the support is $[0,1]$. For $q = -a$ with $0 < a < 1$, it is

$$\left[-\frac{a}{1-a}, \frac{1}{1-a}\right]. \tag{20.7}$$

Proof. The coefficients in (20.1) sum to $(1-q)\sum_{j\ge 0} q^j = 1$. Replacing every U_j by $1 - U_j$ therefore maps X_q to $1 - X_q$ without changing its law, proving reflection and the mean. For $q \ge 0$, all coefficient intervals add to $[0,1]$. For $q = -a$, the positive even coefficients sum to

$$(1+a)\sum_{m\ge 0} a^{2m} = \frac{1}{1-a},$$

while the negative odd coefficients sum to

$$-(1+a)\sum_{m\ge 0} a^{2m+1} = -\frac{a}{1-a}.$$

Minkowski sums of intervals are intervals, and finite partial supports converge to the stated endpoints. □

20.2 Bernoulli cumulants and Bell moments

Theorem 20.3 (Cumulant dictionary). *For $|q| < 1$, the cumulants of X_q are*

$$\boxed{\kappa_1(q) = \frac{1}{2}, \qquad \kappa_n(q) = \frac{B_n}{n}\frac{(1-q)^n}{1-q^n} \quad (n \ge 2).} \tag{20.8}$$

Consequently,

$$\boxed{\mu_n(q) = \mathcal{B}_n\big(\kappa_1(q), \dots, \kappa_n(q)\big).} \tag{20.9}$$

For real $q \in (-1,1)$,

$$\operatorname{Var}(X_q) = \frac{1}{12}\frac{1-q}{1+q}. \tag{20.10}$$

Proof. Take logarithms in (20.3) and use Theorem 19.6. The linear terms sum to

$$\frac{1-q}{2}\sum_{j\ge 0} q^j = \frac{1}{2}.$$

For $n \ge 2$, the coefficient of $t^n/n!$ is

$$\frac{B_n}{n}(1-q)^n \sum_{j\ge 0} q^{jn} = \frac{B_n}{n}\frac{(1-q)^n}{1-q^n}.$$

This proves the cumulant formula. Exponentiating the cumulant series proves the Bell-polynomial formula. Finally $B_2 = 1/6$, and the second cumulant is the variance. □

At $q = 1/2$, (20.8) reduces to $\kappa_n = B_n/[n(2^n - 1)]$, recovering Theorem 19.7.

20.3 Negative parameters as affine positive laws

Theorem 20.4 (Negative-q affine conjugacy)**.** *For* $0 < a < 1$*, put*

$$\lambda_a = \frac{1+a}{1-a}, \qquad \beta_a = \frac{a}{1-a}. \tag{20.11}$$

Then

$$\boxed{X_{-a} \stackrel{d}{=} \lambda_a X_a - \beta_a,} \tag{20.12}$$

and

$$\boxed{M_{-a}(t) = \mathrm{e}^{-\beta_a t} M_a(\lambda_a t).} \tag{20.13}$$

Proof. Put $V_j = U_j - 1/2$. The V_j are independent and symmetric, and

$$X_q - \frac{1}{2} = (1-q) \sum_{j \ge 0} q^j V_j.$$

Changing V_j to $(-1)^j V_j$ preserves their joint law. Hence

$$X_{-a} - \frac{1}{2} = (1+a) \sum_{j \ge 0} (-a)^j V_j \stackrel{d}{=} (1+a) \sum_{j \ge 0} a^j V_j = \lambda_a \left(X_a - \frac{1}{2} \right).$$

Expanding the affine map gives the shift $-\beta_a$; taking moment generating functions gives (20.13). □

For example, $X_{-1/2} \stackrel{d}{=} 3X_{1/2} - 1$ and $X_{-1/4} \stackrel{d}{=} (5/3)X_{1/4} - 1/3$.

20.4 Reciprocal normalized germs

For $|q| > 1$, the random series no longer converges. Nevertheless the Mahler equation has a canonical analytic germ at the origin:

$$M_q(t) = \left[\prod_{j=1}^{\infty} \Phi\big((1-q)q^{-j} t\big) \right]^{-1}. \tag{20.14}$$

The product converges locally uniformly, and its value at 0 is 1.

Theorem 20.5 (Reciprocal Mahler and convolution inversion)**.** *For* $|q| > 1$*, the germ* (20.14) *satisfies the same functional equation*

$$M_q(t) = \Phi((1-q)t) M_q(qt). \tag{20.15}$$

For every q off the unit circle,

$$\boxed{M_q(t) M_{1/q}(-t) = 1} \tag{20.16}$$

as analytic germs at $t = 0$ and meromorphically wherever both sides are defined.

Proof. Let $P_q(t) = \prod_{j \ge 1} \Phi((1-q)q^{-j} t)$, so $M_q = P_q^{-1}$. Then

$$\begin{aligned} P_q(qt) &= \prod_{j \ge 1} \Phi((1-q)q^{1-j} t) \\ &= \Phi((1-q)t) P_q(t). \end{aligned}$$

Taking reciprocals and rearranging proves (20.15).

Now assume $|q| < 1$. By the reciprocal definition at $1/q$,

$$M_{1/q}(-t)^{-1} = \prod_{j=1}^{\infty} \Phi\big((1-q^{-1})q^j(-t)\big)$$
$$= \prod_{j=1}^{\infty} \Phi\big((1-q)q^{j-1}t\big) = M_q(t).$$

This is (20.16). The case $|q| > 1$ follows by interchanging q and $1/q$. □

At the coefficient level, if $M_q(t) = \sum_n \mu_n(q)t^n/n!$, reciprocity is the signed binomial-convolution identity

$$\sum_{k=0}^{n} \binom{n}{k} \mu_k(q)(-1)^{n-k}\mu_{n-k}(1/q) = \delta_{n0}. \tag{20.17}$$

20.5 Spectral q^2-Pochhammer factorization

Euler's product for the hyperbolic sine is

$$\frac{\sinh(z/2)}{z/2} = \prod_{m=1}^{\infty}\left(1 + \frac{z^2}{4\pi^2 m^2}\right). \tag{20.18}$$

Since

$$\Phi(z) = \mathrm{e}^{z/2}\frac{\sinh(z/2)}{z/2}, \tag{20.19}$$

the geometric ratio q in the random series becomes the nome q^2 in the spectral product.

Theorem 20.6 (Spectral Pochhammer factorization)**.** *For $|q| < 1$,*

$$\boxed{M_q(t) = \mathrm{e}^{t/2}\prod_{m=1}^{\infty}(-\frac{(1-q)^2t^2}{4\pi^2m^2}; q^2)_\infty.} \tag{20.20}$$

For $|q| > 1$,

$$\boxed{M_q(t) = \mathrm{e}^{t/2}\prod_{m=1}^{\infty}(-\frac{(1-q^{-1})^2t^2}{4\pi^2m^2}; q^{-2})_\infty^{-1}.} \tag{20.21}$$

Proof. For $|q| < 1$, insert (20.19) and (20.18) into (20.3). The exponential factors combine to

$$\exp\left(\frac{t}{2}(1-q)\sum_{j\geq 0} q^j\right) = \mathrm{e}^{t/2}.$$

For each fixed m, the product over j is

$$\prod_{j\geq 0}\left(1 + \frac{(1-q)^2q^{2j}t^2}{4\pi^2m^2}\right) = (-\frac{(1-q)^2t^2}{4\pi^2m^2}; q^2)_\infty.$$

The double product is normally convergent on compact t-sets because the sum of the absolute quadratic perturbations is bounded by a constant times $\sum_m m^{-2} \sum_j |q|^{2j}$. It may therefore be rearranged, proving (20.20).

For $|q| > 1$, apply the same argument to the inverse product (20.14). The sum of its linear arguments is $-t$, so the inverse exponential factor is again $\mathrm{e}^{t/2}$. Moreover,

$$(1-q)^2 q^{-2j} = (1-q^{-1})^2 q^{-2(j-1)}.$$

Reindexing $j-1 \geq 0$ yields the inverse q^{-2}-Pochhammer products in (20.21). □

For $q = \pm 1/2$ the spectral nome is $1/4$; for $q = \pm 1/4$ it is $1/16$. Reciprocal parameters exchange the corresponding zeros and poles.

20.6 A Pochhammer-cleared moment polynomial

The recurrence (20.5) has apparent poles at roots of unity. A finite q-Pochhammer factor clears them.

Definition 20.7. For $n \geq 0$, define

$$\mathcal{P}_n(q) = \frac{(q;q)_n}{(1-q)^n} \frac{\mu_n(q)}{n!}. \tag{20.22}$$

Theorem 20.8 (Polynomial recurrence)**.** *The functions $\mathcal{P}_n$ are polynomials in $\mathbb{Q}[q]$ satisfying*

$$\mathcal{P}_0(q) = 1 \tag{20.23}$$

and, for $n \geq 1$,

$$\boxed{\mathcal{P}_n(q) = \sum_{k=0}^{n-1} \frac{q^k}{(n-k+1)!} \frac{(q;q)_{n-1}}{(q;q)_k} \mathcal{P}_k(q).} \tag{20.24}$$

Moreover,

$$\deg \mathcal{P}_n \leq \binom{n}{2}, \qquad \mathcal{P}_n(0) = \frac{1}{(n+1)!}. \tag{20.25}$$

Proof. Multiply (20.5) by $(q;q)_{n-1}/(1-q)^n$ and substitute

$$a_k(q) = \frac{(1-q)^k}{(q;q)_k} \mathcal{P}_k(q).$$

This gives (20.24). The quotient $(q;q)_{n-1}/(q;q)_k = \prod_{j=k+1}^{n-1}(1-q^j)$ is polynomial. If $\deg \mathcal{P}_k \leq \binom{k}{2}$, the kth summand has degree at most

$$k + \sum_{j=k+1}^{n-1} j + \binom{k}{2} = \binom{n}{2}.$$

Induction proves polynomiality and the degree bound. At $q = 0$, every term with $k > 0$ vanishes, while the $k = 0$ term is $1/(n+1)!$. □

The first cases are

$$\mathcal{P}_0(q) = 1, \qquad \mathcal{P}_1(q) = \frac{1}{2},$$

$$\mathcal{P}_2(q) = \frac{q+2}{12}, \qquad \mathcal{P}_3(q) = \frac{q^2+q+1}{24}.$$

At $q = 1/2$, the polynomial is exactly the Pochhammer-cleared Fabius moment:

$$\mathcal{P}_n(1/2) = 2^{\binom{n+1}{2}} \Pi_n \mathsf{F}(2^{-n}). \tag{20.26}$$

This follows immediately from Theorem 18.4.

20.7 Inverse-geometric endpoint values

The dyadic simplex proof extends to every positive contraction small enough that the first omitted coefficient dominates the entire normalized tail.

Theorem 20.9 (Geometric endpoint formula)**.** *Let* $0 < q \le 1/2$ *and let* F_q *be the distribution function of* X_q*. Then, for every* $n \ge 0$*,*

$$\boxed{F_q(q^n) = \frac{q^{\binom{n+1}{2}}}{(q;q)_n}\mathcal{P}_n(q) = \frac{q^{\binom{n+1}{2}}}{(1-q)^n}\frac{\mu_n(q)}{n!}.} \tag{20.27}$$

Proof. The case $n = 0$ is $F_q(1) = 1$. For $n \ge 1$, split the first n coordinates and scale by q^{-n}:

$$q^{-n}X_q = \sum_{j=0}^{n-1}(1-q)q^{j-n}U_j + X_q',$$

where X_q' is an independent copy of X_q. Because $q \le 1/2$, every coefficient $c_j = (1-q)q^{j-n}$ is at least $(1-q)/q \ge 1$. Conditional on $X_q' = x$, the event $X_q \le q^n$ is therefore a simplex exactly as in Theorem 18.4, with volume

$$\frac{(1-x)^n}{n!\prod_{j=0}^{n-1}(1-q)q^{j-n}}.$$

Reflection gives $\mathbb{E}(1-X_q')^n = \mu_n(q)$, while

$$\prod_{j=0}^{n-1}(1-q)q^{j-n} = (1-q)^n q^{-\binom{n+1}{2}}.$$

This proves the second expression in (20.27). The first is the definition (20.22) rearranged. $\square$

At $q = 1/2$, Theorem 20.9 is precisely Theorem 18.4; the finite factor $(q;q)_n$ becomes the Mersenne-normalized product Π_n. Thus the q-Pochhammer denominator in the endpoint formula is not imported notation: it is the exact pole-clearing factor of the moment recurrence.

Part VI

Central sums, limits, and effective methods

Chapter 21

Central q-binomial sums and Lambert series

Central Gaussian coefficients frequently become tractable only after a seemingly extraneous factor is inserted. The elementary identity in the next lemma shows why the factor $(-q;q)_{2k}$ is natural: it converts a central coefficient at base q^2 into a single quotient of shifted factorials. This chapter develops both a regular q-beta evaluation and a deleted-singularity identity proved recently by Chern, Dilcher, and Jiu [7].

21.1 The central-factor reduction

Lemma 21.1. *For every $k \geq 0$,*

$$\begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{1}{(-q;q)_{2k}} = \frac{(q;q^2)_k}{(q^2;q^2)_k}. \tag{18.1}$$

Proof. The elementary product identity

$$(q;q)_{2k}(-q;q)_{2k} = (q^2;q^2)_{2k}$$

gives

$$\begin{aligned} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{1}{(-q;q)_{2k}} &= \frac{(q^2;q^2)_{2k}}{(q^2;q^2)_k^2} \frac{(q;q)_{2k}}{(q^2;q^2)_{2k}} \\ &= \frac{(q;q^2)_k(q^2;q^2)_k}{(q^2;q^2)_k^2}, \end{aligned}$$

which is the claimed quotient. □

For a complex parameter z and $0 < q < 1$, we use the analytic extension

$$[z]_q = \frac{1-q^z}{1-q},$$

with the real logarithm in $q^z = \exp(z \log q)$.

21.2 A q-beta evaluation

Theorem 21.2 (A regular central q-binomial sum)**.** *Let $0 < q < 1$, and let $\alpha \in \mathbb{C}$ be such that none of the denominators below vanishes. Then*

$$\begin{aligned}\sum_{k=0}^{\infty} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^k}{(-q;q)_{2k}\,[2k+1+\alpha]_q} \\ = \frac{\Gamma_{q^2}(3/2)\,\Gamma_{q^2}((\alpha+1)/2)}{\Gamma_{q^2}((\alpha+2)/2)}.\end{aligned} \tag{18.2}$$

In particular, for real α it is enough to exclude $\alpha = -1, -3, -5, \ldots$.

Proof. By Theorem 21.1 and the telescoping quotient

$$\frac{1}{[2k+1+\alpha]_q} = \frac{1}{[\alpha+1]_q} \frac{(q^{\alpha+1};q^2)_k}{(q^{\alpha+3};q^2)_k},$$

the left side is

$$\frac{1}{[\alpha+1]_q}\, {}_2\phi_1\left[\begin{matrix} q, q^{\alpha+1} \\ q^{\alpha+3} \end{matrix}; q^2, q\right].$$

The q-Gauss sum from Theorem 9.4, with base q^2, gives

$$\frac{1}{[\alpha+1]_q} \frac{(q^{\alpha+2}, q^2; q^2)_\infty}{(q^{\alpha+3}, q; q^2)_\infty}. \tag{18.3}$$

Now

$$(q^{\alpha+1};q^2)_\infty = (1-q^{\alpha+1})(q^{\alpha+3};q^2)_\infty, \qquad (q;q^2)_\infty = (1-q)(q^3;q^2)_\infty.$$

Using these two relations to simplify (18.3), we obtain

$$\frac{(q^2;q^2)_\infty (q^{\alpha+2};q^2)_\infty}{(q^3;q^2)_\infty (q^{\alpha+1};q^2)_\infty}. \tag{18.4}$$

Finally insert

$$\Gamma_{q^2}(z) = (1-q^2)^{1-z} \frac{(q^2;q^2)_\infty}{(q^{2z};q^2)_\infty}.$$

In the quotient on the right of (18.2), all powers of $1-q^2$ cancel, and the remaining products are exactly (18.4). □

Corollary 21.3 (Classical limit of the regular sum)**.** *For $\alpha \notin \{-1, -3, -5, \ldots\}$,*

$$\sum_{k=0}^{\infty} \binom{2k}{k} \frac{1}{4^k(2k+1+\alpha)} = \frac{\Gamma(3/2)\Gamma((\alpha+1)/2)}{\Gamma((\alpha+2)/2)}. \tag{18.5}$$

Proof. The series is

$$\frac{1}{\alpha+1}\, {}_2F_1\left(\begin{matrix} 1/2, (\alpha+1)/2 \\ (\alpha+3)/2 \end{matrix}; 1\right).$$

Gauss's classical summation gives

$$\frac{1}{\alpha+1} \frac{\Gamma((\alpha+3)/2)\Gamma(1/2)}{\Gamma((\alpha+2)/2)}.$$

Since $\Gamma((\alpha+3)/2) = ((\alpha+1)/2)\Gamma((\alpha+1)/2)$ and $\Gamma(1/2)/2 = \Gamma(3/2)$, this is (18.5). The same identity is also the $q \to 1^-$ limit of Theorem 21.2 by the q-gamma limit in Theorem 12.5. □

21.3 Deleting a singular term

The regular formula becomes singular when $\alpha = -2m - 1$. Removing the offending $k = m$ term does not merely produce a finite part of the same gamma quotient; it yields an alternating Lambert-series tail.

Theorem 21.4 (Deleted-singularity central identity)**.** *Let* $m \geq 0$ *and* $0 < q < 1$*. Then*

$$\begin{aligned}\sum_{\substack{k\geq 0\\ k\neq m}} \begin{bmatrix}2k\\ k\end{bmatrix}_{q^2} \frac{q^k}{(-q;q)_{2k}[k-m]_{q^2}} \\ = \begin{bmatrix}2m\\ m\end{bmatrix}_{q^2} \frac{(1+q)q^m}{(-q;q)_{2m}} \sum_{r=2m+1}^{\infty} \frac{(-1)^{r-1}q^r}{[r]_q}.\end{aligned} \tag{18.6}$$

Proof. By Theorem 21.1, the left side is

$$(1-q^2)\sum_{\substack{k\geq 0\\ k\neq m}} \frac{(q;q^2)_k q^k}{(q^2;q^2)_k(1-q^{2k-2m})}. \tag{18.7}$$

Introduce an auxiliary parameter a and define

$$\begin{aligned}F(a,q) := \frac{1}{1-aq^{-2m}}\Bigg({}_2\phi_1\left[\begin{matrix}q, aq^{-2m}\\ aq^{-2m+2}\end{matrix}; q^2, q\right] \\ - \frac{(q, aq^{-2m};q^2)_m q^m}{(q^2, aq^{-2m+2};q^2)_m}\Bigg).\end{aligned} \tag{18.8}$$

The elementary cancellation

$$\frac{(aq^{-2m};q^2)_k}{(aq^{-2m+2};q^2)_k} = \frac{1-aq^{-2m}}{1-aq^{2k-2m}}$$

shows that

$$F(a,q) = \sum_{\substack{k\geq 0\\ k\neq m}} \frac{(q;q^2)_k q^k}{(q^2;q^2)_k(1-aq^{2k-2m})}. \tag{18.9}$$

Thus (18.7) equals $(1-q^2)\lim_{a\to 1} F(a,q)$.

Apply q-Gauss with base q^2 to the ${}_2\phi_1$ in (18.8). We obtain

$$\begin{aligned}F(a,q) &= \frac{1}{1-aq^{-2m}}\left(\frac{(q^2, aq^{-2m+1};q^2)_\infty}{(q, aq^{-2m+2};q^2)_\infty} - \frac{(q, aq^{-2m};q^2)_m q^m}{(q^2, aq^{-2m+2};q^2)_m}\right) \\ &= \frac{H(a)-C_m}{1-a},\end{aligned} \tag{18.10}$$

where

$$\begin{aligned}H(a) &= \frac{(q^2, aq;q^2)_\infty (aq^{-2m+1};q^2)_m}{(q, aq^2;q^2)_\infty (aq^{-2m};q^2)_m}, \\ C_m &= \frac{(q;q^2)_m q^m}{(q^2;q^2)_m}.\end{aligned}$$

The reversal identity for finite shifted factorials gives

$$\frac{(q^{-2m+1};q^2)_m}{(q^{-2m};q^2)_m} = q^m \frac{(q;q^2)_m}{(q^2;q^2)_m},$$

so $H(1) = C_m$. Hence L'Hospital's rule yields

$$\lim_{a\to 1} F(a,q) = -H'(1) = -H(1)\,\frac{\partial}{\partial a}\log H(a)\bigg|_{a=1}. \tag{18.11}$$

Differentiating the logarithms of the product factors gives

$$\lim_{a\to 1} F(a,q) = C_m\left(\sum_{r=1}^{\infty}\frac{(-1)^{r-1}q^r}{1-q^r} + \sum_{r=1}^{2m}\frac{(-1)^{r-1}q^{-r}}{1-q^{-r}}\right). \tag{18.12}$$

For $r \ge 1$,

$$\frac{q^{-r}}{1-q^{-r}} = -\frac{1}{1-q^r}.$$

Consequently, the two summands in (18.12) combine, for $1 \le r \le 2m$, to $(-1)^r$. Their sum is 0 because $2m$ is even. Therefore

$$\lim_{a\to 1} F(a,q) = C_m \sum_{r=2m+1}^{\infty}\frac{(-1)^{r-1}q^r}{1-q^r}. \tag{18.13}$$

Multiply by $1-q^2 = (1+q)(1-q)$, use $(1-q)/(1-q^r) = 1/[r]_q$, and finally use (18.1) at $k = m$ to rewrite

$$C_m = q^m\frac{(q;q^2)_m}{(q^2;q^2)_m} = q^m\begin{bmatrix}2m\\ m\end{bmatrix}_{q^2}\frac{1}{(-q;q)_{2m}}.$$

The result is exactly (18.6). □

Remark 21.5 (The mechanism). The deleted term is encoded by the auxiliary parameter a. The q-Gauss product has a removable zero-over-zero singularity at $a = 1$, and its logarithmic derivative turns products into Lambert series. This parameter-differentiation mechanism recurs throughout the theory of harmonic-number and Lambert-series refinements of hypergeometric summations.

21.4 Separate finite and infinite components

The deleted sum in Theorem 21.4 naturally splits at the missing index. Each component has a simpler one-sided Lambert-series evaluation.

Theorem 21.6 (Infinite and finite components)**.** *Let $m \ge 0$ and $0 < q < 1$, and put*

$$P_m(q) = \begin{bmatrix}2m\\ m\end{bmatrix}_{q^2}\frac{(1+q)q^m}{(-q;q)_{2m}}.$$

Then

$$\sum_{k=m+1}^{\infty}\begin{bmatrix}2k\\ k\end{bmatrix}_{q^2}\frac{q^k}{(-q;q)_{2k}[k-m]_{q^2}} = P_m(q)\left(\sum_{r=1}^{\infty}\frac{q^{2r-1}}{[2r-1]_q} - \sum_{r=1}^{\infty}\frac{q^{2r+2m}}{[2r+2m]_q}\right), \tag{18.14}$$

and

$$\sum_{k=0}^{m-1}\begin{bmatrix}2k\\ k\end{bmatrix}_{q^2}\frac{q^k}{(-q;q)_{2k}[k-m]_{q^2}} = -P_m(q)\sum_{r=1}^{m}\frac{q^{2r-1}}{[2r-1]_q}. \tag{18.15}$$

For $m = 0$, the finite sum and the sum on its right are both empty.

Proof. By Theorem 21.1, shifting $k \mapsto k+m$ in the infinite component gives

$$\sum_{k=m+1}^{\infty} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^k}{(-q;q)_{2k}[k-m]_{q^2}} = (1-q^2)q^m \frac{(q;q^2)_m}{(q^2;q^2)_m} \sum_{k=1}^{\infty} \frac{(q^{2m+1};q^2)_k q^k}{(q^{2m+2};q^2)_k(1-q^{2k})}.$$

Introduce

$$U(a,q) = \frac{1}{1-a}\left({}_2\phi_1\begin{bmatrix} a, q^{2m+1} \\ q^{2m+2} \end{bmatrix};q^2, a^{-1}q\right] - 1\right).$$

For a near 1, absolute convergence permits termwise passage to the limit. Since

$$\frac{(a;q^2)_k}{1-a} = (aq^2;q^2)_{k-1} \qquad (k \geq 1),$$

we obtain

$$\lim_{a\to 1} U(a,q) = \sum_{k=1}^{\infty} \frac{(q^{2m+1};q^2)_k q^k}{(q^{2m+2};q^2)_k(1-q^{2k})}.$$

The q-Gauss summation, at base q^2, evaluates the auxiliary series as

$$U(a,q) = \frac{1}{1-a}\left(\frac{(q, a^{-1}q^{2m+2};q^2)_\infty}{(a^{-1}q, q^{2m+2};q^2)_\infty} - 1\right).$$

The product quotient tends to 1 as $a \to 1$. L'Hospital's rule and termwise logarithmic differentiation, justified locally uniformly, give

$$\lim_{a\to 1} U(a,q) = \sum_{r=1}^{\infty} \frac{q^{2r-1}}{1-q^{2r-1}} - \sum_{r=1}^{\infty} \frac{q^{2r+2m}}{1-q^{2r+2m}}.$$

Multiplication by the prefactor above, followed by $(1-q)/(1-q^j) = 1/[j]_q$ and Theorem 21.1 at $k=m$, proves (18.14).

To obtain the finite component, subtract (18.14) from the full deleted sum (18.6). Indeed,

$$\sum_{j=2m+1}^{\infty} \frac{(-1)^{j-1}q^j}{[j]_q} = \sum_{r=m+1}^{\infty} \frac{q^{2r-1}}{[2r-1]_q} - \sum_{r=m+1}^{\infty} \frac{q^{2r}}{[2r]_q} = \sum_{r=1}^{\infty} \frac{q^{2r-1}}{[2r-1]_q} - \sum_{r=1}^{\infty} \frac{q^{2r+2m}}{[2r+2m]_q} - \sum_{r=1}^{m} \frac{q^{2r-1}}{[2r-1]_q}.$$

The first two sums constitute the infinite component, leaving exactly (18.15). $\square$

21.5 The classical finite part

Define the alternating harmonic number

$$\overline{H}_N = \sum_{r=1}^{N} \frac{(-1)^{r-1}}{r}.$$

Theorem 21.7 (Classical deleted-singularity identity)**.** *For every $m \geq 0$,*

$$\sum_{\substack{k\geq 0\\ k\neq m}} \binom{2k}{k} \frac{1}{4^k(k-m)} = \binom{2m}{m} \frac{\log 2 - \overline{H}_{2m}}{2^{2m-1}}. \tag{18.16}$$

Proof. Put $c_k = (1/2)_k/k! = \binom{2k}{k}/4^k$. For a away from the nonpositive integers, the series

$$T(a) = \sum_{k=0}^{\infty} \frac{c_k}{k+a}$$

converges locally uniformly and defines a meromorphic function. For $\Re a > 0$, termwise integration of $(1-x)^{-1/2} = \sum_{k\geq 0} c_k x^k$ gives

$$T(a) = \int_0^1 x^{a-1}(1-x)^{-1/2}\, \mathrm{d}x = B(a, 1/2) = \frac{\Gamma(a)\Gamma(1/2)}{\Gamma(a+1/2)}. \tag{18.17}$$

Analytic continuation makes (18.17) valid wherever both sides are defined. The desired deleted sum is the finite part at $a = -m$:

$$\lim_{a\to -m} \left(T(a) - \frac{c_m}{a+m} \right). \tag{18.18}$$

Write $a = -m + \varepsilon$. The standard expansions of the gamma function at a negative integer and at a regular point are

$$\Gamma(-m+\varepsilon) = \frac{(-1)^m}{m!\,\varepsilon} \left(1 + (H_m - \gamma)\varepsilon + O(\varepsilon^2) \right),$$
$$\Gamma(1/2 - m + \varepsilon) = \Gamma(1/2-m) \left(1 + \psi(1/2-m)\varepsilon + O(\varepsilon^2) \right).$$

Moreover,

$$\frac{(-1)^m \Gamma(1/2)}{m!\Gamma(1/2-m)} = c_m.$$

Thus the finite part in (18.18) is

$$c_m \big(H_m - \gamma - \psi(1/2-m) \big). \tag{18.19}$$

Using $\psi(1/2) = -\gamma - 2\log 2$ and the recurrence $\psi(z+1) = \psi(z) + 1/z$, we find

$$\psi(1/2-m) = -\gamma - 2\log 2 + 2\sum_{r=1}^{m} \frac{1}{2r-1}.$$

Since

$$\overline{H}_{2m} = \sum_{r=1}^{m} \frac{1}{2r-1} - \frac{1}{2} H_m,$$

the bracket in (18.19) is $2(\log 2 - \overline{H}_{2m})$. Hence the finite part equals

$$2\frac{\binom{2m}{m}}{4^m}(\log 2 - \overline{H}_{2m}),$$

which is (18.16). □

Proposition 21.8 (Limit of the Lambert tail)**.** *For fixed $m \geq 0$,*

$$\lim_{q\to 1^-} \sum_{r=2m+1}^{\infty} \frac{(-1)^{r-1}q^r}{[r]_q} = \log 2 - \overline{H}_{2m}. \tag{18.20}$$

Proof. For $0 < q < 1$, set $a_r(q) = q^r/[r]_q$. These numbers are positive and decrease with r, since

$$\frac{a_{r+1}(q)}{a_r(q)} = q\frac{[r]_q}{[r+1]_q} < 1.$$

Moreover,

$$a_r(q) = \frac{q^r}{1+q+\cdots+q^{r-1}} \leq \frac{1}{r},$$

because every summand in the denominator is at least q^{r-1}. The alternating-series remainder after N terms therefore has absolute value at most $a_{N+1}(q) \leq 1/(N+1)$, uniformly in q. For each fixed r, $a_r(q) \to 1/r$ as $q \to 1^-$. First pass to the limit in a finite partial sum and then let $N \to \infty$. The result is the tail of the alternating harmonic series,

$$\sum_{r=2m+1}^{\infty} \frac{(-1)^{r-1}}{r} = \log 2 - \overline{H}_{2m}.$$

□

Chapter 22

Asymptotic regimes and Bernoulli expansions

A q-analogue should explain both the deformation near $q = 1$ and behavior far from it. Gaussian coefficients have especially clean asymptotic descriptions because their logarithms are finite sums of a single elementary function.

22.1 A complete expansion as $q \to 1$

Let the Bernoulli numbers be defined by

$$\frac{x}{e^x - 1} = \sum_{r=0}^{\infty} B_r \frac{x^r}{r!}.$$

Lemma 22.1 (The basic logarithmic expansion)**.** *As $x \to 0$,*

$$\log \frac{1 - e^{-x}}{x} = -\frac{x}{2} + \sum_{r=1}^{\infty} \frac{B_{2r}}{2r(2r)!} x^{2r}. \tag{19.1}$$

The series converges for $|x| < 2\pi$.

Proof. Differentiate the left side:

$$\frac{\mathrm{d}}{\mathrm{d}x} \log \frac{1 - e^{-x}}{x} = \frac{1}{e^x - 1} - \frac{1}{x}.$$

By the Bernoulli generating function,

$$\frac{1}{e^x - 1} - \frac{1}{x} = -\frac{1}{2} + \sum_{r=1}^{\infty} B_{2r} \frac{x^{2r-1}}{(2r)!}.$$

Integrate term by term and use the value 0 at $x = 0$. The nearest nonzero singularities are at $x = \pm 2\pi i$, giving radius 2π. □

Theorem 22.2 (Bernoulli expansion of a Gaussian coefficient)**.** *Let $0 \le k \le n$, put $D = k(n-k)$, and set $q = e^{-t}$. As $t \to 0$,*

$$\log \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{e^{-t}}}{\binom{n}{k}} = -\frac{D}{2} t + \sum_{r=1}^{\infty} \frac{B_{2r} t^{2r}}{2r(2r)!} \sum_{j=1}^{k} \left((n-k+j)^{2r} - j^{2r} \right). \tag{19.2}$$

For fixed n, k, the displayed series converges whenever $|t| < 2\pi/n$.

Proof. From the product formula,

$$\begin{aligned}\begin{bmatrix} n \\ k \end{bmatrix}_{e^{-t}} &= \prod_{j=1}^{k} \frac{1 - e^{-(n-k+j)t}}{1 - e^{-jt}} \\ &= \binom{n}{k} \prod_{j=1}^{k} \frac{(1 - e^{-(n-k+j)t})/((n-k+j)t)}{(1 - e^{-jt})/(jt)}.\end{aligned}$$

Take logarithms and apply Theorem 22.1 to every factor. The linear term is

$$-\frac{t}{2}\sum_{j=1}^{k}(n-k) = -\frac{1}{2}k(n-k)t = -\frac{D}{2}t,$$

and the even terms give the sum in (19.2). Every argument of the basic series has modulus below 2π when $|t| < 2\pi/n$, so the finite sum converges there. □

Corollary 22.3 (First terms)**.** *With $D = k(n-k)$,*

$$\begin{bmatrix} n \\ k \end{bmatrix}_{e^{-t}} = \binom{n}{k} \exp\left(-\frac{D}{2}t + \frac{D(n+1)}{24}t^2 + O(t^4)\right), \tag{19.3}$$

$$= \binom{n}{k}\left(1 - \frac{D}{2}t + \left(\frac{D^2}{8} + \frac{D(n+1)}{24}\right)t^2 + O(t^3)\right). \tag{19.4}$$

Proof. For $r = 1$ in (19.2),

$$\begin{aligned}\sum_{j=1}^{k}\left((n-k+j)^2 - j^2\right) &= k(n-k)^2 + (n-k)k(k+1) \\ &= k(n-k)(n+1) = D(n+1).\end{aligned}$$

Since $B_2/(2 \cdot 2!) = 1/24$, the logarithmic formula follows. Expanding the exponential gives the second formula. □

Remark 22.4 (Reciprocity and centering). The factor $e^{-Dt/2}$ is forced by reciprocity:

$$\begin{bmatrix} n \\ k \end{bmatrix}_{e^{-t}} = e^{-Dt}\begin{bmatrix} n \\ k \end{bmatrix}_{e^{t}}.$$

Therefore the centered quantity $e^{Dt/2}\left[\begin{smallmatrix} n \\ k \end{smallmatrix}\right]_{e^{-t}}$ is an even analytic function of t. This explains, before any calculation, why all odd powers beyond the linear term vanish from the logarithm.

22.2 Fixed q and growing parameters

Theorem 22.5 (Fixed-column limit)**.** *If $0 < |q| < 1$ and k is fixed, then*

$$\lim_{n\to\infty}\begin{bmatrix} n \\ k \end{bmatrix}_{q} = \frac{1}{(q;q)_k}. \tag{19.5}$$

More precisely,

$$\begin{bmatrix} n \\ k \end{bmatrix}_{q} = \frac{1}{(q;q)_k}\left(1 + O(|q|^{n-k+1})\right). \tag{19.6}$$

Proof. The product formula gives

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q^{n-k+1};q)_k}{(q;q)_k}.$$

The numerator tends to 1. Since the number of factors is fixed,

$$\prod_{j=0}^{k-1}(1-q^{n-k+1+j}) = 1 + O(|q|^{n-k+1}),$$

which proves both assertions. □

Theorem 22.6 (Central fixed-q limit)**.** *For* $0 < |q| < 1$*,*

$$\lim_{m\to\infty} \begin{bmatrix} 2m \\ m \end{bmatrix}_q = \frac{1}{(q;q)_\infty}. \tag{19.7}$$

Proof. By the factorial form,

$$\begin{bmatrix} 2m \\ m \end{bmatrix}_q = \frac{(q;q)_{2m}}{(q;q)_m^2}.$$

Every finite shifted factorial tends to $(q;q)_\infty$ as its index tends to infinity. The quotient therefore tends to $(q;q)_\infty/(q;q)_\infty^2$. □

Proposition 22.7 (Uniform elementary bound)**.** *For* $0 < q < 1$ *and* $0 \le k \le n$*,*

$$1 \le \begin{bmatrix} n \\ k \end{bmatrix}_q \le \frac{1}{(q;q)_\infty}. \tag{19.8}$$

Proof. The lower bound follows from the partition interpretation: the constant term is 1 and every coefficient is nonnegative. The same interpretation says that $\left[\begin{smallmatrix} n \\ k \end{smallmatrix}\right]_q$ is the generating function of partitions fitting in a k by $(n-k)$ rectangle. Dropping the rectangle restriction embeds this set into all partitions, whose generating function is $(q;q)_\infty^{-1}$. Evaluating at a positive $q < 1$ gives the upper bound. □

22.3 The regime $q > 1$

Corollary 22.8 (Transfer through reciprocity)**.** *Let* $q > 1$ *and* $D = k(n-k)$*. Then*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = q^D \begin{bmatrix} n \\ k \end{bmatrix}_{q^{-1}}. \tag{19.9}$$

Consequently, for fixed k*,*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{q^{k(n-k)}}{(q^{-1};q^{-1})_k}\left(1 + O(q^{-n+k-1})\right), \tag{19.10}$$

and

$$\begin{bmatrix} 2m \\ m \end{bmatrix}_q \sim \frac{q^{m^2}}{(q^{-1};q^{-1})_\infty}. \tag{19.11}$$

Proof. The reciprocity identity was proved in Theorem 2.5. Apply Theorems 22.5 and 22.6 with q^{-1} in place of q. □

22.4 Extreme specializations

Proposition 22.9. *For* $0 \le k \le n$*,*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q \Bigg|_{q=0} = 1, \qquad \lim_{q \to 1} \begin{bmatrix} n \\ k \end{bmatrix}_q = \binom{n}{k},$$

and the leading term as $q \to \infty$ *is* $q^{k(n-k)}$ *with coefficient* 1.

Proof. The partition generating polynomial has constant term 1, proving the first assertion. The second follows by replacing every ratio $(1 - q^a)/(1 - q^b)$ in the product formula by its limit a/b. Palindromicity and degree $k(n - k)$ show that the leading coefficient equals the constant coefficient, namely 1. □

Chapter 23

Computation, certification, and formal proof architecture

The identities in this book are well suited to exact computation because their natural objects are polynomials, rational functions, and formal power series. They are also well suited to proof assistants, provided algebraic identities are separated cleanly from analytic convergence. Lau, Lee, and Ono's 2026 formalization project [25] demonstrates this architecture for q-Pochhammer symbols, Bailey pairs, Jacobi's triple product, and the Rogers–Ramanujan identities.

23.1 Dynamic programming for Gaussian polynomials

A polynomial is represented here by its coefficient vector. Multiplication by q^h is therefore a shift by h places.

Algorithm 23.1 (Pascal dynamic program)**.** Given $n \geq 0$ and $0 \leq k \leq n$:

1. Replace k by $\min(k, n-k)$, using symmetry.
2. Initialize $G_0(q) = 1$ and $G_s(q) = 0$ for $1 \leq s \leq k$.
3. For $r = 1, 2, \ldots, n$, update in descending order

$$G_s(q) \leftarrow G_s(q) + q^{r-s} G_{s-1}(q) \qquad \big(s = \min(r,k), \ldots, 1\big).$$

4. Return $G_k(q)$.

Theorem 23.2 (Correctness and complexity)**.** *Theorem 23.1 returns $\begin{bmatrix} n \\ k \end{bmatrix}_q$. If $D = k(n-k)$ after the symmetry reduction, its coefficient-level arithmetic cost is $O(nkD)$ and its storage cost is $O(kD)$.*

Proof. Let $G_s^{(r)}$ denote the value after row r. We prove by induction on r that

$$G_s^{(r)} = \begin{bmatrix} r \\ s \end{bmatrix}_q \qquad (0 \leq s \leq \min(r,k)).$$

The assertion is true at $r = 0$. Descending updates ensure that both G_s and G_{s-1} on the right are still values from row $r-1$. Hence the update is

$$G_s^{(r)} = \begin{bmatrix} r-1 \\ s \end{bmatrix}_q + q^{r-s} \begin{bmatrix} r-1 \\ s-1 \end{bmatrix}_q = \begin{bmatrix} r \\ s \end{bmatrix}_q$$

by the second q-Pascal recurrence. At $r = n$ this gives the desired output.

After replacing k by $\min(k, n-k)$, we have $k \le n/2$. Every intermediate polynomial $\left[\begin{smallmatrix} r \\ s \end{smallmatrix}\right]_q$ with $s \le k$ has degree $s(r-s) \le k(n-k) = D$. There are at most nk shift-and-add updates, each touching at most $D+1$ coefficients. Storing $k+1$ such vectors uses $O(kD)$ coefficients. □

Remark 23.3 (Numerical specialization). When only a numerical value is required, one should not first expand the polynomial. For positive q near 1, the stable formula is

$$\log \begin{bmatrix} n \\ k \end{bmatrix}_q = \sum_{j=1}^{k} \left[\log\bigl(-\operatorname{expm1}((n-k+j)\log q)\bigr) - \log\bigl(-\operatorname{expm1}(j\log q)\bigr)\right],$$

where `expm1` avoids cancellation in $e^x - 1$. For exact symbolic work, the Pascal algorithm or the cyclotomic factorization is preferable.

23.2 A factorization algorithm

Algorithm 23.4 (Cyclotomic factor list)**.** For each $1 \le d \le n$, compute

$$e_d = \lfloor n/d \rfloor - \lfloor k/d \rfloor - \lfloor (n-k)/d \rfloor .$$

Return the list of d for which $e_d = 1$.

Theorem 23.5 (Correctness of the factor list)**.** *The output of Theorem 23.4 is exactly the set of cyclotomic factors of* $\left[\begin{smallmatrix} n \\ k \end{smallmatrix}\right]_q$*, each with multiplicity one.*

Proof. This is Theorem 16.1: the exponent of $\Phi_d(q)$ is e_d, and the floor calculation in that theorem proves $e_d \in \{0, 1\}$. □

Remark 23.6. The factor-list algorithm takes only $O(n)$ integer floor operations. It is therefore far cheaper than expanding $\left[\begin{smallmatrix} n \\ k \end{smallmatrix}\right]_q$ when the purpose is root-of-unity evaluation, divisibility testing, or congruence work. Polynomial expansion may be postponed until the selected cyclotomic factors are actually multiplied.

23.3 Telescoping certificates for finite sums

A term $T(n,k)$ is called *q-hypergeometric* if the ratios

$$\frac{T(n,k+1)}{T(n,k)}, \qquad \frac{T(n+1,k)}{T(n,k)}$$

are rational functions of q^n and q^k (and of any fixed parameters). Shifted-factorial terms have this property because adjacent factors cancel.

Theorem 23.7 (Finite telescoping certificate)**.** *Let*

$$S_n = \sum_{k=L(n)}^{U(n)} T(n,k).$$

Suppose there are functions $A_0(n), \ldots, A_r(n)$ and a certificate $G(n,k)$ such that

$$\sum_{j=0}^{r} A_j(n) T(n+j,k) = G(n,k+1) - G(n,k) \tag{20.1}$$

for all relevant k, after extending T by zero outside its natural finite range. If $G(n,k)$ vanishes at both ends of the resulting finite summation interval, then

$$\sum_{j=0}^{r} A_j(n) S_{n+j} = 0. \tag{20.2}$$

Proof. Sum (20.1) over an interval containing the support of every $T(n+j,k)$. The left side becomes $\sum_j A_j(n)S_{n+j}$. The right side telescopes to the terminal value of G minus its initial value, which is 0 by hypothesis. □

Corollary 23.8 (Identity certification). *If two finite sums satisfy the same recurrence of order r and agree for r consecutive initial values at every nonsingular parameter choice, then they are equal wherever the recurrence determines subsequent values.*

Proof. Subtract the two sequences. The difference satisfies the homogeneous recurrence and has r consecutive zero initial values. Induction through the recurrence gives zero thereafter. □

Remark 23.9 (Creative telescoping). The q-Zeilberger method searches algorithmically for a rational multiple $G(n,k) = R(q^n, q^k)T(n,k)$ satisfying (20.1). Once found, the certificate is independently checkable by rational-function arithmetic. The search may be sophisticated, but verification is elementary: divide by $T(n,k)$ and clear denominators.

23.4 Denominator clearing and specialization

Lemma 23.10 (Polynomial identity principle). *Let $R(q), S(q) \in \mathbb{Q}(q)$. If $R(q) = S(q)$ at infinitely many complex values of q where both sides are defined, then $R = S$ in $\mathbb{Q}(q)$.*

Proof. Write $R - S = A/B$ with $A, B \in \mathbb{Q}[q]$ and $B \neq 0$. The hypothesis says that A has infinitely many zeros. A nonzero polynomial has only finitely many zeros, so $A = 0$. □

Corollary 23.11 (Safe generic specialization). *A rational q-identity proved under generic nonvanishing assumptions remains valid after any specialization at which the denominator-cleared identity is defined. Apparent excluded values may therefore be recovered by cancellation or by taking limits.*

Proof. Clear all denominators to obtain a polynomial identity. Polynomial evaluation is valid at every specialization. If the original rational expressions have removable singularities, divide out common factors or take their uniquely determined limits. □

This principle is particularly important at roots of unity. One should first rewrite a Gaussian coefficient as its polynomial, not substitute a root into the factorial quotient and accept an indeterminate $0/0$.

23.5 Formal series versus analytic series

Theorem 23.12 (Formal-to-analytic transfer). *Suppose*

$$\sum_{n\geq 0} a_n z^n = \sum_{n\geq 0} b_n z^n$$

coefficientwise in $\mathbb{C}[[z]]$. If both series converge for $|z| < R$, then they define the same analytic function on that disk. Conversely, if two power series converge in a disk and define the same function on a subset with an accumulation point in the disk, then their coefficients are equal.

Proof. The forward implication is immediate because the two partial sums agree term by term. For the converse, the identity theorem makes the analytic functions equal throughout the disk; differentiating n times at 0 gives $n!a_n = n!b_n$. □

Principle 23.13 (A scalable proof architecture)**.** *For a mixed algebraic-analytic q-identity, the following order minimizes hidden assumptions:*

1. *define finite shifted factorials and prove their algebraic laws;*
2. *prove terminating identities after denominator clearing;*
3. *define formal infinite products by coefficient stabilization when possible;*
4. *separately establish analytic convergence for $|q| < 1$;*
5. *transfer the formal coefficient identity to analytic functions;*
6. *only then take limits, differentiate parameters, or specialize at roots of unity.*

Proof. Each step is justified by results already proved: finite product algebra in Chapter 1; polynomial identity and specialization by Theorems 23.10 and 23.11; formal-to-analytic transfer by Theorem 23.12; and the convergence theorems in Chapters 8 to 10. The ordering prevents an analytic operation from being used before its convergence hypothesis has been supplied. □

Historical note 23.14. Formal verification forces distinctions that handwritten mathematics often suppresses: a finite product indexed by a natural number is not the same object as a bilateral product; a polynomial Gaussian coefficient is not definitionally the same as a rational factorial quotient; and a formal power-series identity does not automatically contain a proof of analytic convergence. The Lean development of Lau, Lee, and Ono [25] turns these distinctions into reusable interfaces rather than obstacles.

Appendix A

Formula atlas

This appendix is a navigation table, not a substitute for hypotheses and proofs. Each formula points back to the theorem where its exact domain and derivation are given.

Name	Formula	Location
Concatenation	$(a;q)_{m+n} = (a;q)_m (aq^m;q)_n$	Theorem 1.2
Dissection	$(a;q)_{rn} = \prod_{j=0}^{r-1}(aq^j;q^r)_n$	Theorem 1.3
Base reversal	$(a;q)_n = (-a)^n q^{\binom{n}{2}}(a^{-1}q^{1-n};q)_n$	Theorem 1.4
Negative index	$(a;q)_{-n} = 1/(aq^{-n};q)_n$	Theorem 1.6
Infinite logarithm	$\log(a;q)_\infty = -\sum_{r\ge 1} a^r/(r(1-q^r))$	Theorem 1.14
Gaussian product	$\begin{bmatrix} n \\ k \end{bmatrix}_q = (q;q)_n/((q;q)_k(q;q)_{n-k})$	Theorem 2.2
Pascal I	$\begin{bmatrix} n \\ k \end{bmatrix}_q = q^k \begin{bmatrix} n-1 \\ k \end{bmatrix}_q + \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q$	Theorem 2.3
Pascal II	$\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n-1 \\ k \end{bmatrix}_q + q^{n-k}\begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q$	Theorem 2.3
Reciprocity	$\begin{bmatrix} n \\ k \end{bmatrix}_q = q^{k(n-k)}\begin{bmatrix} n \\ k \end{bmatrix}_{q^{-1}}$	Theorem 2.5
Finite theorem	$(z;q)_n = \sum_k (-1)^k q^{\binom{k}{2}}\begin{bmatrix} n \\ k \end{bmatrix}_q z^k$	Theorem 3.1
Reciprocal theorem	$(z;q)_{n+1}^{-1} = \sum_{k\ge 0}\begin{bmatrix} n+k \\ k \end{bmatrix}_q z^k$	Theorem 3.4
q-Vandermonde	$\begin{bmatrix} m+n \\ r \end{bmatrix}_q = \sum_k q^{(m-k)(r-k)}\begin{bmatrix} m \\ k \end{bmatrix}_q \begin{bmatrix} n \\ r-k \end{bmatrix}_q$	Theorem 3.5
Infinite theorem	$\sum_{n\ge 0}(a;q)_n z^n/(q;q)_n = (az;q)_\infty/(z;q)_\infty$	Theorem 8.1
q-Gauss	${}_2\phi_1(a,b;c;q,c/(ab)) = (c/a,c/b;q)_\infty/(c,c/(ab);q)_\infty$	Theorem 9.4
q-Pfaff–Saalschütz	Terminating balanced ${}_3\phi_2$ product evaluation	Theorem 9.6
q-Chu–Vandermonde	Two terminating ${}_2\phi_1$ evaluations	Theorem 9.7
Ramanujan ${}_1\psi_1$	${}_1\psi_1(a;b;q,z) = (q,b/a,az,q/(az);q)_\infty/(b,q/a,z,b/(az);q)_\infty$	Theorem 10.2
Jacobi triple product	$\sum_{n\in\mathbb{Z}}(-1)^n q^{\binom{n}{2}} z^n = (q,z,q/z;q)_\infty$	Theorem 13.1
Pentagonal theorem	$(q;q)_\infty = \sum_{n\in\mathbb{Z}}(-1)^n q^{n(3n-1)/2}$	Theorem 13.3
q-derivative	$D_q f(x) = (f(x)-f(qx))/((1-q)x)$	Theorem 11.1
Jackson integral	$\int_0^a f(x)\,\mathrm{d}_q x = (1-q)a\sum_{n\ge 0} q^n f(aq^n)$	Theorem 11.9
q-gamma	$\Gamma_q(x) = (1-q)^{1-x}(q;q)_\infty/(q^x;q)_\infty$	Theorem 12.2
Cyclotomic factorization	$\begin{bmatrix} n \\ k \end{bmatrix}_q = \prod_d \Phi_d(q)^{\lfloor n/d\rfloor - \lfloor k/d\rfloor - \lfloor (n-k)/d\rfloor}$	Theorem 16.1
q-Lucas	$\begin{bmatrix} ad+b \\ rd+s \end{bmatrix}_q \equiv \binom{a}{r}\begin{bmatrix} b \\ s \end{bmatrix}_q \pmod{\Phi_d(q)}$	Theorem 16.5
q-Babbage	$\begin{bmatrix} an \\ bn \end{bmatrix}_q \equiv \begin{bmatrix} a \\ b \end{bmatrix}_{q^{n^2}} \pmod{\Phi_n(q)^2}$	Theorem 16.8
Negative upper index	$\begin{bmatrix} -m \\ k \end{bmatrix}_q = (-1)^k q^{-mk-\binom{k}{2}}\begin{bmatrix} m+k-1 \\ k \end{bmatrix}_q$	Theorem 17.2
Geometric Newton basis	$P(x) = \sum_k c_k (-1)^k q^{\binom{k}{2}}(x;q^{-1})_k$	Theorem 17.10
Central reduction	$\begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2}/(-q;q)_{2k} = (q;q^2)_k/(q^2;q^2)_k$	Theorem 21.1
Bernoulli expansion	All-orders expansion of $\log(\begin{bmatrix} n \\ k \end{bmatrix}_{e^{-t}}/\binom{n}{k})$	Theorem 22.2

Appendix B

Limit and specialization dictionary

B.1 Elementary limits

For fixed nonnegative integers, the following transitions are direct consequences of the product formulas:

$$\lim_{q\to 1}[n]_q = n, \qquad \lim_{q\to 1}(1-q)^{-n}(q^a;q)_n = (a)_n,$$

$$\lim_{q\to 1}\begin{bmatrix} n \\ k \end{bmatrix}_q = \binom{n}{k}, \qquad \lim_{n\to\infty}\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{1}{(q;q)_k} \quad (|q|<1),$$

$$\lim_{q\to 1^-}\Gamma_q(x) = \Gamma(x), \qquad \lim_{q\to 0}\begin{bmatrix} n \\ k \end{bmatrix}_q = 1 \quad (0 \le k \le n).$$

Each has been proved in the main text; the relevant references are Theorems 1.17, 2.6, 12.5, 22.5 and 22.9.

B.2 Classical shadows of the main summations

Under the standard scaling $a = q^A$, $b = q^B$, and $q \to 1^-$, terminating basic hypergeometric sums tend to ordinary hypergeometric sums. The mechanism is

$$\frac{(q^A;q)_n}{(1-q)^n} \longrightarrow (A)_n, \qquad \frac{(q;q)_n}{(1-q)^n} \longrightarrow n!.$$

Thus:

1. the finite q-binomial theorem tends to Newton's binomial theorem;
2. q-Chu–Vandermonde tends to Chu–Vandermonde;
3. q-Pfaff–Saalschütz tends to Pfaff–Saalschütz;
4. the Jackson q-beta integral tends to Euler's beta integral;
5. Theorem 21.2 tends to Theorem 21.3;
6. Theorem 21.4 tends, at the level of its Lambert tail and prefactor, to Theorem 21.7.

B.3 Roots of unity

At a root of unity, factorial quotients should be replaced by polynomial formulas before evaluation. The basic dictionary is:

$$\Phi_d(q) \mid \begin{bmatrix} n \\ k \end{bmatrix}_q \iff k \bmod d > n \bmod d,$$

$$\begin{bmatrix} ad+b \\ rd+s \end{bmatrix}_q \equiv \binom{a}{r} \begin{bmatrix} b \\ s \end{bmatrix}_q \pmod{\Phi_d(q)},$$

$$\begin{bmatrix} n \\ k \end{bmatrix}_q \Bigg|_{q=-1} \text{ is given by Theorem 16.6.}$$

These statements turn apparently singular substitutions into finite enumerations and are the algebraic basis of the subset cyclic-sieving theorem.

Appendix C

Proof-dependency guide

The monograph can be read linearly, but several shorter routes are useful.

Route to Gaussian combinatorics

$$\text{finite products} \longrightarrow q\text{-Pascal} \longrightarrow \text{finite } q\text{-binomial theorem} \longrightarrow \text{partitions, paths, subspaces.}$$

Read Chapters 1 to 3, 5 and 6.

Route to product–sum identities

$$\text{infinite } q\text{-binomial theorem} \longrightarrow \text{Heine transformation} \longrightarrow q\text{-Gauss},$$
$$\text{negative indices and bilateral convergence} \longrightarrow {}_1\psi_1,$$
$$\text{Euler expansions} \longrightarrow \text{Jacobi triple product} \longrightarrow \text{Bailey pairs.}$$

Read Chapters 8 to 10, 13 and 15.

Route to arithmetic

$$1 - q^n = \prod_{d|n} \Phi_d(q) \longrightarrow \text{carry criterion} \longrightarrow q\text{-Lucas} \longrightarrow \text{cyclic sieving and } q\text{-Babbage.}$$

Read Chapter 16.

Route to analytic deformation

$$\log(a;q)_\infty \longrightarrow \text{Lambert series and dilogarithms} \longrightarrow q\text{-gamma}$$
$$\longrightarrow \text{central sums and Bernoulli asymptotics.}$$

Read Chapters 1, 12, 21 and 22.

Route to dyadic self-similarity

$$\text{finite } q\text{-binomial theorem at } q = \tfrac{1}{2} \longrightarrow \text{geometric nodal annihilation} \longrightarrow \text{Thue–Morse jets}$$
$$\longrightarrow \text{exact Fabius values} \longrightarrow \text{Toeplitz and spectral products.}$$

Read Chapters 3 and 18 to 20.

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